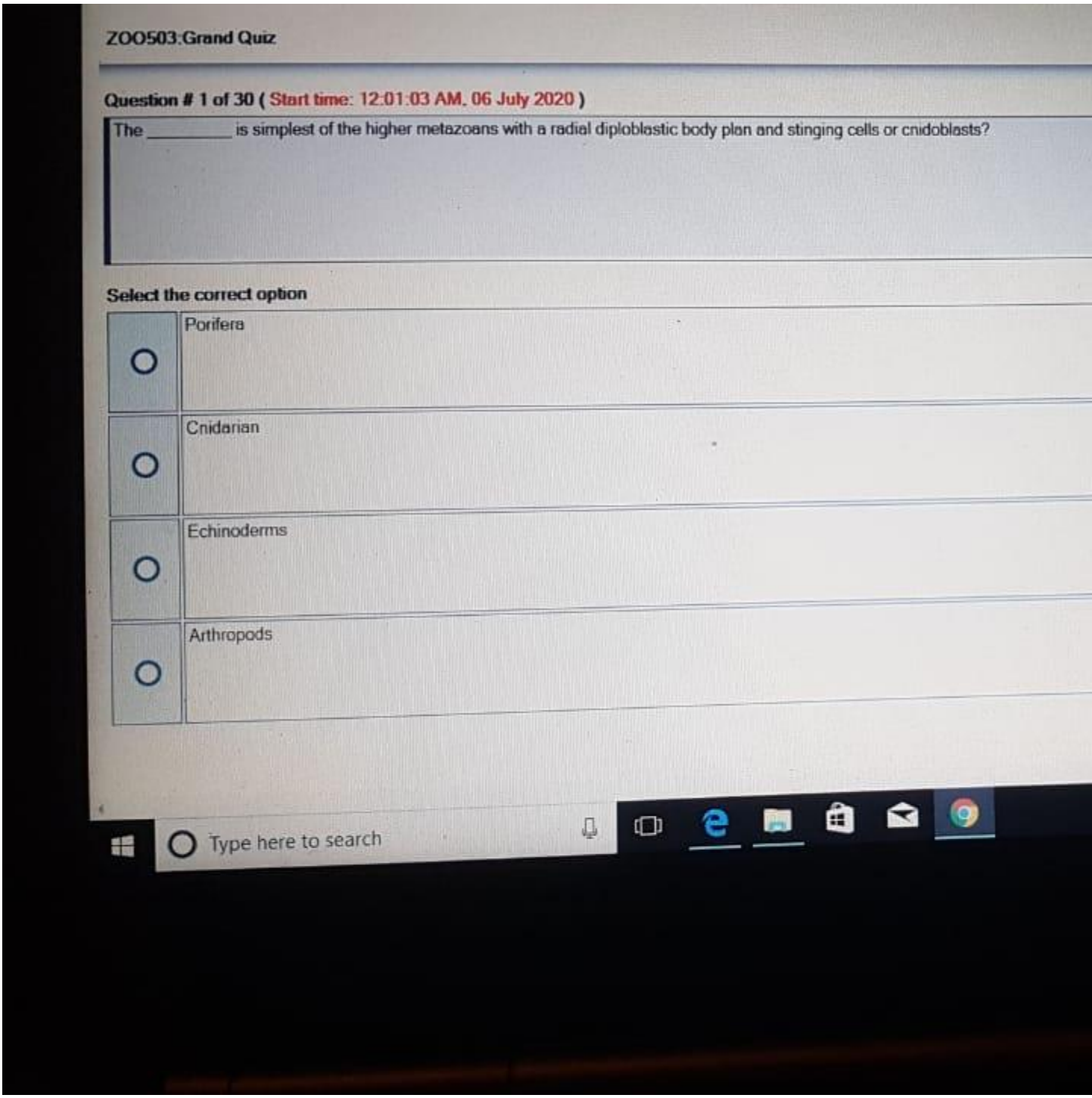
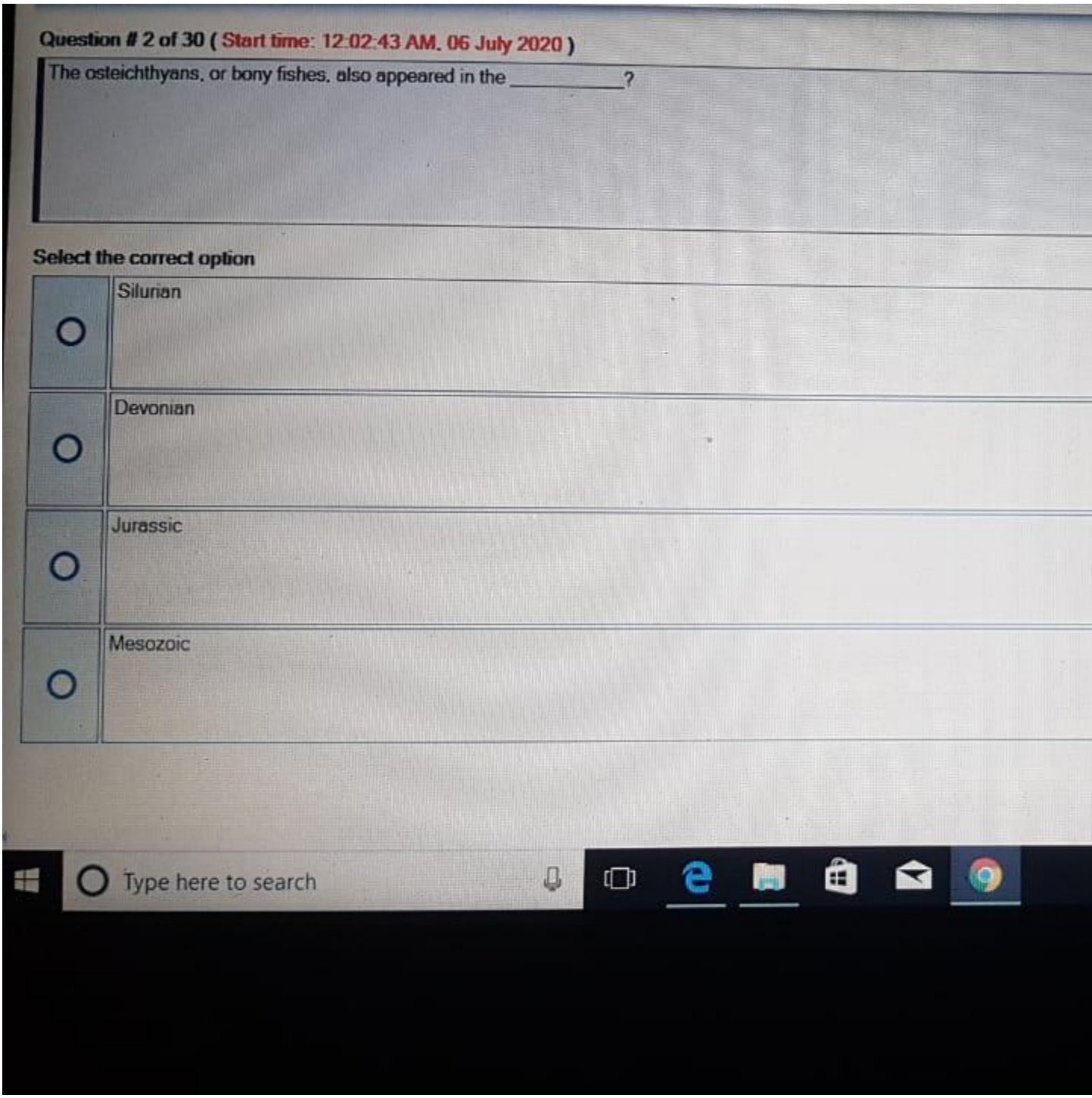

ZOO503:Grand Quiz
Question # 1 of 30 (Start time: 12:01:03 AM, 06 July 2020)
The _____ is simplest of the higher metazoans with a radial diploblastic body plan and stinging cells or cnidoblasts?
Select the correct option
Porifera
Cnidarian
Echinoderms
Arthropods
Type here to search


Question # 2 of 30 (Start time: 12:02:43 AM, 06 July 2020)
The osteichthyans, or bony fishes, also appeared in the _____?
Select the correct option
Silurian
Devonian
Jurassic
Mesozoic
Type here to search

In porifera, _____ have Sacs with a single chamber lined by flagellate cells?

Select the correct option

- Ascon
- Sycon
- Leucon
- None

Type here to search

72% 12:04

Google

K

bony fishes appeared in

ALL IMAGES NEWS VIDEOS MAPS

Silurian



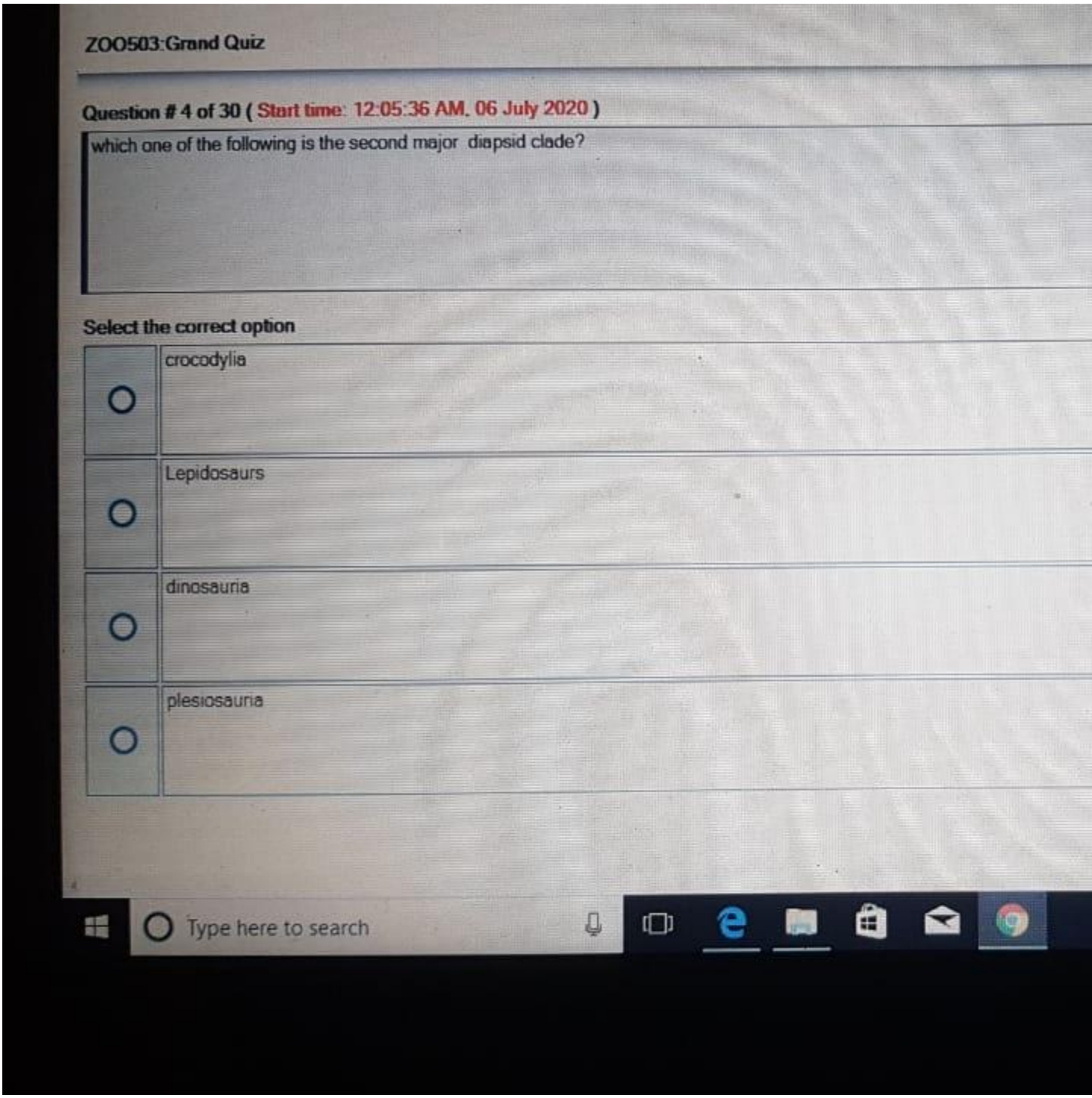
Bony fishes, class **Osteichthyes** are characterised by **bony** skeleton rather than cartilage. They **appeared** in the late Silurian, about 419 million years ago.

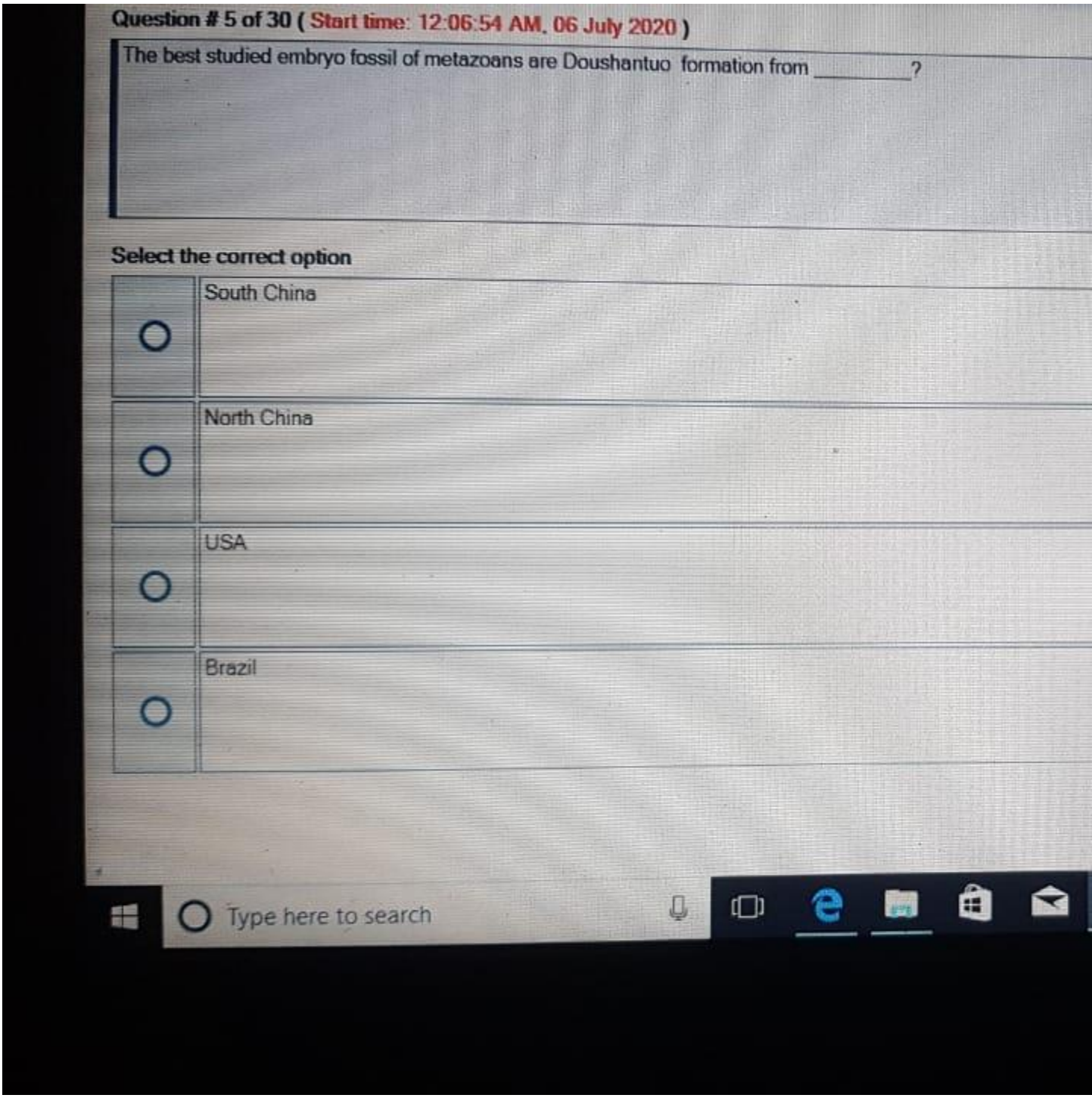
https://en.m.wikipedia.org › wiki

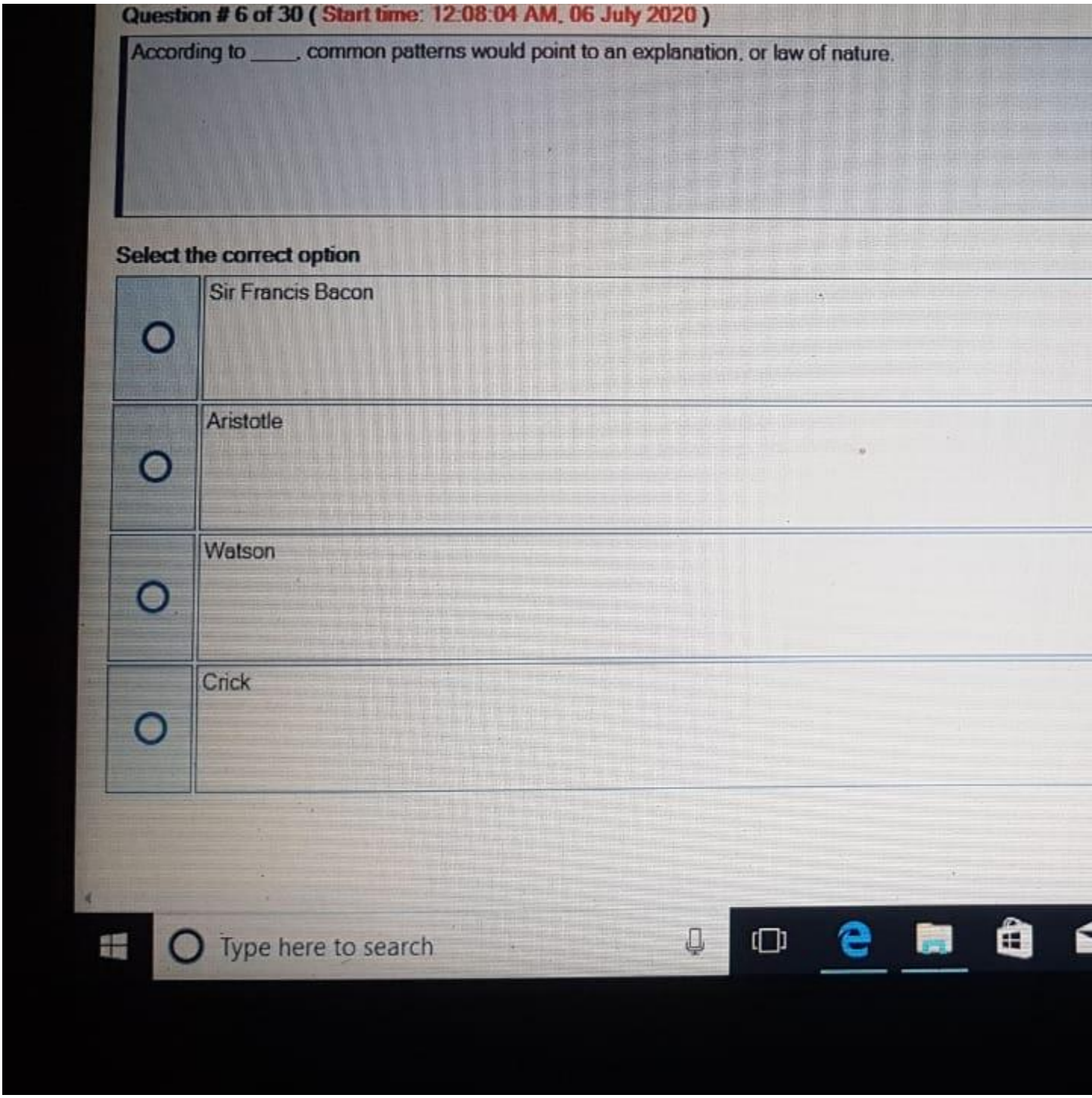
Evolution of fish - Wikipedia

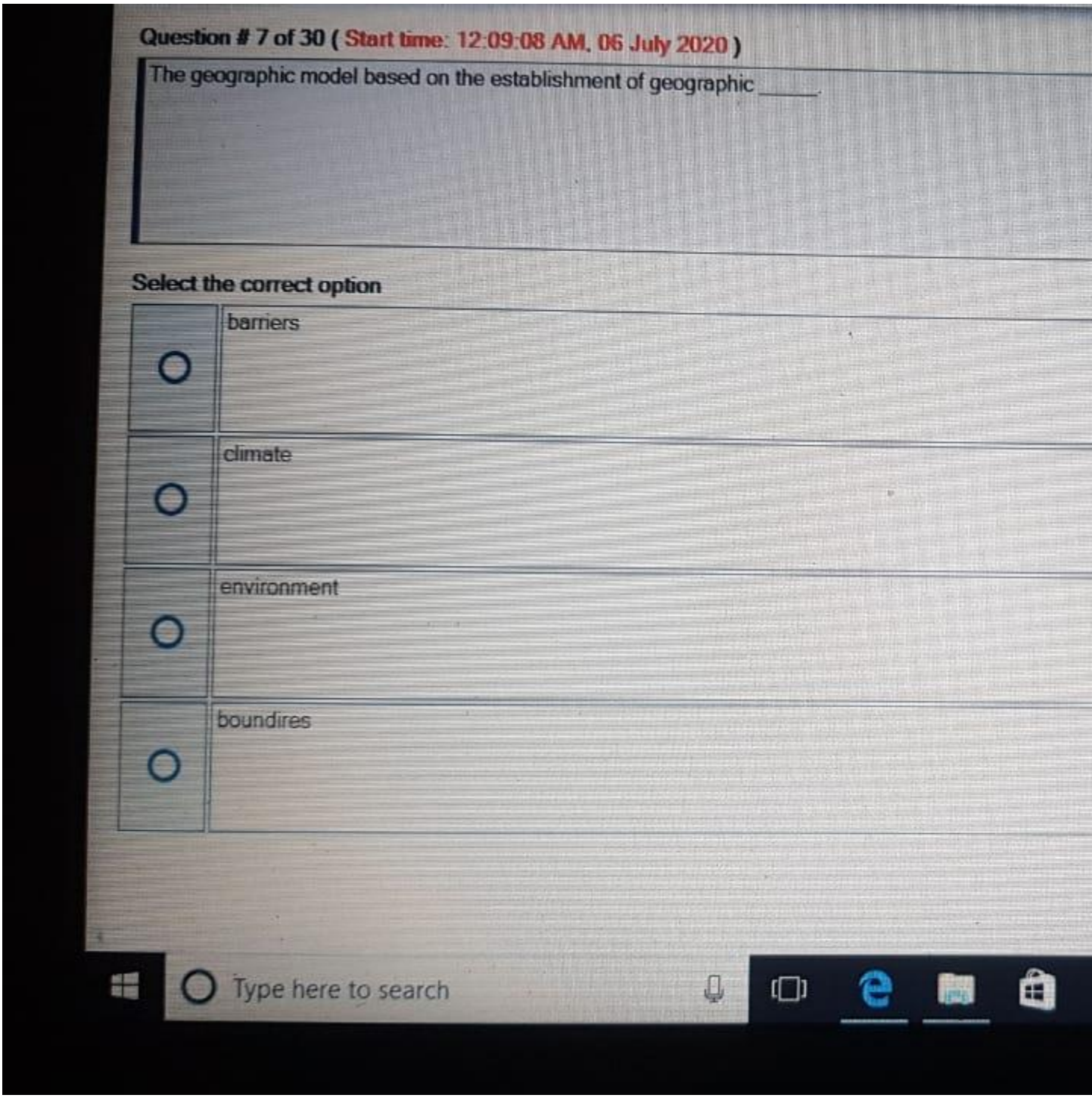
More about Silurian

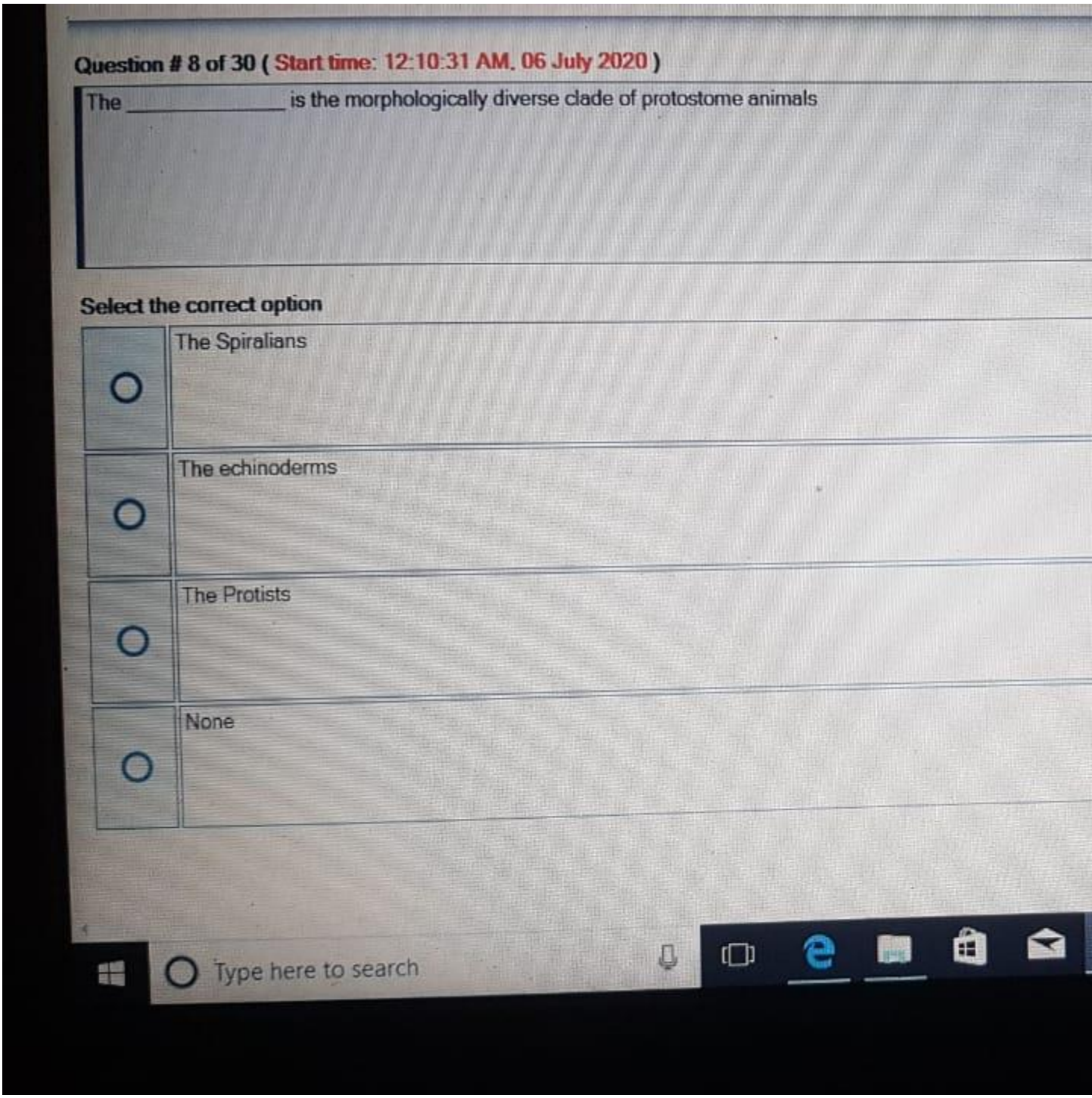
PEOPLE ALSO ASK

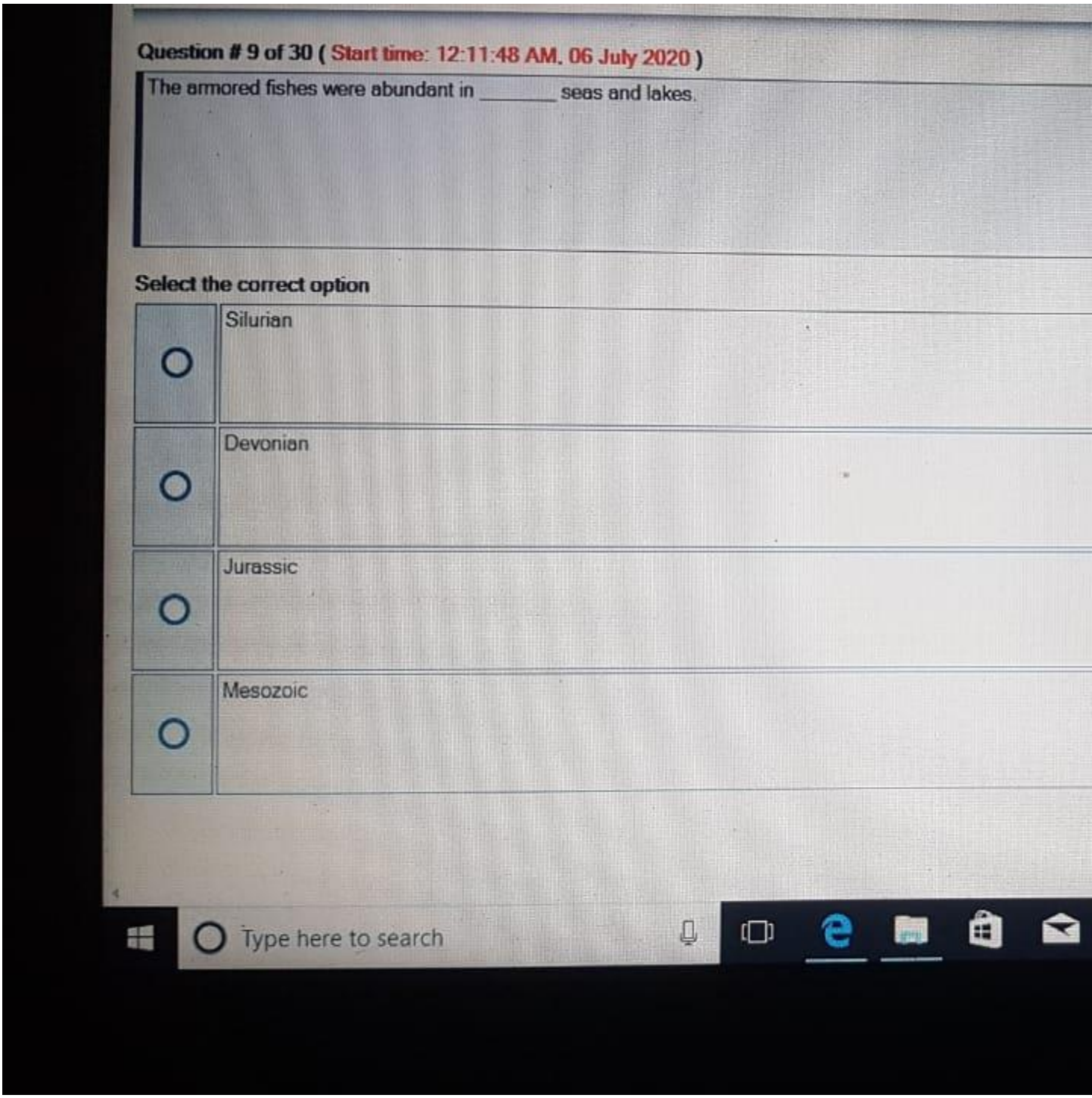

ZOO503:Grand Quiz
Question # 4 of 30 (Start time: 12:05:36 AM, 06 July 2020)
which one of the following is the second major diapsid clade?
Select the correct option
crocodylia
Lepidosaurs
dinosauria
plesiosauria
Type here to search


Question # 5 of 30 (Start time: 12:06:54 AM, 06 July 2020)
The best studied embryo fossil of metazoans are Doushantuo formation from _____?
Select the correct option
South China
North China
USA
Brazil
Type here to search


Question # 6 of 30 (Start time: 12:08:04 AM, 06 July 2020)
According to ____, common patterns would point to an explanation, or law of nature.
Select the correct option
Sir Francis Bacon
Aristotle
Watson
Crick
Type here to search


Question # 7 of 30 (Start time: 12:09:08 AM, 06 July 2020)
The geographic model based on the establishment of geographic _____.
Select the correct option
barriers
climate
environment
boundires
Type here to search


Question # 8 of 30 (Start time: 12:10:31 AM, 06 July 2020)
The _____ is the morphologically diverse clade of protostome animals
Select the correct option
The Spiralians
The echinoderms
The Protists
None
Type here to search



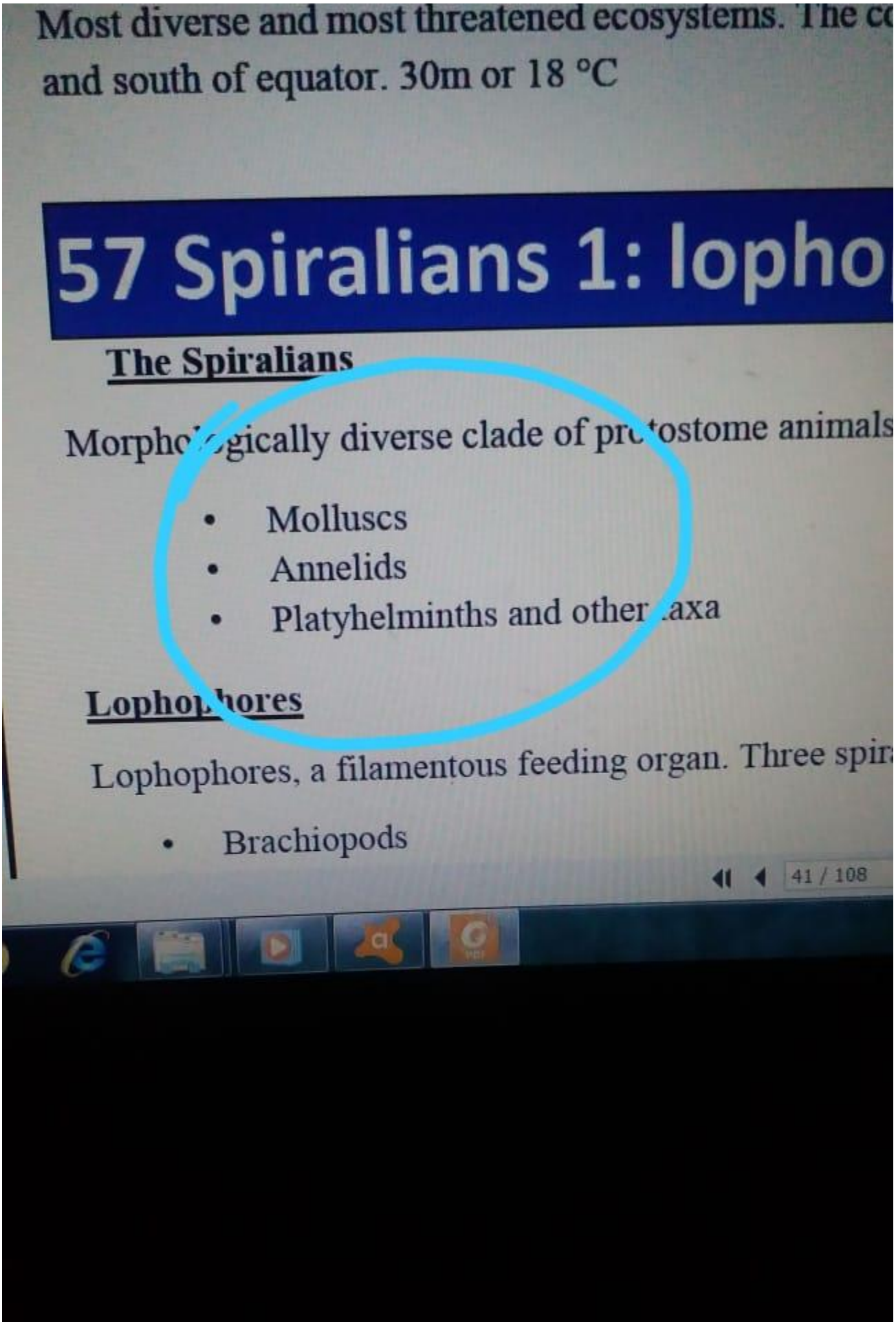
Question # 9 of 30 (Start time: 12:11:48 AM, 06 July 2020)

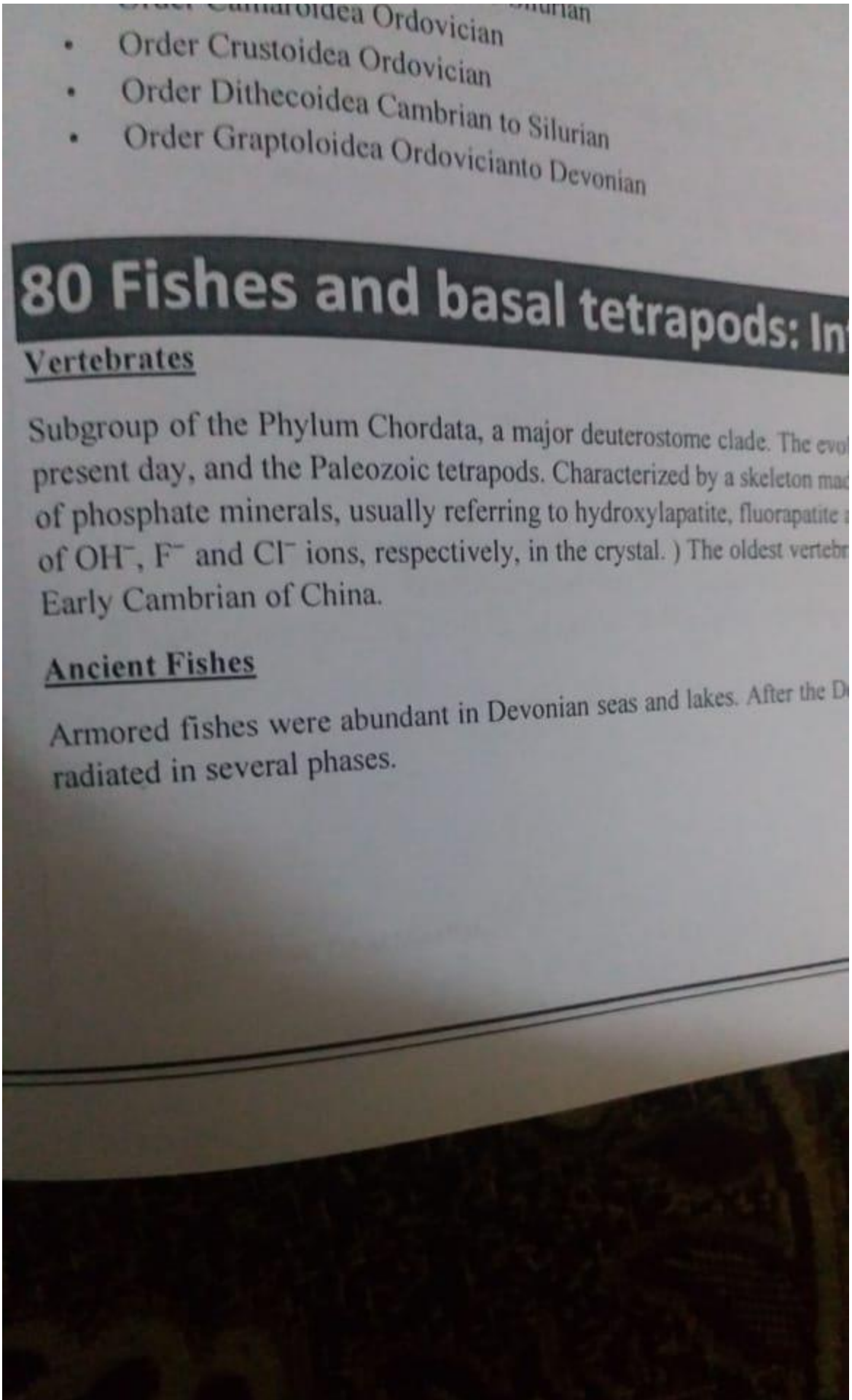
The armored fishes were abundant in _____ seas and lakes.

Select the correct option

- Silurian
- Devonian
- Jurassic
- Mesozoic

Type here to search


Most diverse and most threatened ecosystems. The c
and south of equator. 30m or 18 °C
57 Spiralians 1: lopho
The Spiralians
Morpho gically diverse clade of pr tostome animals
Molluscs
Annelids
Platyhelminths and other axa
Lophophores
Lophophores, a filamentous feeding organ. Three spir
Brachiopods
41 / 108



- Order Crustoidea Ordovician
- Order Dithecoidea Cambrian to Silurian
- Order Graptoloidea Ordovicianto Devonian

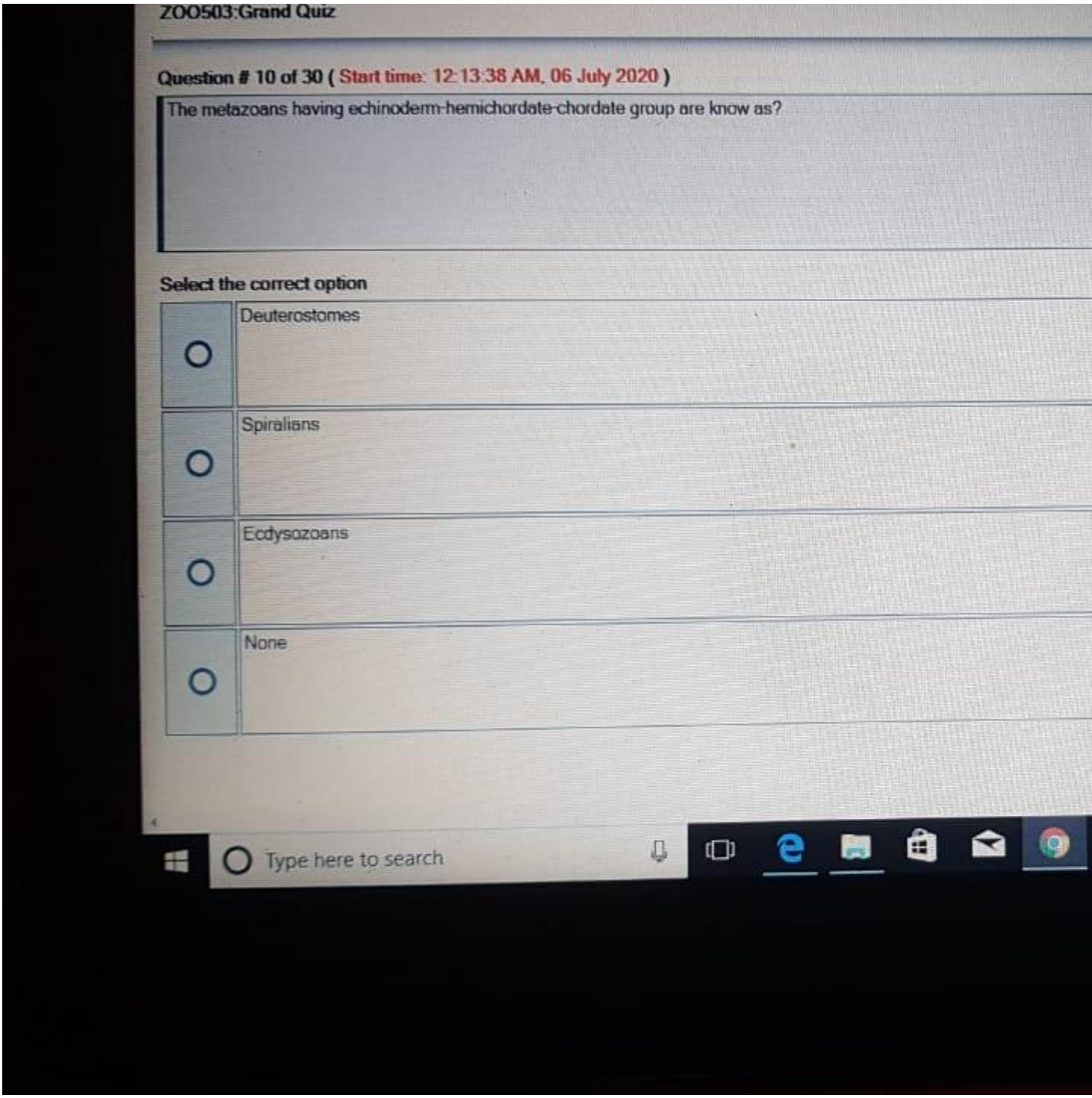
80 Fishes and basal tetrapods: In

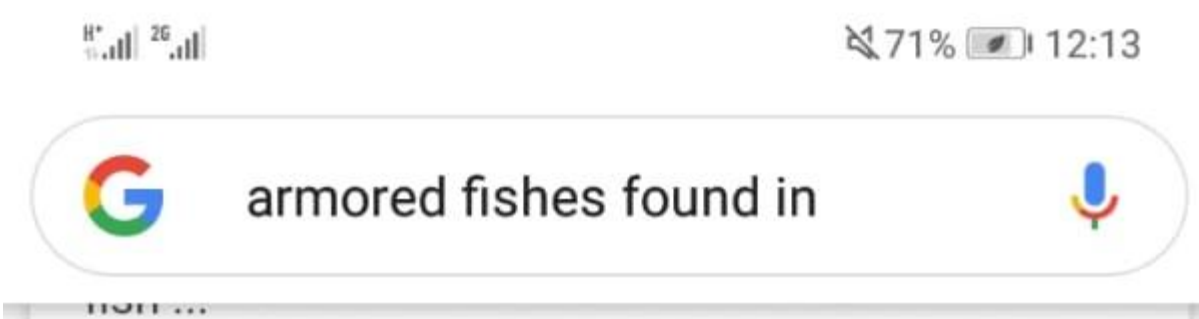
Vertebrates

Subgroup of the Phylum Chordata, a major deuterostome clade. The evo present day, and the Paleozoic tetrapods. Characterized by a skeleton mad of phosphate minerals, usually referring to hydroxylapatite, fluorapatite of OH^-, F^- and Cl^- ions, respectively, in the crystal.) The oldest vertebr Early Cambrian of China.

Ancient Fishes

Armored fishes were abundant in Devonian seas and lakes. After the D radiated in several phases.


ZOO503:Grand Quiz
Question # 10 of 30 (Start time: 12:13:38 AM, 06 July 2020)
The metazoans having echinoderm-hemichordate-chordate group are know as?
Select the correct option
Deuterostomes
Spiralians
Ecdysozoans
None
Type here to search



PEOPLE ALSO ASK

When did armored fishes become extinct?

Diverse **Armored Fishes**. Placoderms are arguably the most diverse groups of the early jawed **fishes**. They first appeared in the Early Silurian and diversified dramatically during the Devonian, but **became extinct** at the end of that period. Jul 9, 2005

www.devoniantimes.org › placoderm

More about Placoderms - Devonian Times

More results

Do Placoderms still exist?

Origin of the metazoans

lence

body plans have appeared in the fossil record, most animals have tr
al body layers.

Data

main groupings of animals:

Deuterostomes
Spiralians
Ecdysozoans

and Ecdysozoans:

stomes. Three Groups

The deuterostomes (echinoderm–hemichordate–chordate group)
The Spiralians (mollusk–annelid–brachiopod–bryozoans group)
The ecdysozoans (arthropod–nematode–priapulid plus other taxa grou

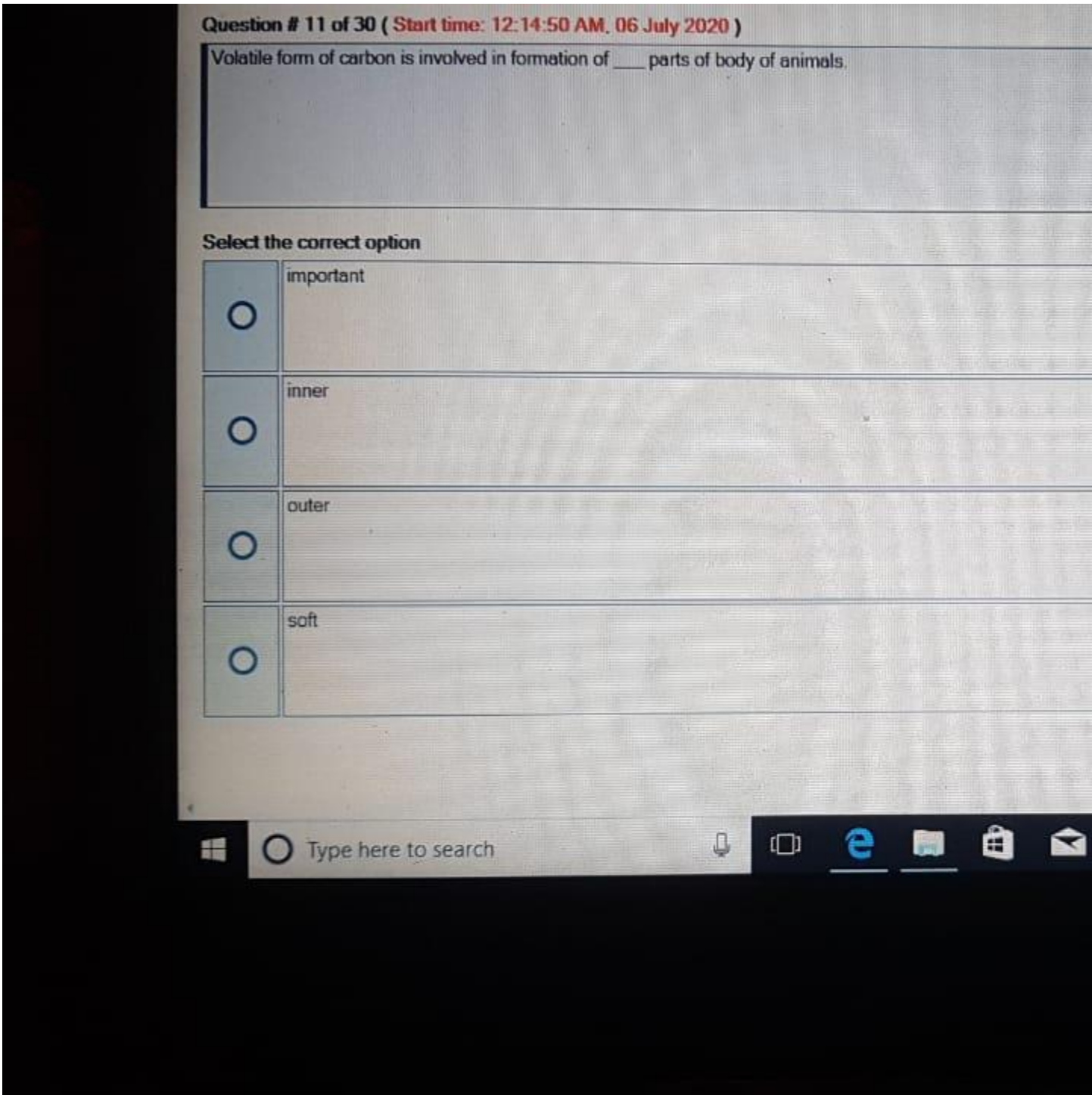
of evidence

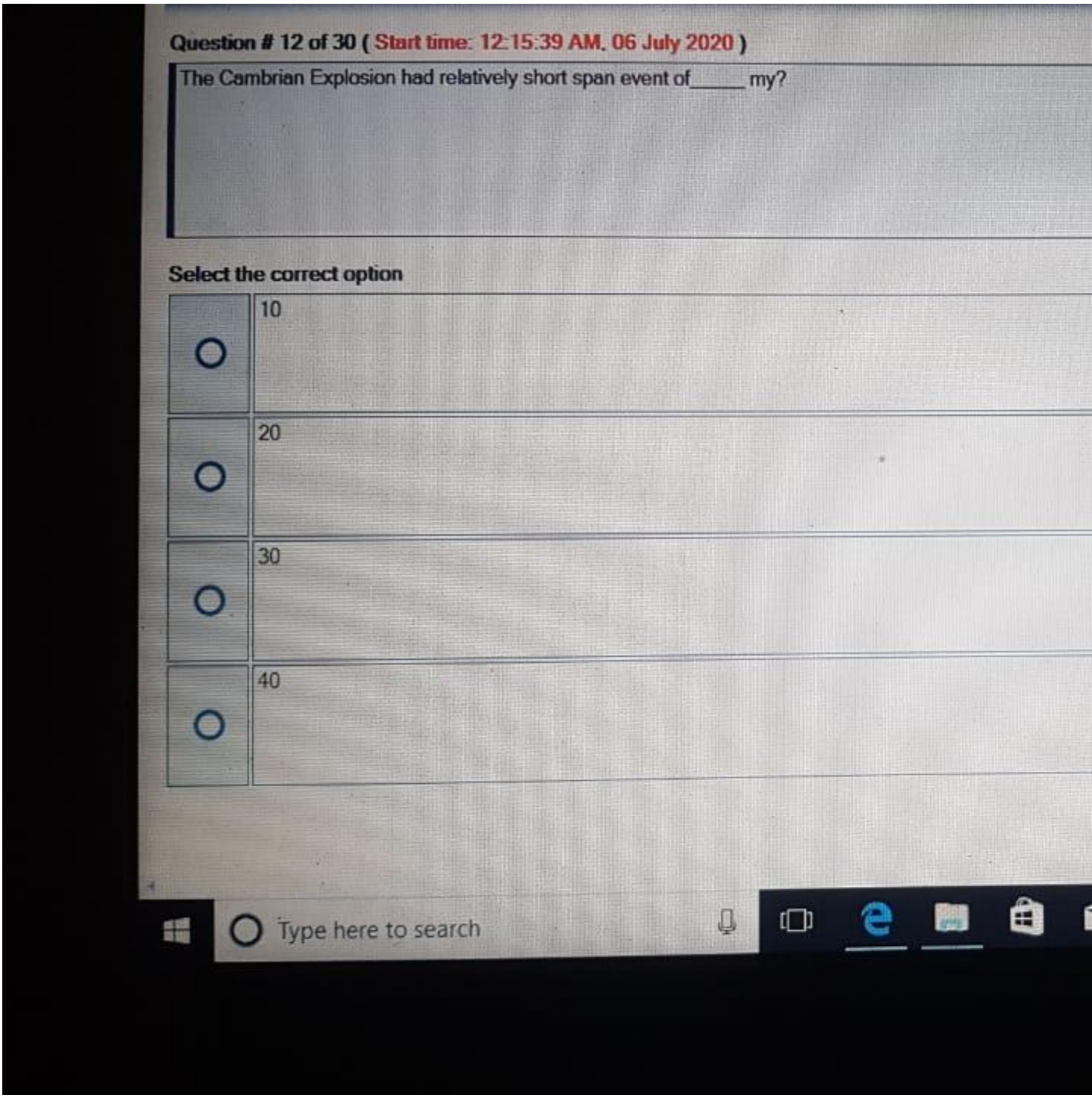
Body fossils
Trace fossils
fossil embryos
The mol. Clock
Biomarkers

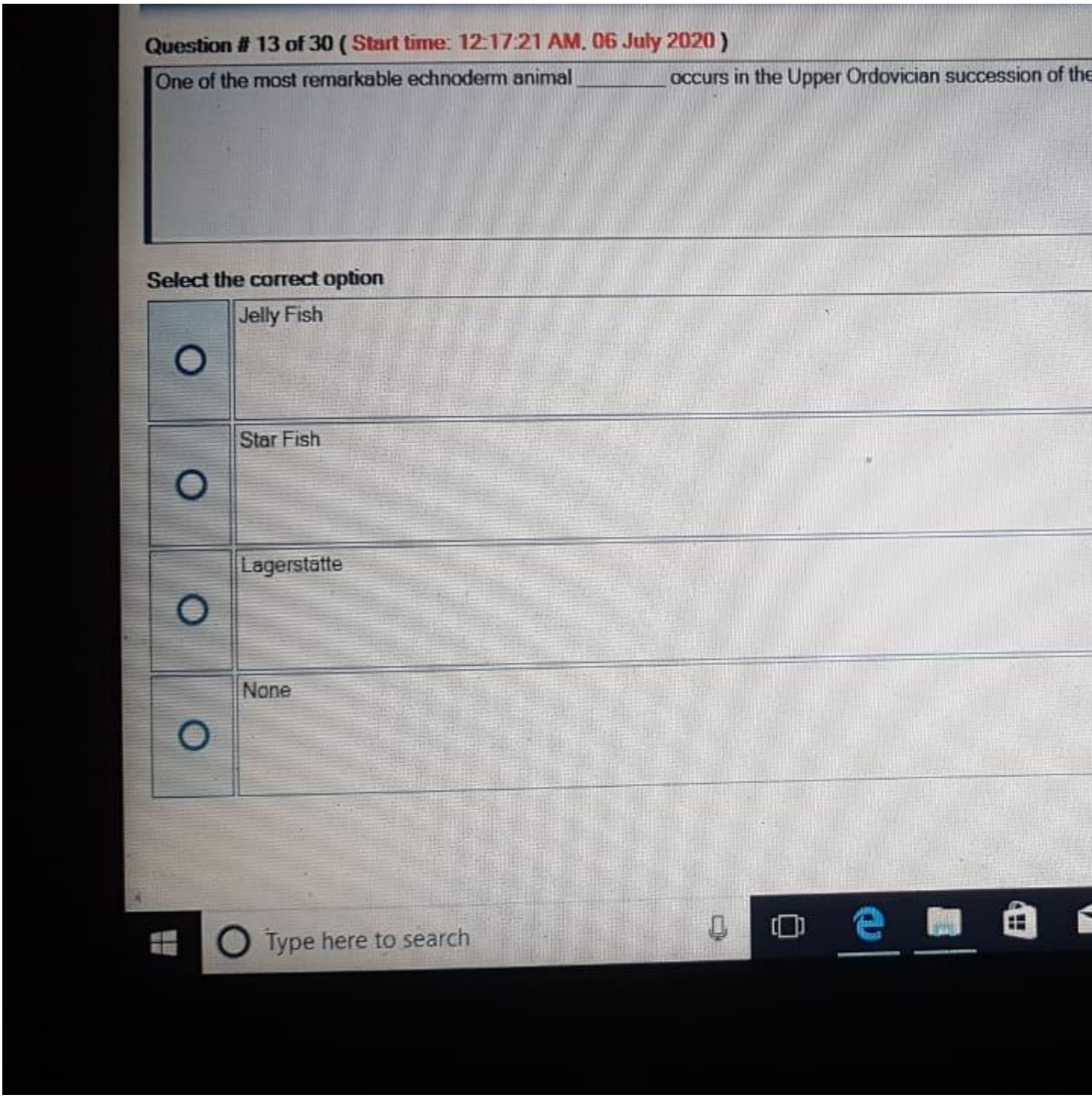
netazoans originated before Ediacaran 600 Ma

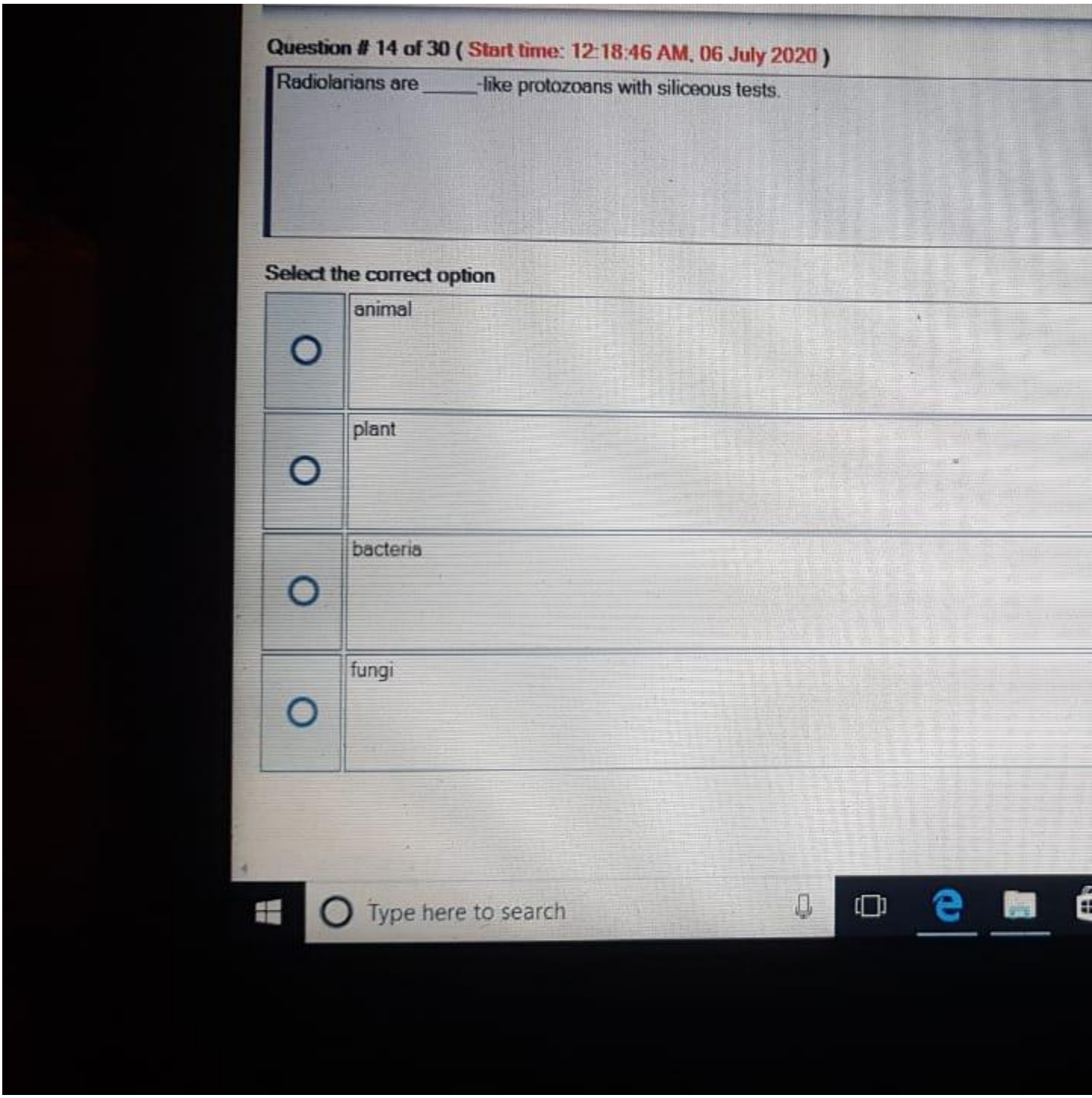
nce or Design

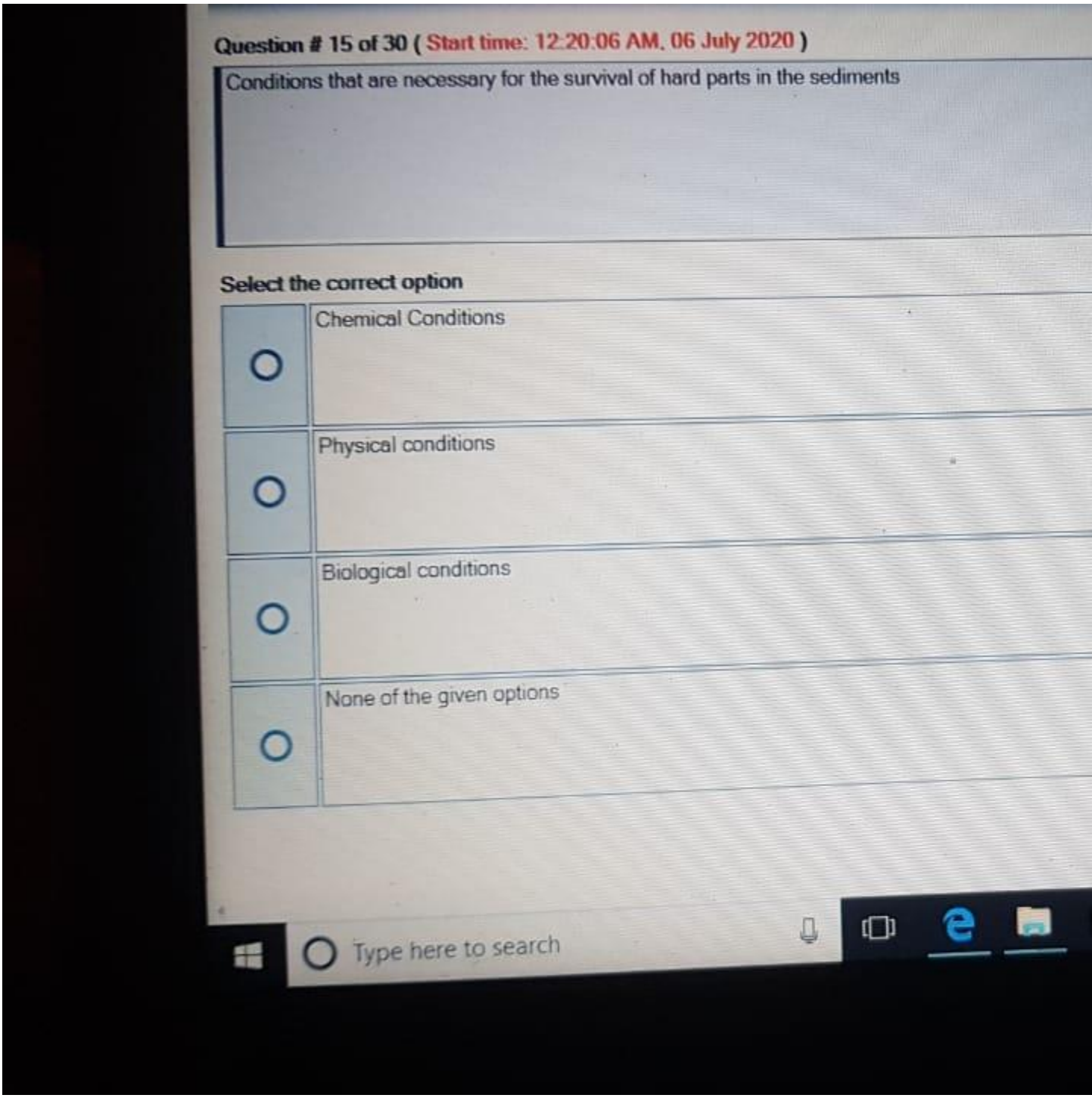
Earth by coincidence or design was a pivotal event in metazoan hi
n glaciation.

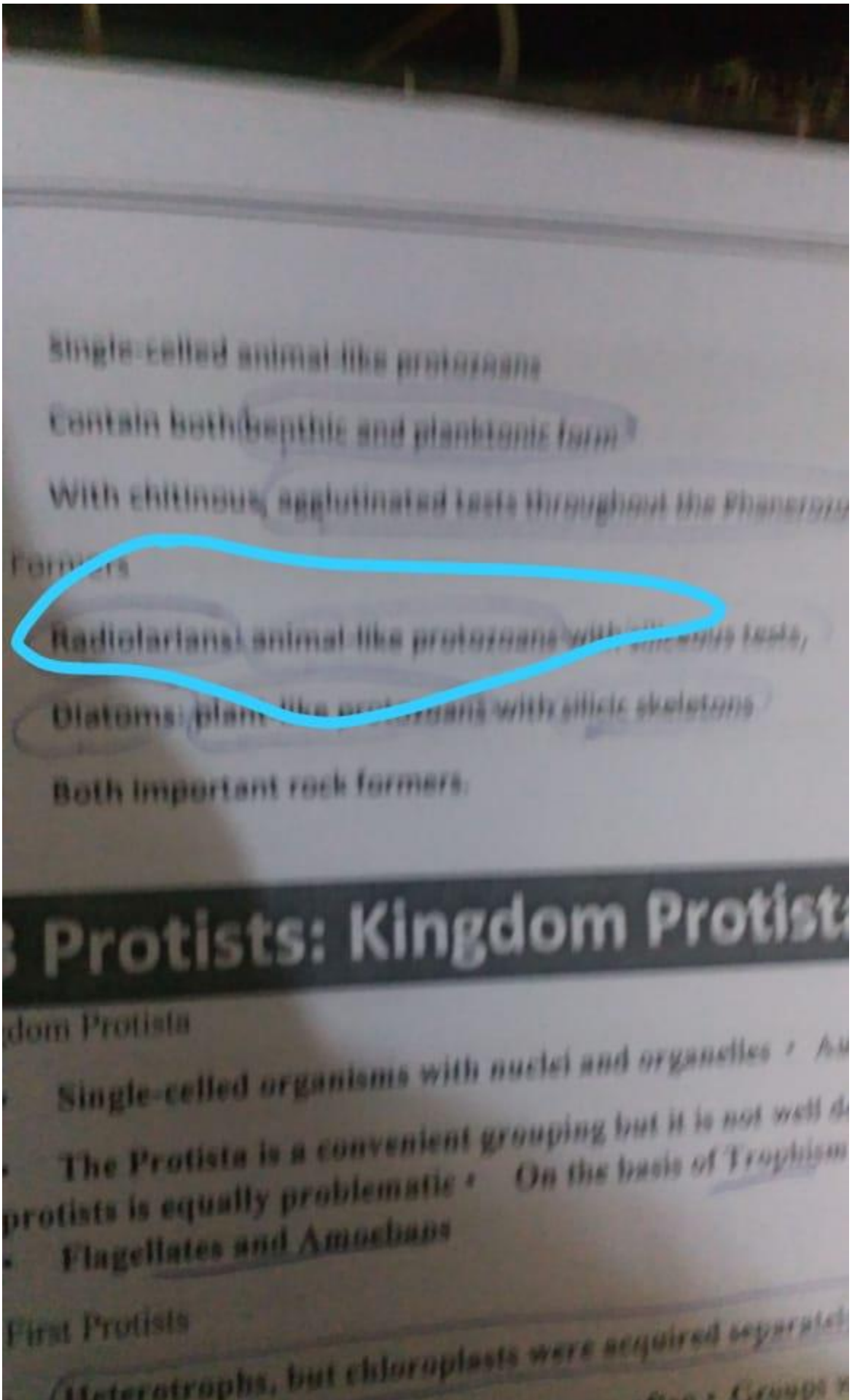

Question # 11 of 30 (Start time: 12:14:50 AM, 06 July 2020)
Volatile form of carbon is involved in formation of ___ parts of body of animals.
Select the correct option
important
inner
outer
soft
Type here to search

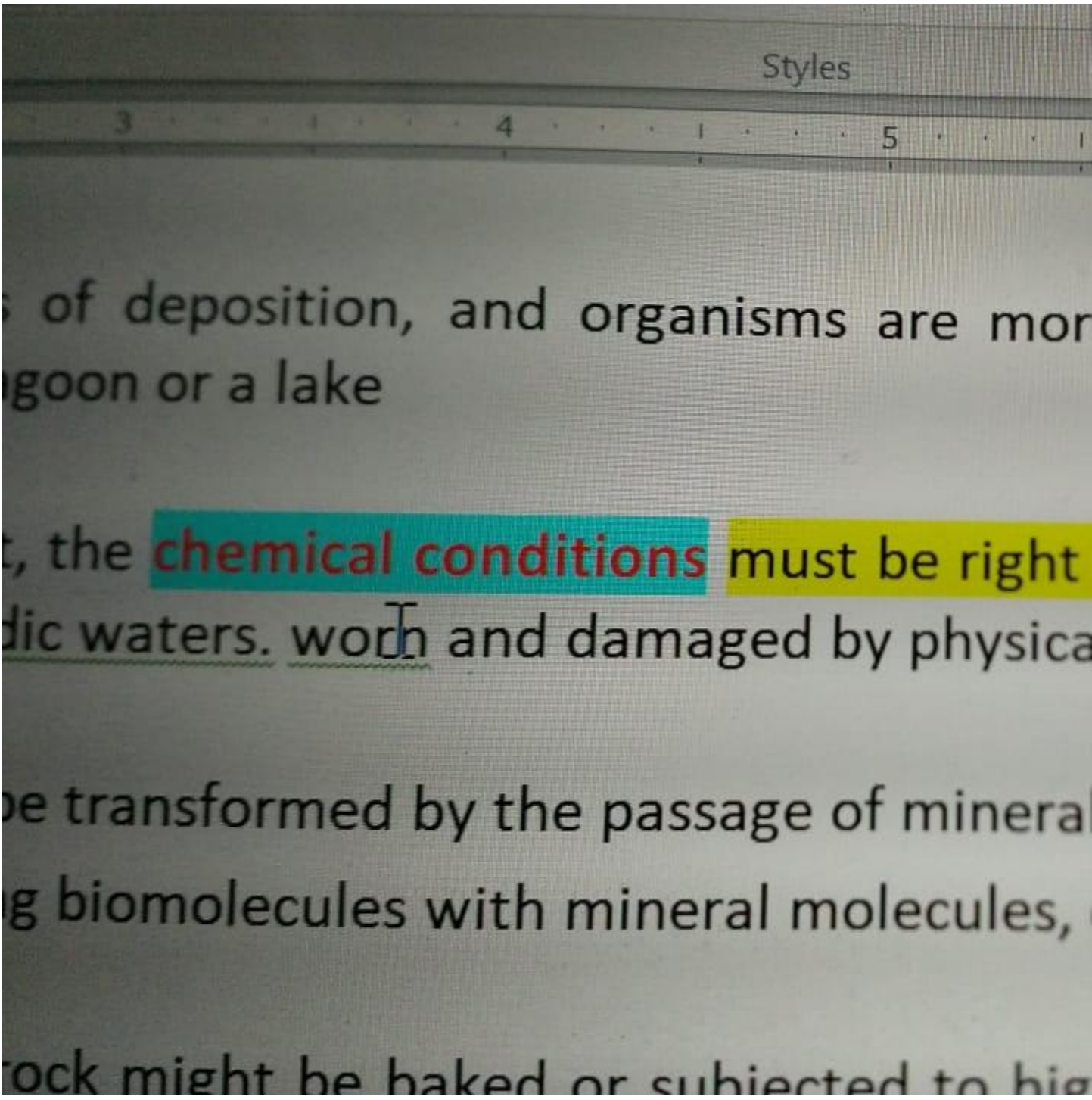

Question # 12 of 30 (Start time: 12:15:39 AM, 06 July 2020)
The Cambrian Explosion had relatively short span event of_____ my?
Select the correct option
10
20
30
40
Type here to search

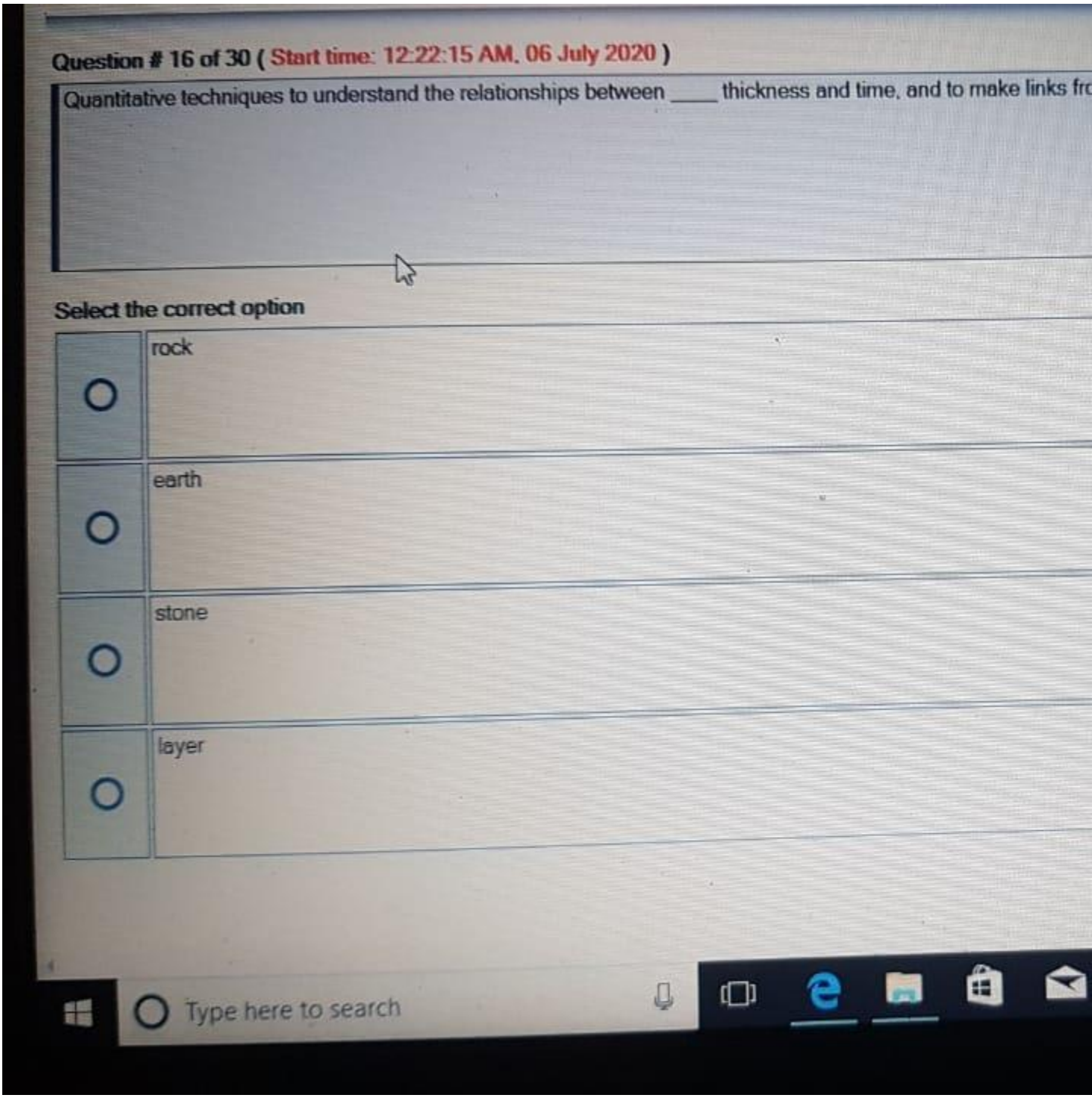

Question # 13 of 30 (Start time: 12:17:21 AM, 06 July 2020)
One of the most remarkable echnoderm animal _____ occurs in the Upper Ordovician succession of the
Select the correct option
Jelly Fish
Star Fish
Lagerstätte
None
Type here to search


Question # 14 of 30 (Start time: 12:18:46 AM, 06 July 2020)
Radiolarians are _____-like protozoans with siliceous tests.
Select the correct option
animal
plant
bacteria
fungi
Type here to search


Question # 15 of 30 (Start time: 12:20:06 AM, 06 July 2020)
Conditions that are necessary for the survival of hard parts in the sediments
Select the correct option
Chemical Conditions
Physical conditions
Biological conditions
None of the given options
Type here to search


Single-celled animal-like protozoans
Contain both benthic and planktonic form
With chitinous, agglutinated tests throughout the Phanerozoic
Foraminifera
Radiolarians: animal-like protozoans with siliceous tests,
Diatoms: plant-like protozoans with silicic skeletons
Both important rock formers.
Protists: Kingdom Protista
dom Protista
Single-celled organisms with nuclei and organelles
The Protista is a convenient grouping but it is not well d
protists is equally problematic
On the basis of Trophism
Flagellates and Amoebans
First Protists


Styles
of deposition, and organisms are mor
goon or a lake
, the chemical conditions must be right
dic waters. worn and damaged by physica
e transformed by the passage of mineral
g biomolecules with mineral molecules,
ock might be baked or subjected to hig

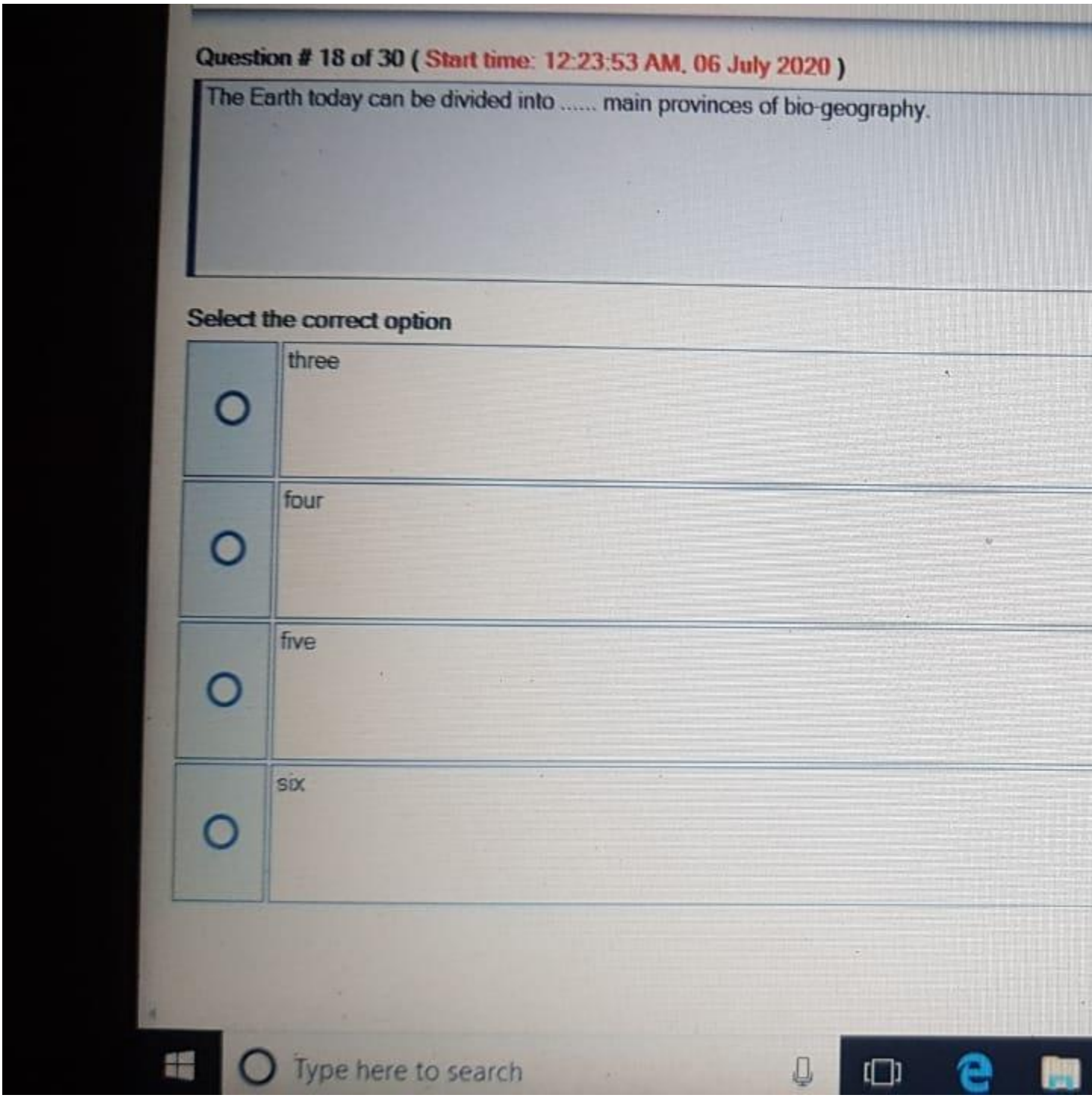

Question # 16 of 30 (Start time: 12:22:15 AM, 06 July 2020)
Quantitative techniques to understand the relationships between ____ thickness and time, and to make links fro
Select the correct option
rock
earth
stone
layer
Type here to search

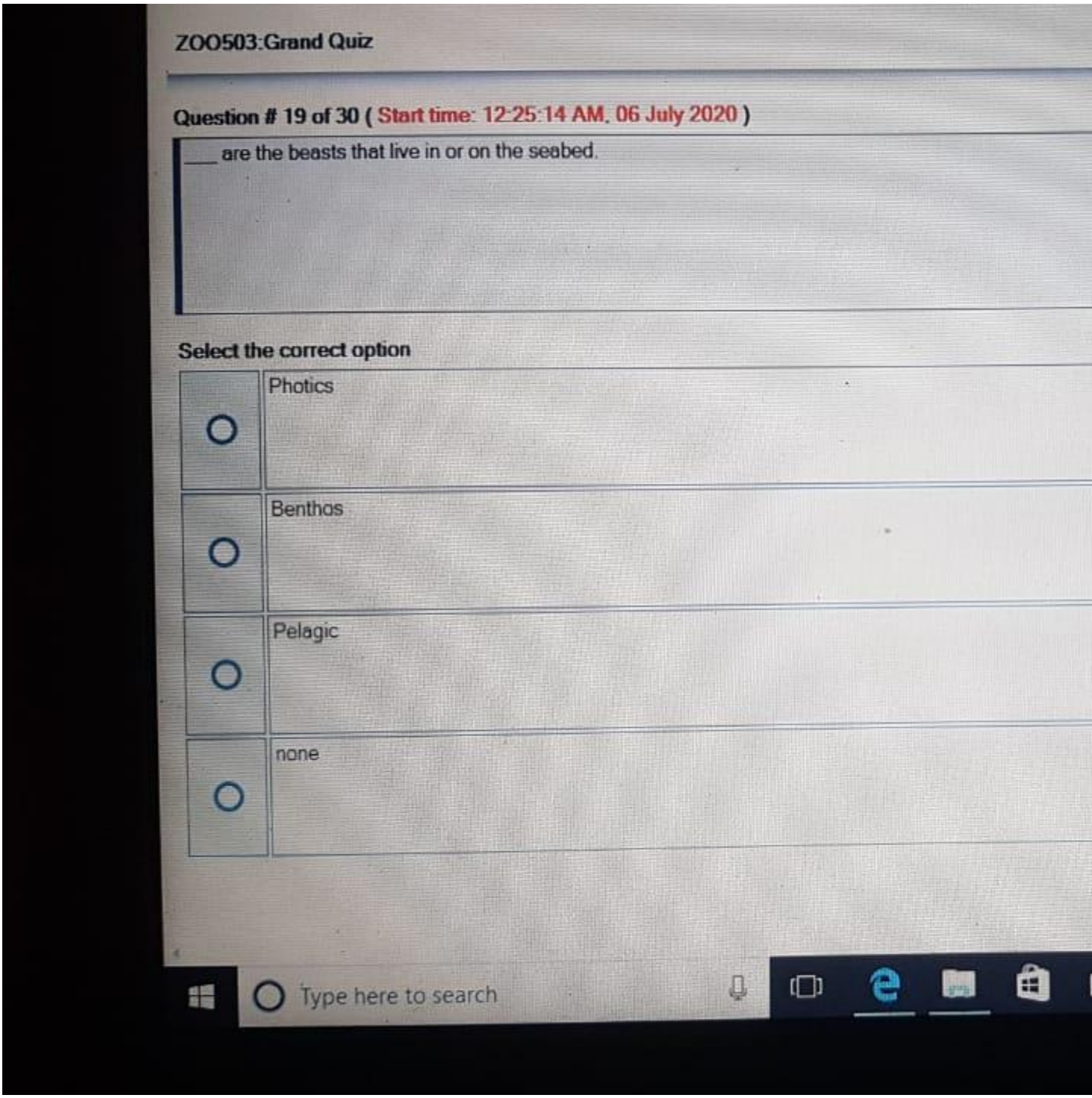
Question # 17 of 30 (Start time: 12:23:15 AM, 06 July 2020)

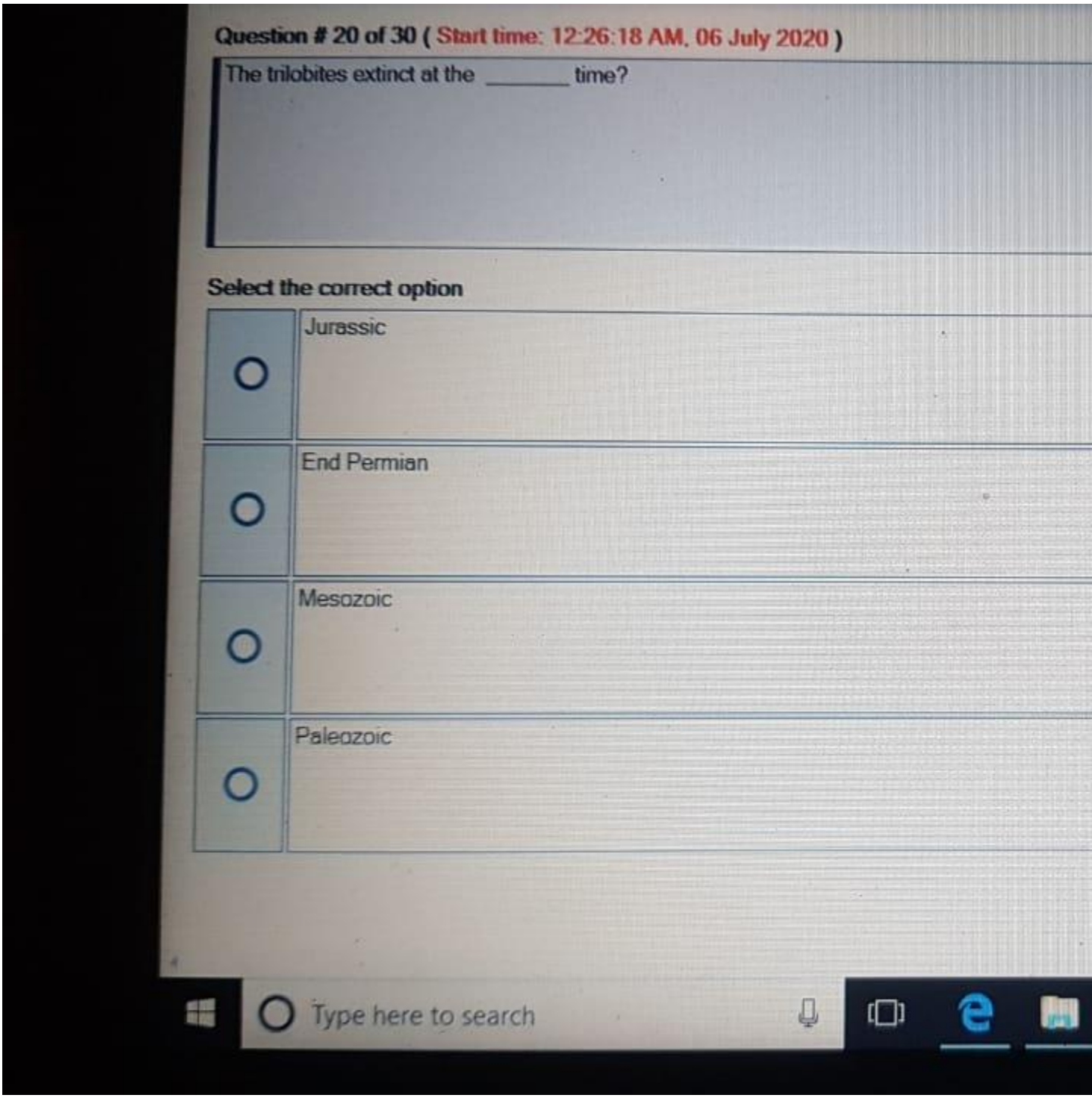
The class insecta of hexopods have _____limbs on their body?

Select the correct option

- 4
- 5
- 6
- 7


Question # 18 of 30 (Start time: 12:23:53 AM, 06 July 2020)
The Earth today can be divided into ..... main provinces of bio-geography.
Select the correct option
three
four
five
six
Type here to search


ZOO503:Grand Quiz
Question # 19 of 30 (Start time: 12:25:14 AM, 06 July 2020)
___ are the beasts that live in or on the seabed.
Select the correct option
Photics
Benthos
Pelagic
none
Type here to search


Question # 20 of 30 (Start time: 12:26:18 AM, 06 July 2020)
The trilobites extinct at the _____ time?
Select the correct option
Jurassic
End Permian
Mesozoic
Paleozoic
Type here to search

69% 12:28

trilobites extinct in





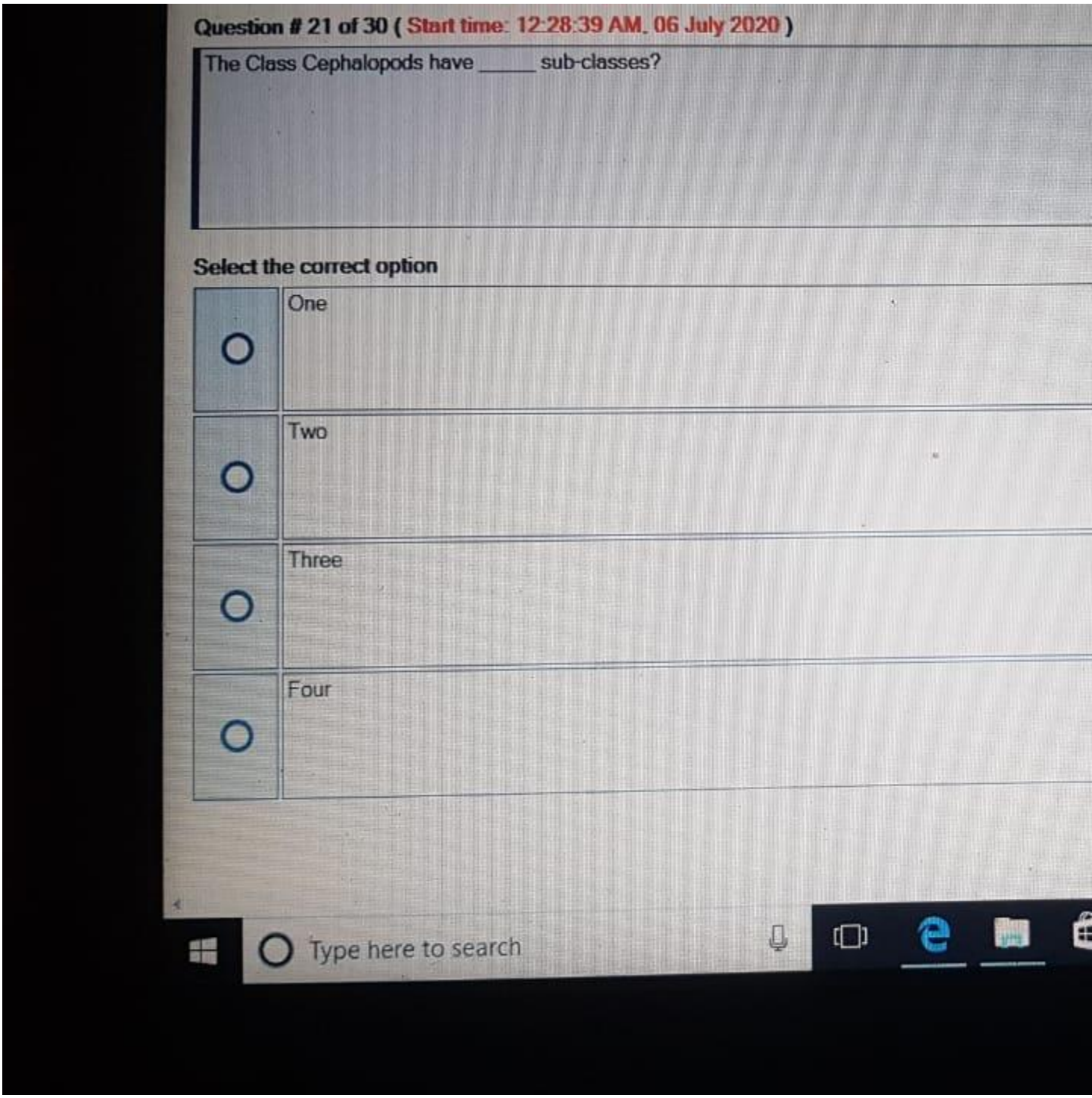
Permian about 252 million years ago

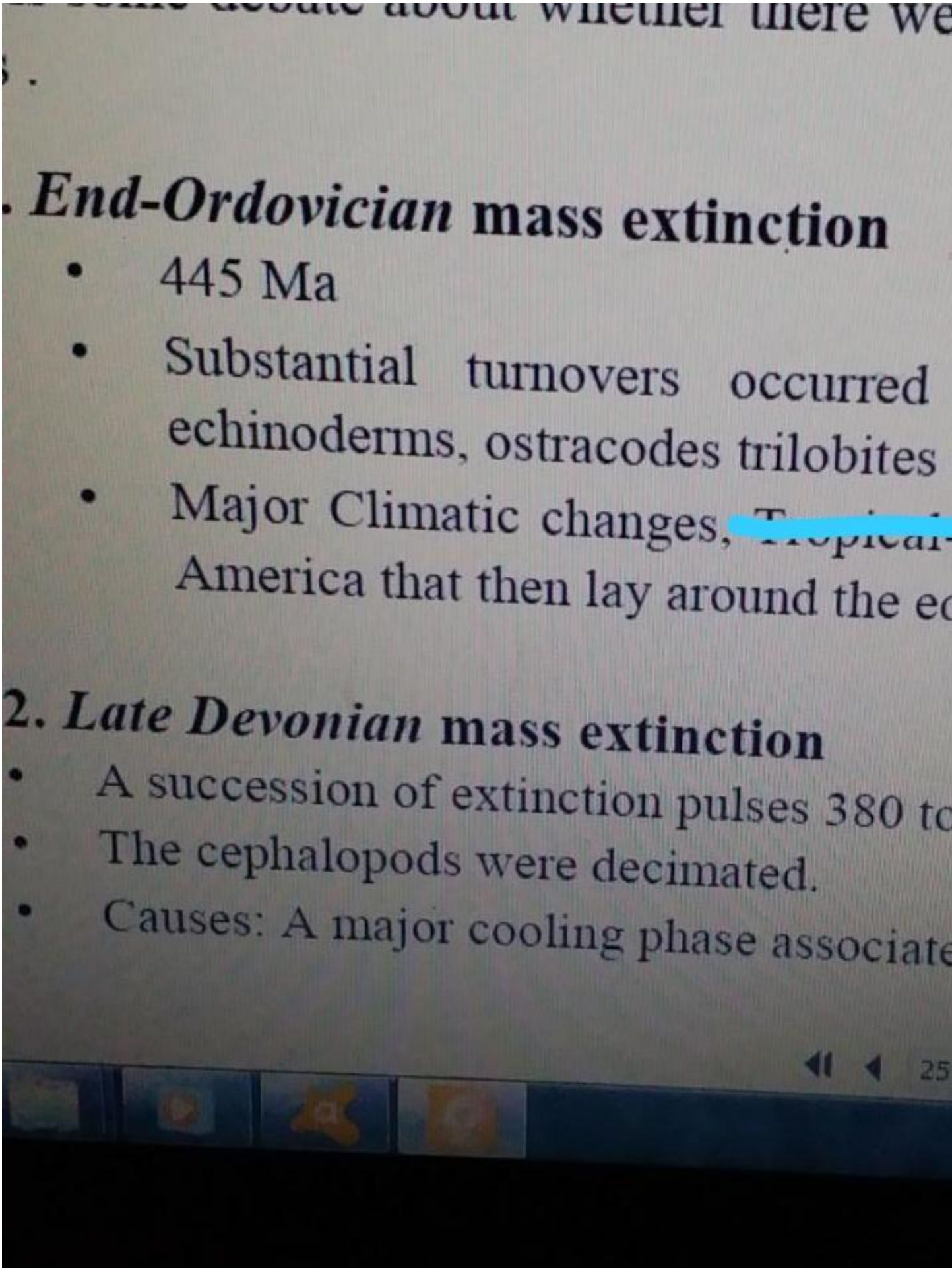
The last extant **trilobites** finally disappeared in the mass **extinction** at the end of the Permian about 252 million years ago. **Trilobites** were among the most successful of all early animals, existing in oceans for almost 300 million years.

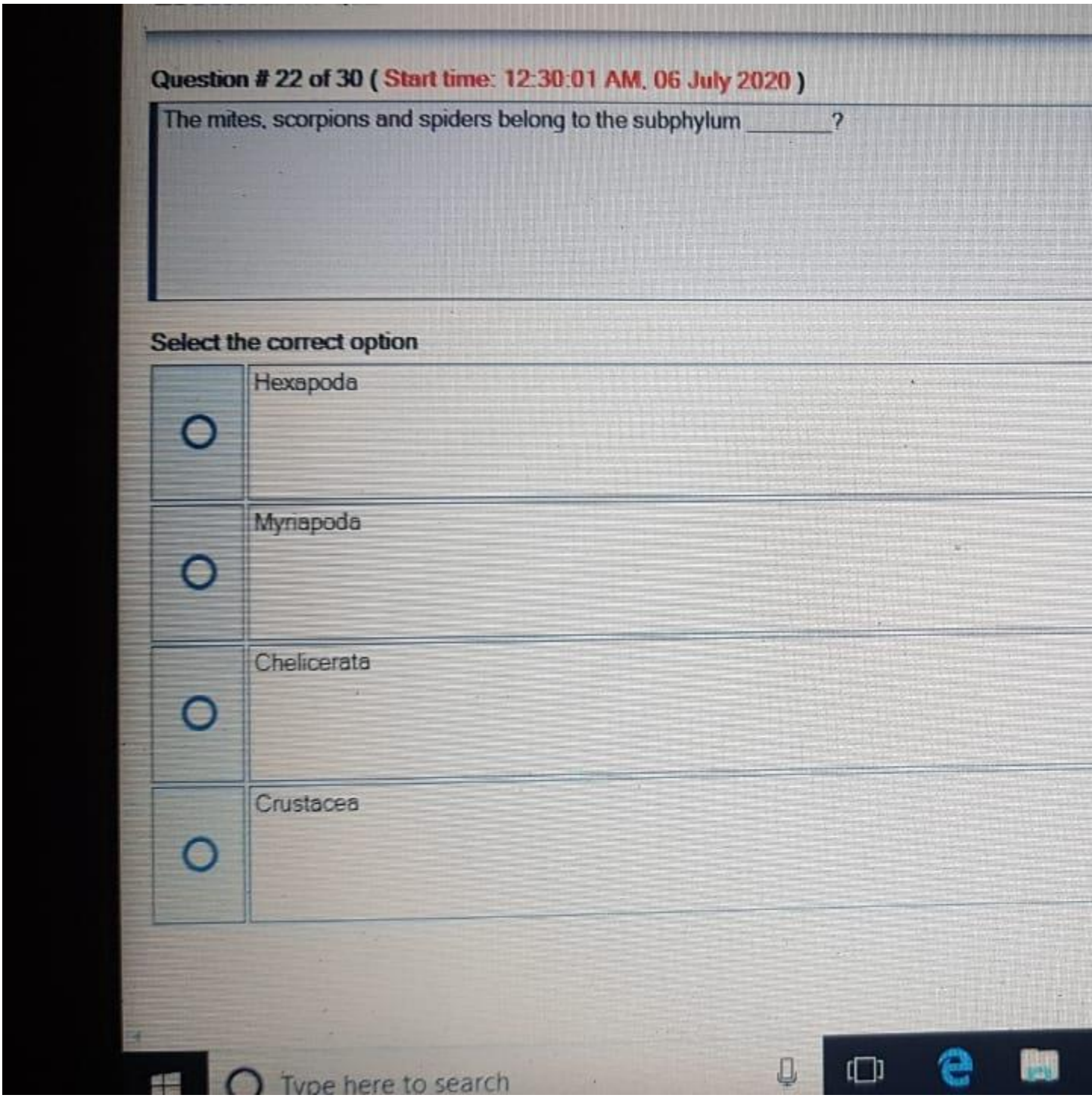
https://en.m.wikipedia.org › wiki

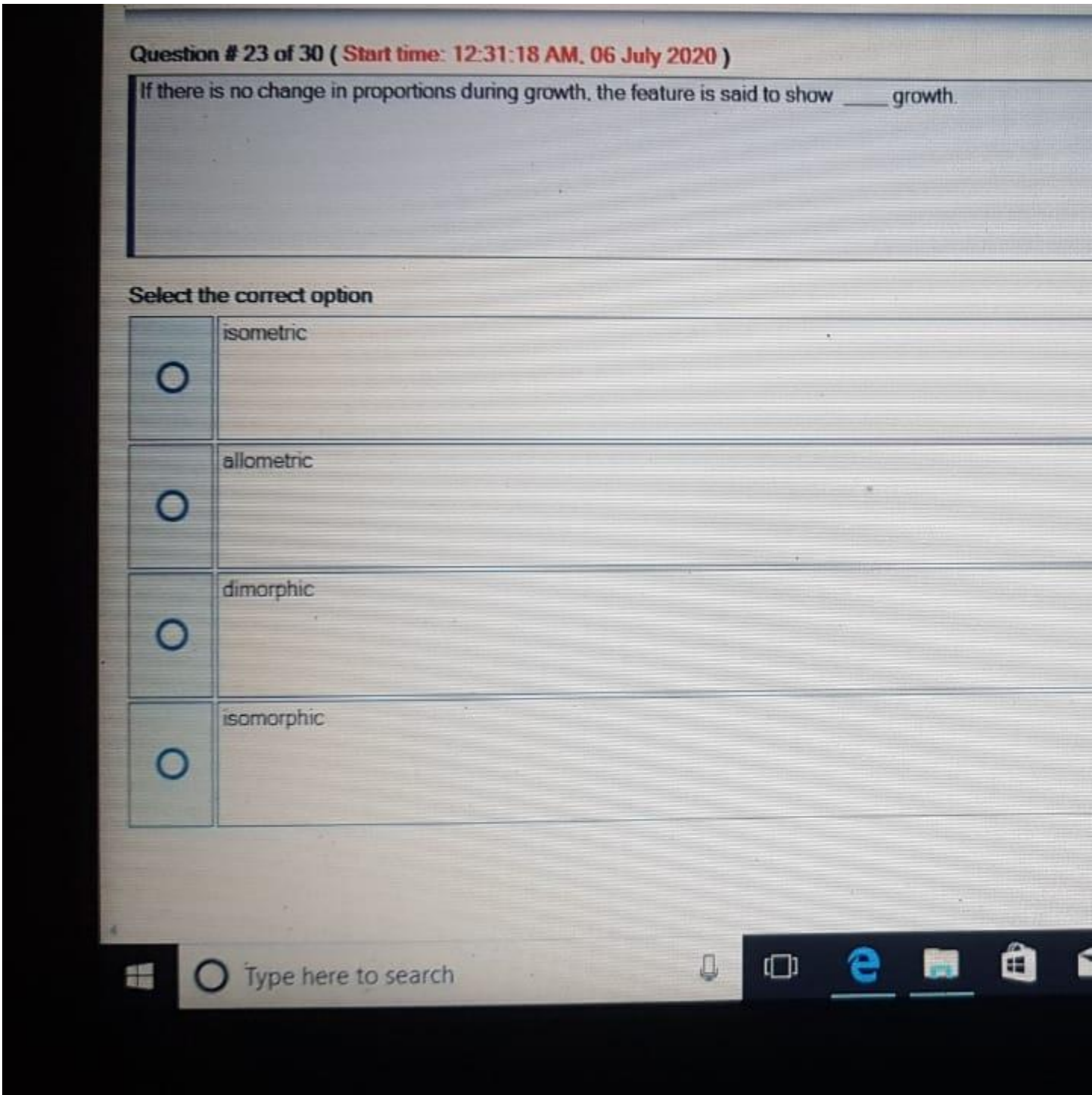
Trilobite - Wikipedia

About Featured Snippets Feedback


Question # 21 of 30 (Start time: 12:28:39 AM, 06 July 2020)
The Class Cephalopods have _____ sub-classes?
Select the correct option
One
Two
Three
Four
Type here to search


. End-Ordovician mass extinction
• 445 Ma
• Substantial turnovers occurred
echinoderms, ostracodes trilobites
• Major Climatic changes,
America that then lay around the ec
2. Late Devonian mass extinction
• A succession of extinction pulses 380 to
• The cephalopods were decimated.
• Causes: A major cooling phase associate


Question # 22 of 30 (Start time: 12:30:01 AM, 06 July 2020)
The mites, scorpions and spiders belong to the subphylum _____?
Select the correct option
Hexapoda
Myriapoda
Chelicerata
Crustacea
Type here to search

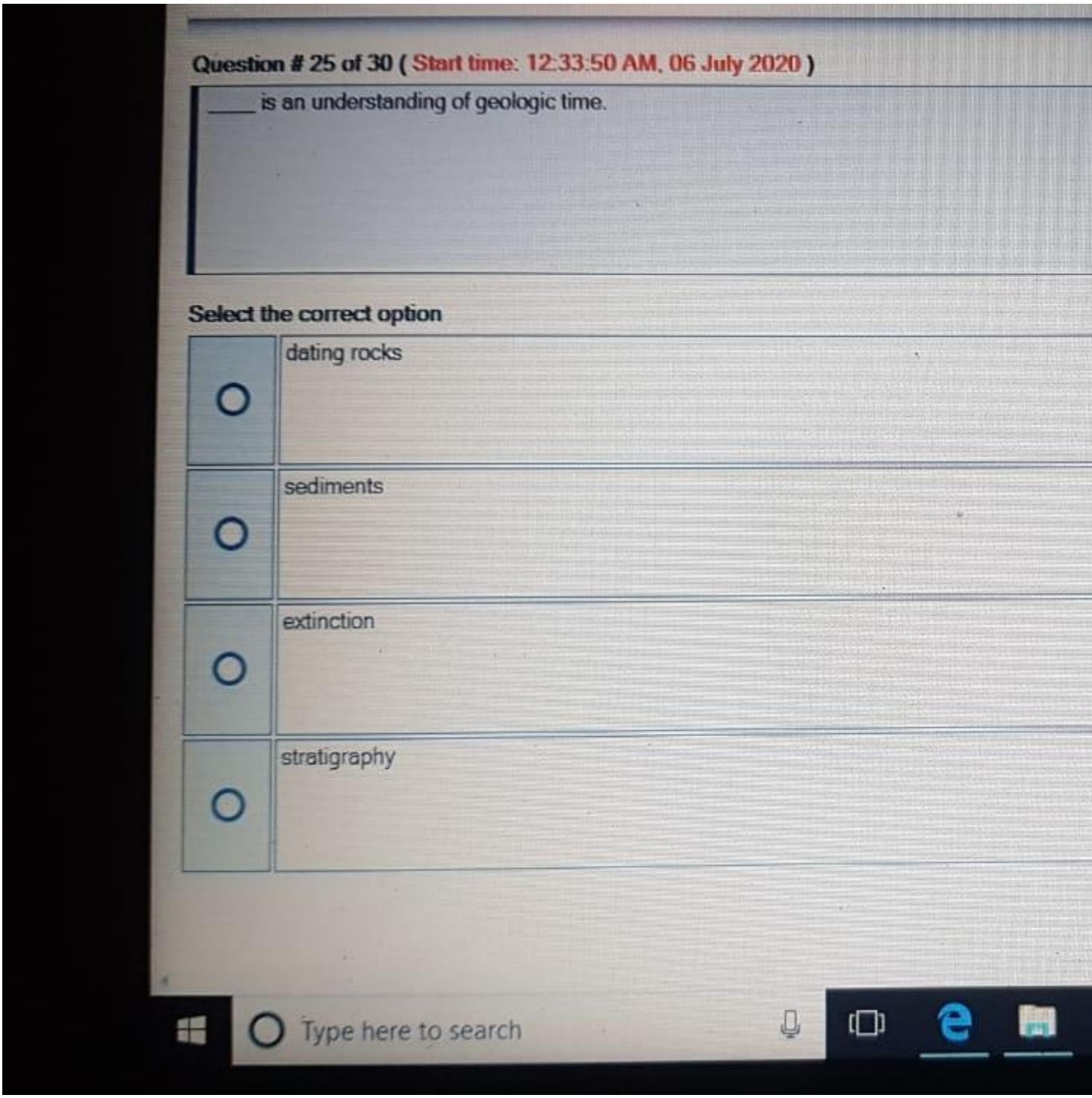

Question # 23 of 30 (Start time: 12:31:18 AM, 06 July 2020)
If there is no change in proportions during growth, the feature is said to show ____ growth.
Select the correct option
isometric
allometric
dimorphic
isomorphic
Type here to search

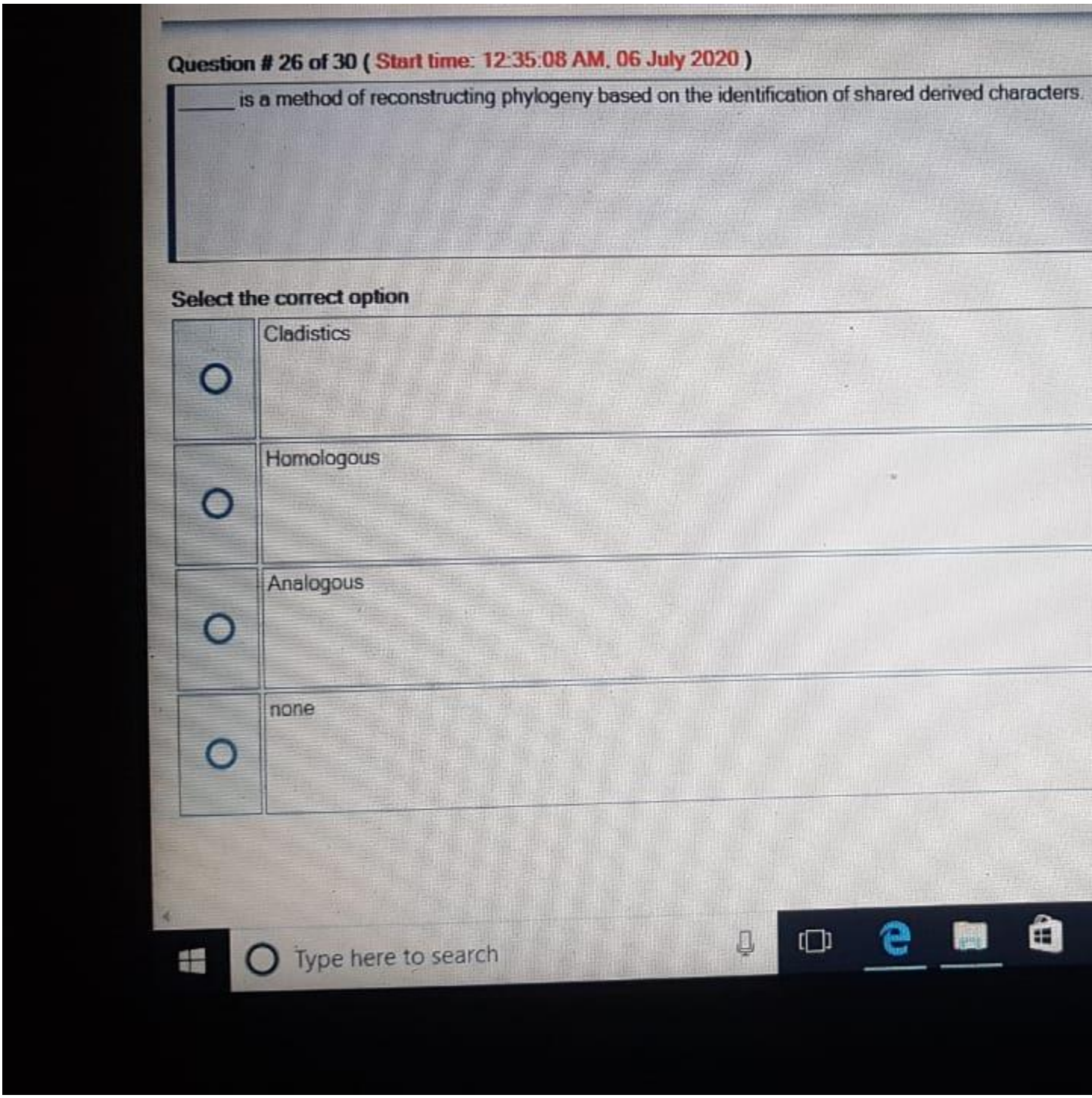
Question # 24 of 30 (Start time: 12:32:14 AM, 06 July 2020)

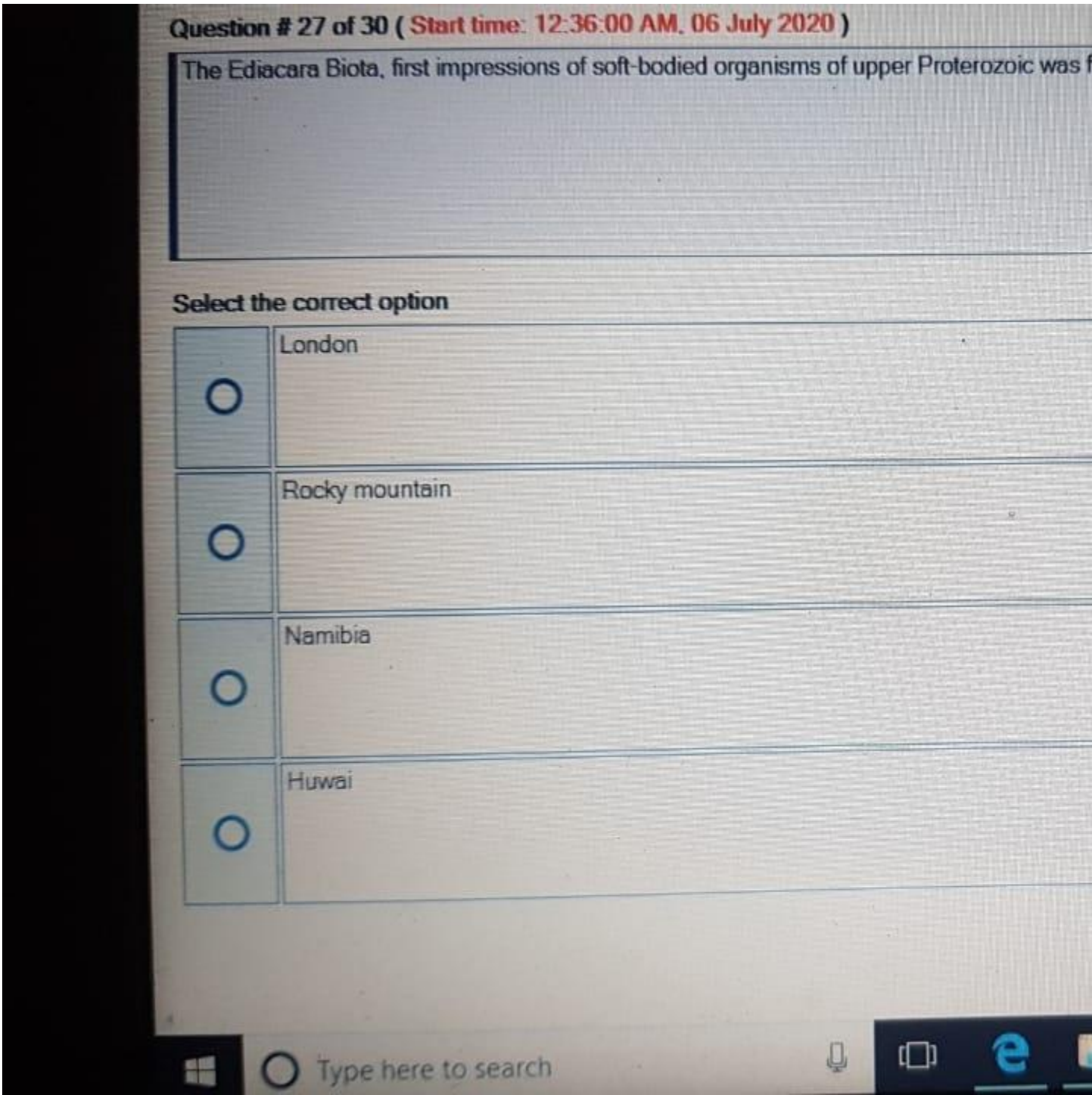
"Proof is impossible" said _____.

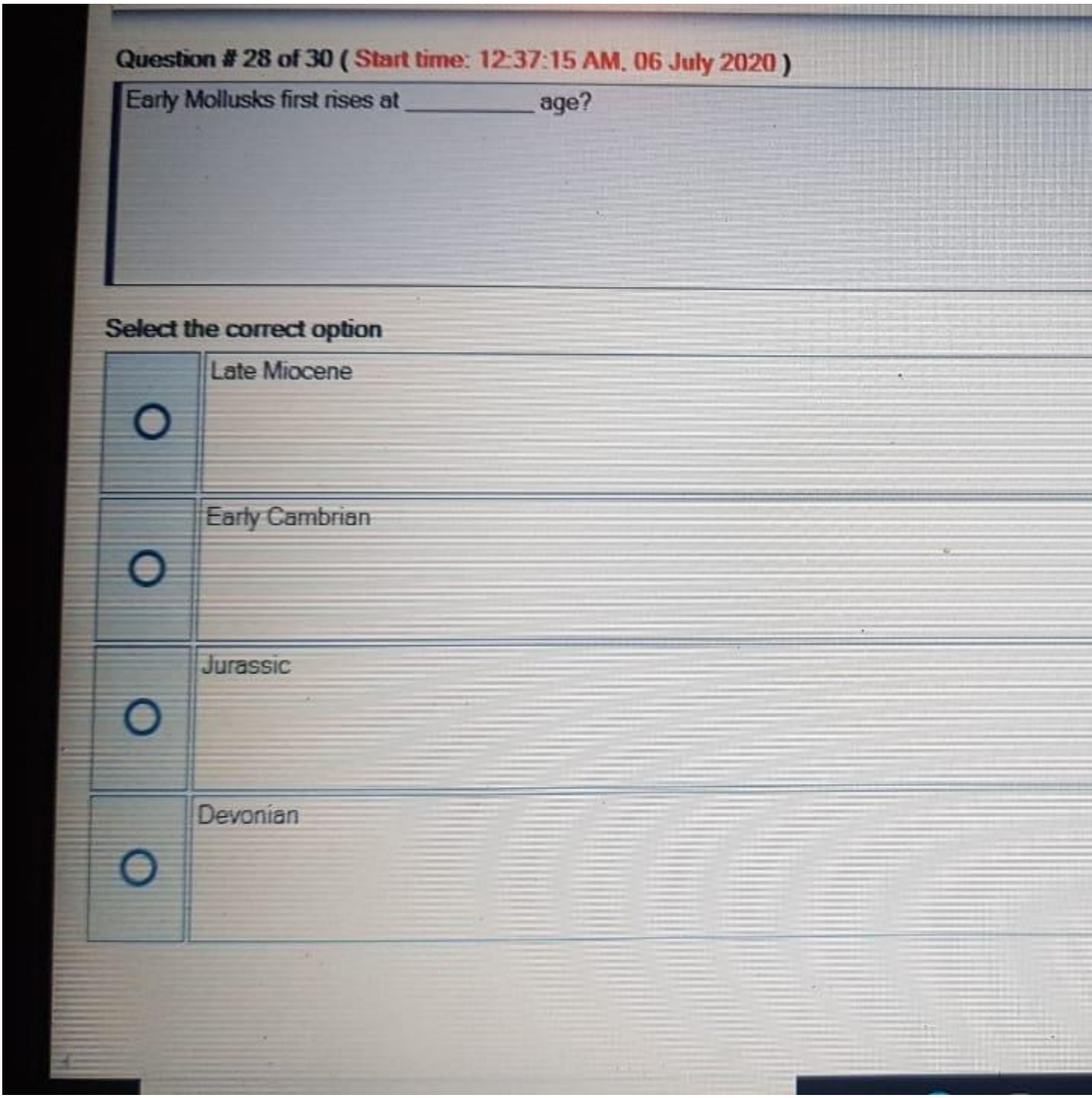
Select the correct option

- Aristotle
- Karl Popper
- Bacon
- Lamarck


Question # 25 of 30 (Start time: 12:33:50 AM, 06 July 2020)
____ is an understanding of geologic time.
Select the correct option
dating rocks
sediments
extinction
stratigraphy
Type here to search


Question # 26 of 30 (Start time: 12:35:08 AM, 06 July 2020)
_____ is a method of reconstructing phylogeny based on the identification of shared derived characters.
Select the correct option
Cladistics
Homologous
Analogous
none
Type here to search


Question # 27 of 30 (Start time: 12:36:00 AM, 06 July 2020)
The Ediacara Biota, first impressions of soft-bodied organisms of upper Proterozoic was
Select the correct option
London
Rocky mountain
Namibia
Huwai
Type here to search

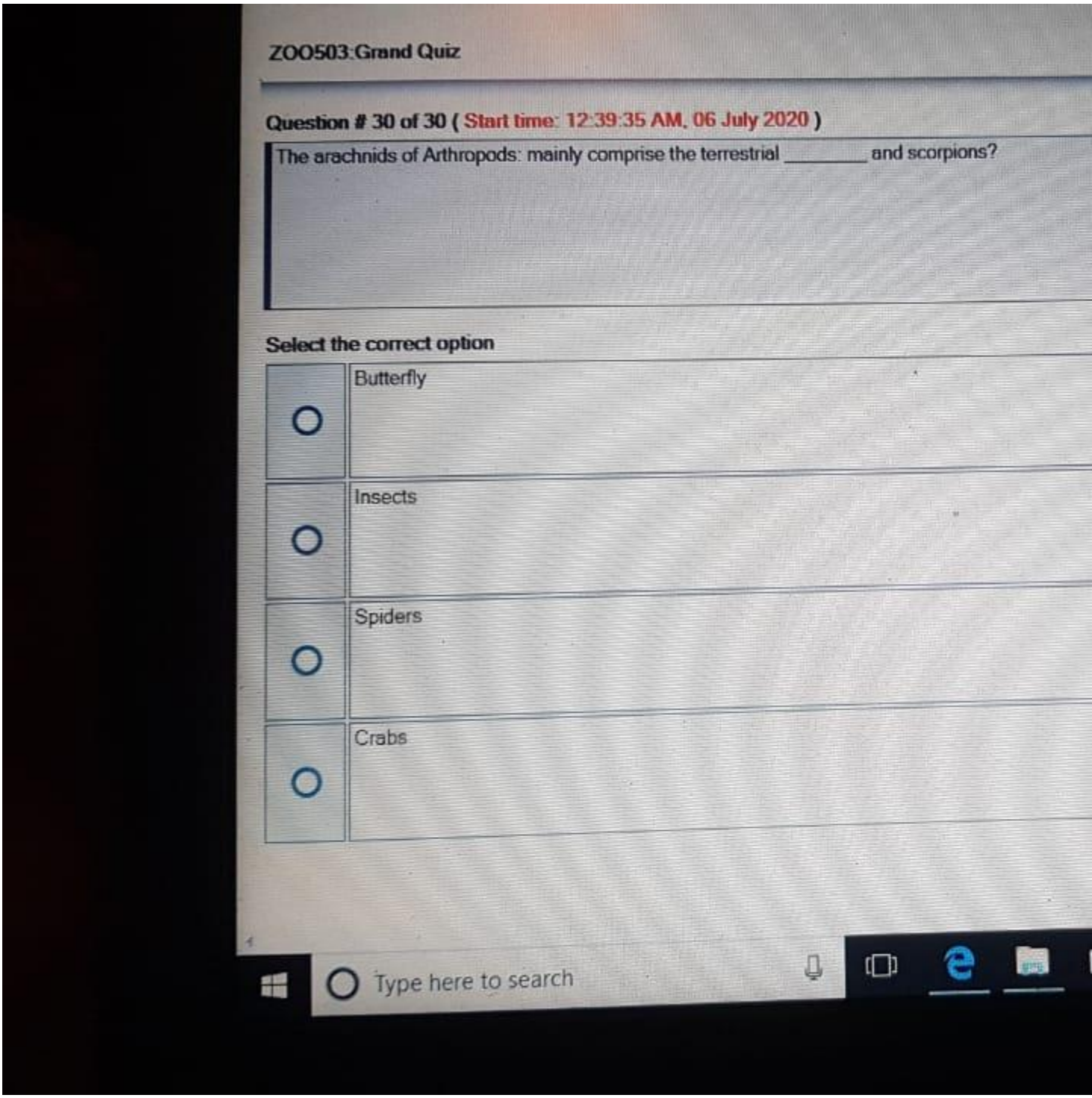


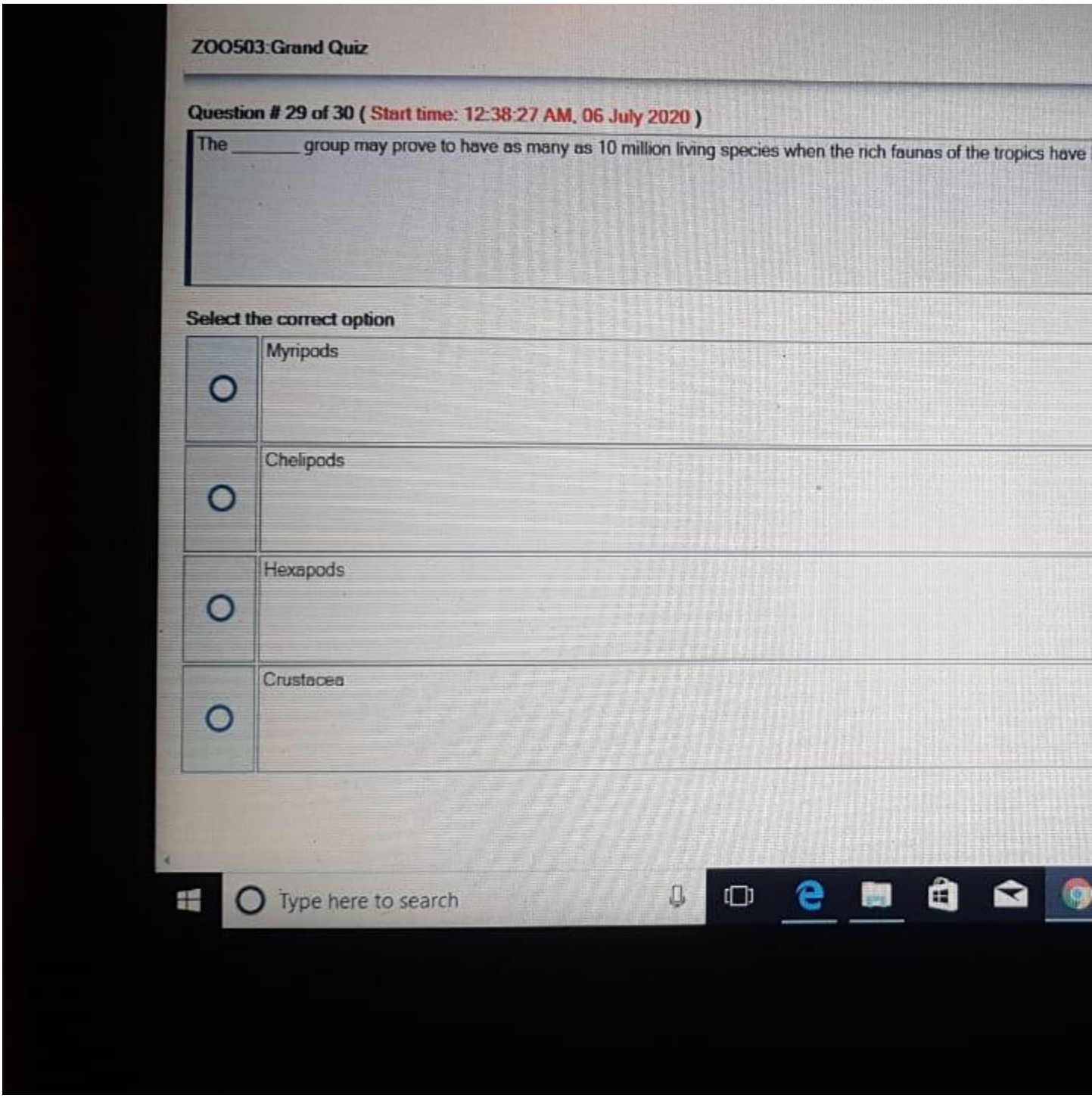
Question # 28 of 30 (Start time: 12:37:15 AM, 06 July 2020)

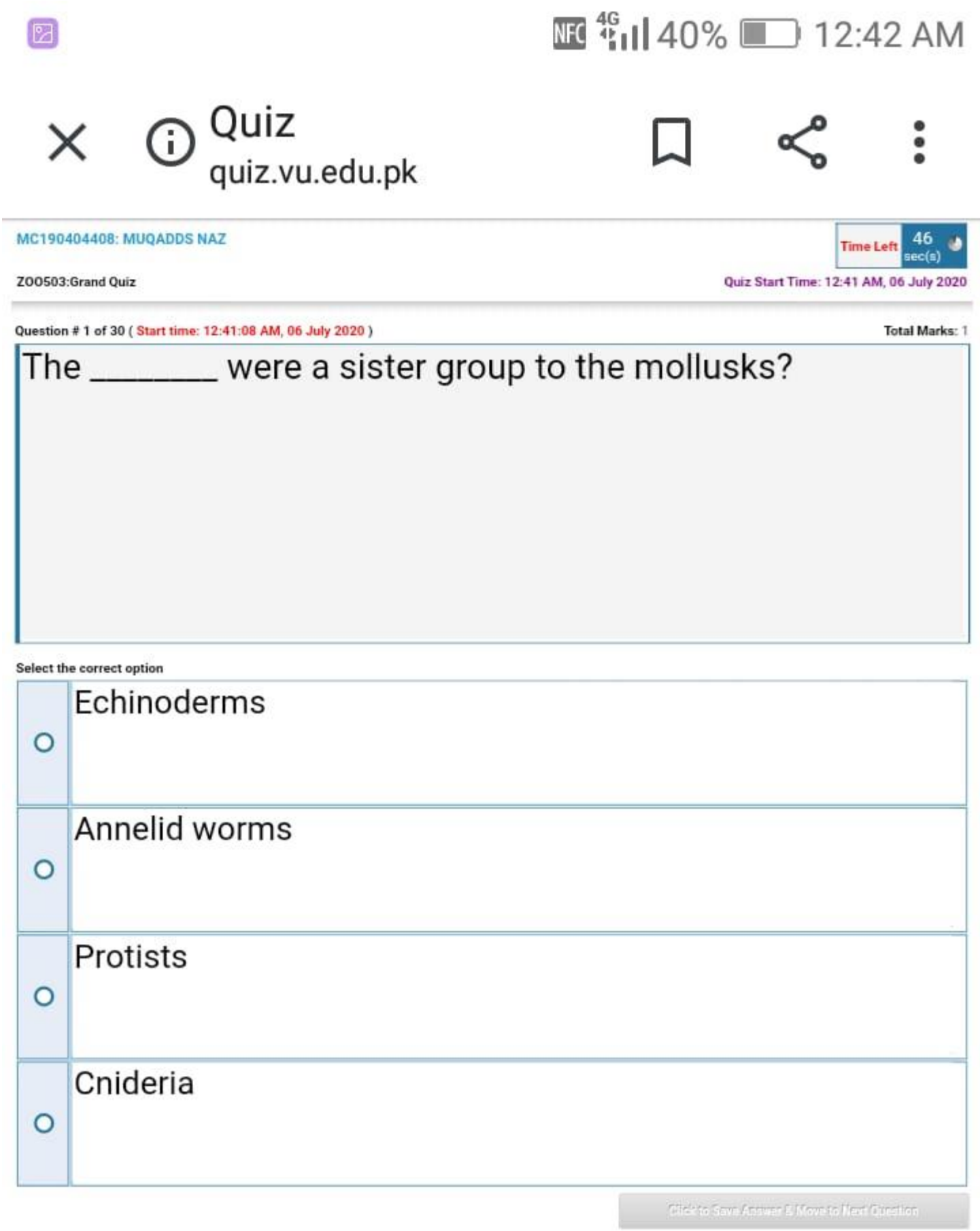
Early Mollusks first rises at _____ age?

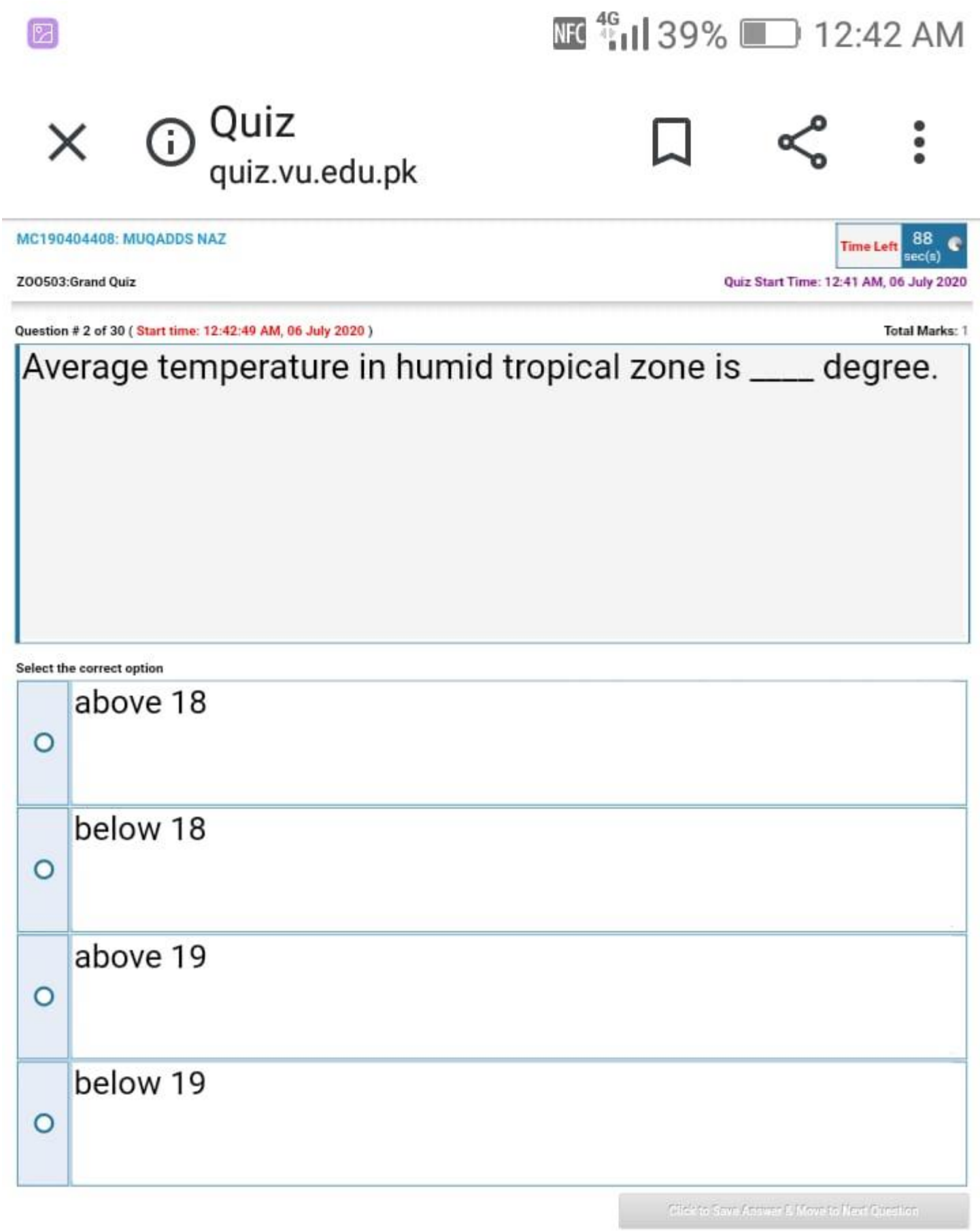
Select the correct option

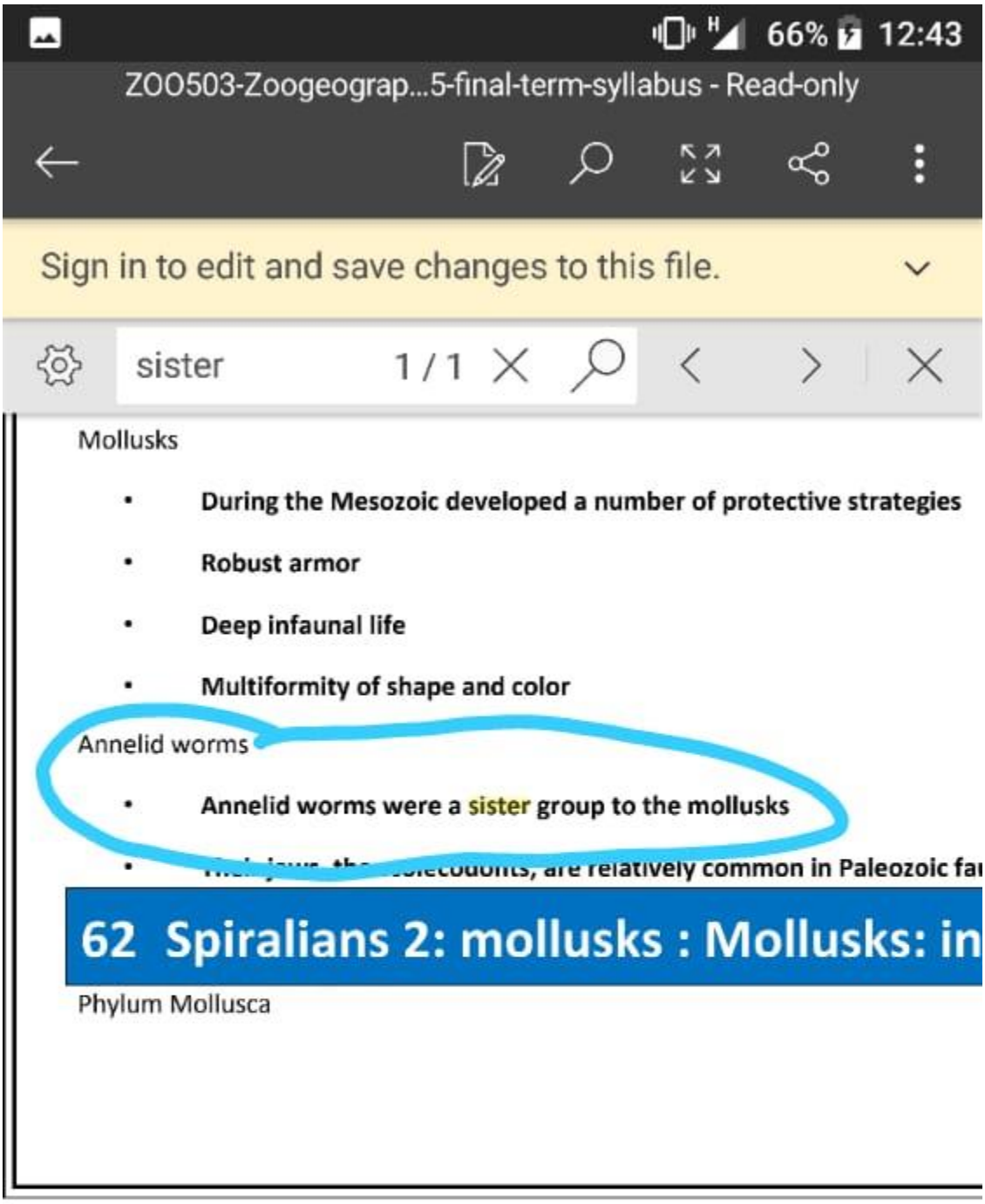
- ○ Late Miocene
- ○ Early Cambrian
- ○ Jurassic
- ○ Devonian

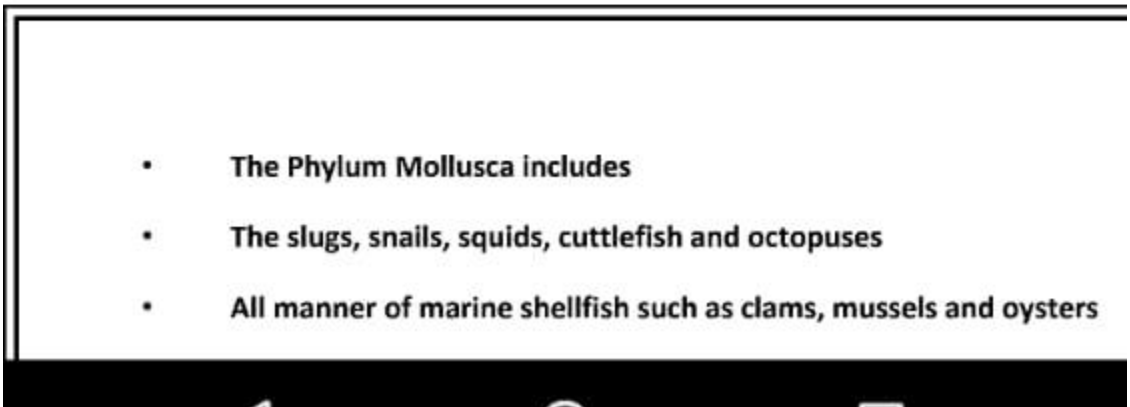

ZOO503:Grand Quiz
Question # 30 of 30 (Start time: 12:39:35 AM, 06 July 2020)
The arachnids of Arthropods: mainly comprise the terrestrial _____ and scorpions?
Select the correct option
Butterfly
Insects
Spiders
Crabs
Type here to search

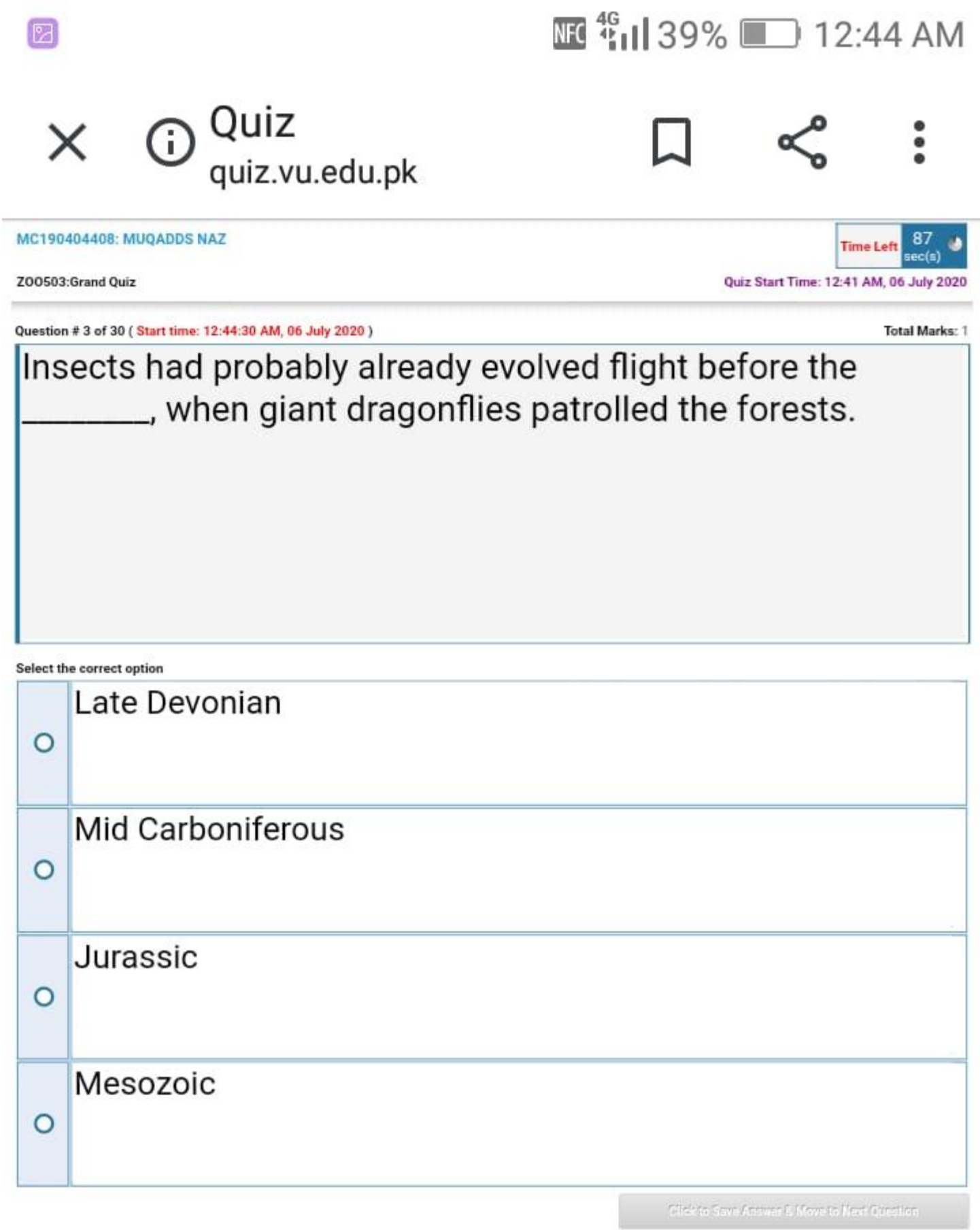

ZOO503:Grand Quiz
Question # 29 of 30 (Start time: 12:38:27 AM, 06 July 2020)
The _____ group may prove to have as many as 10 million living species when the rich faunas of the tropics have
Select the correct option
Myripods
Chelipods
Hexapods
Crustacea
Type here to search

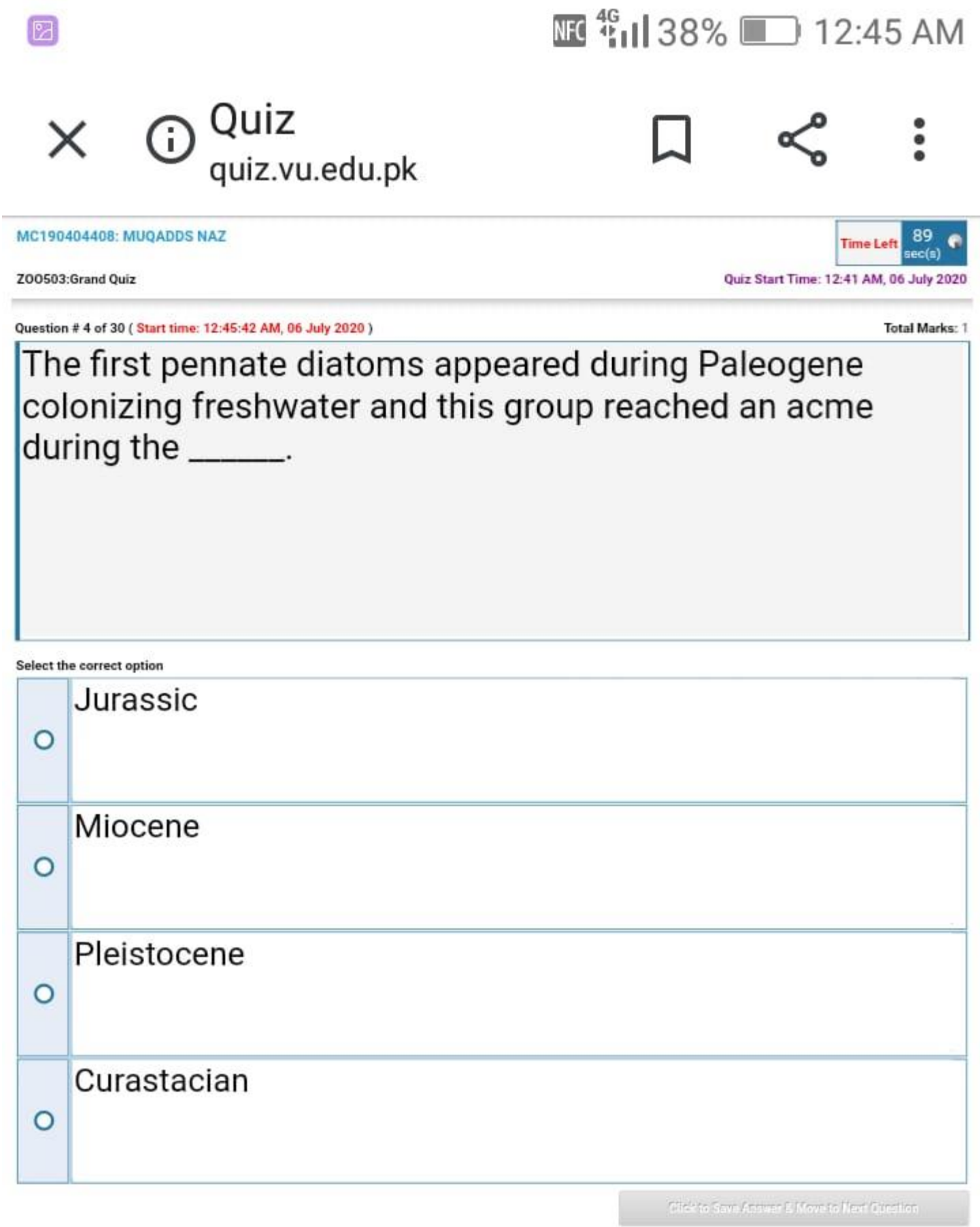

NFC 4G 40% 12:42 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 46 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 1 of 30 (Start time: 12:41:08 AM, 06 July 2020)
Total Marks: 1
The _____ were a sister group to the mollusks?
Select the correct option
Echinoderms
Annelid worms
Protists
Cnideria
Click to Save Answer & Move to Next Question

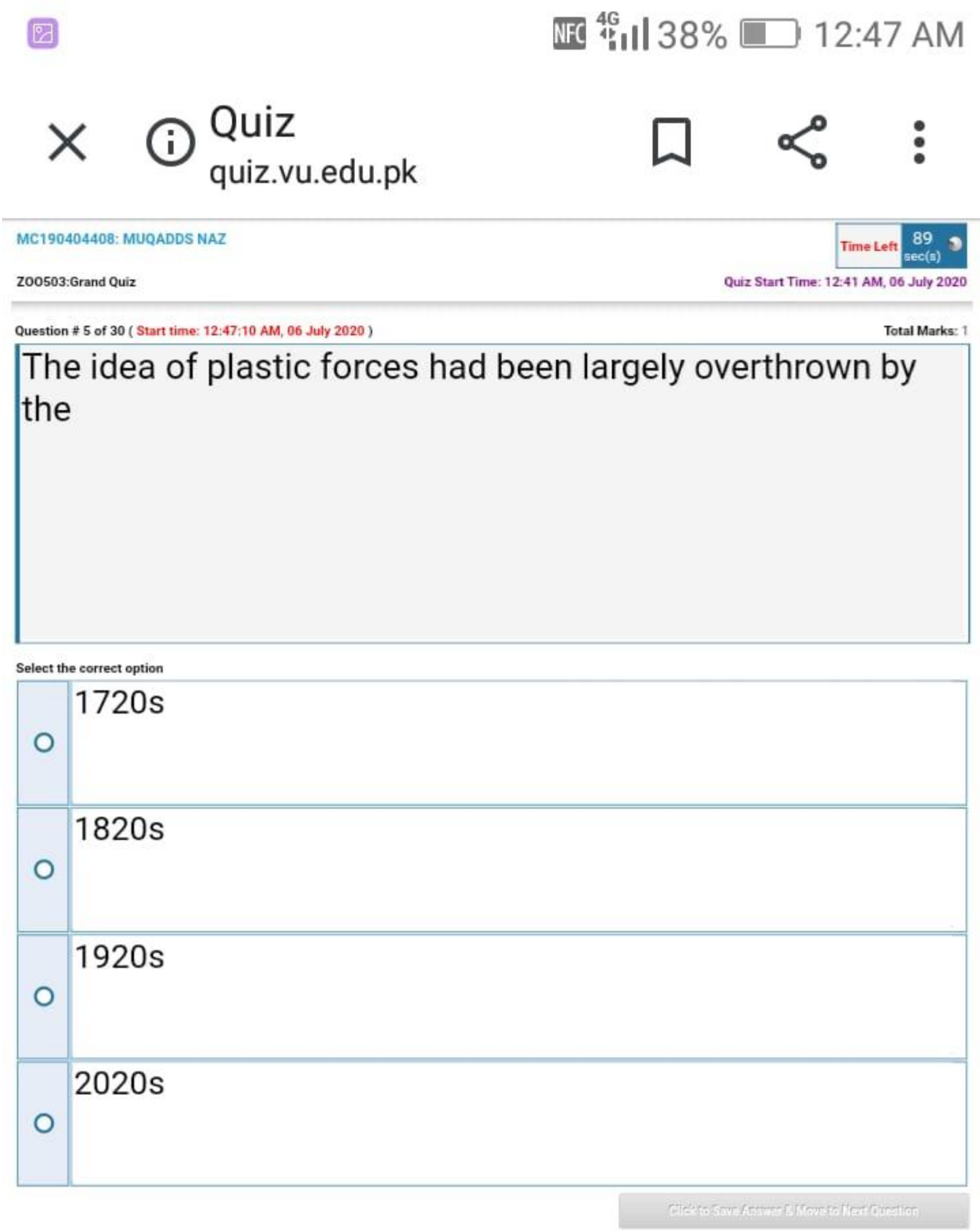

NFC 4G 39% 12:42 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 2 of 30 (Start time: 12:42:49 AM, 06 July 2020)
Total Marks: 1
Average temperature in humid tropical zone is ____ degree.
Select the correct option
above 18
below 18
above 19
below 19
Click to Save Answer & Move to Next Question

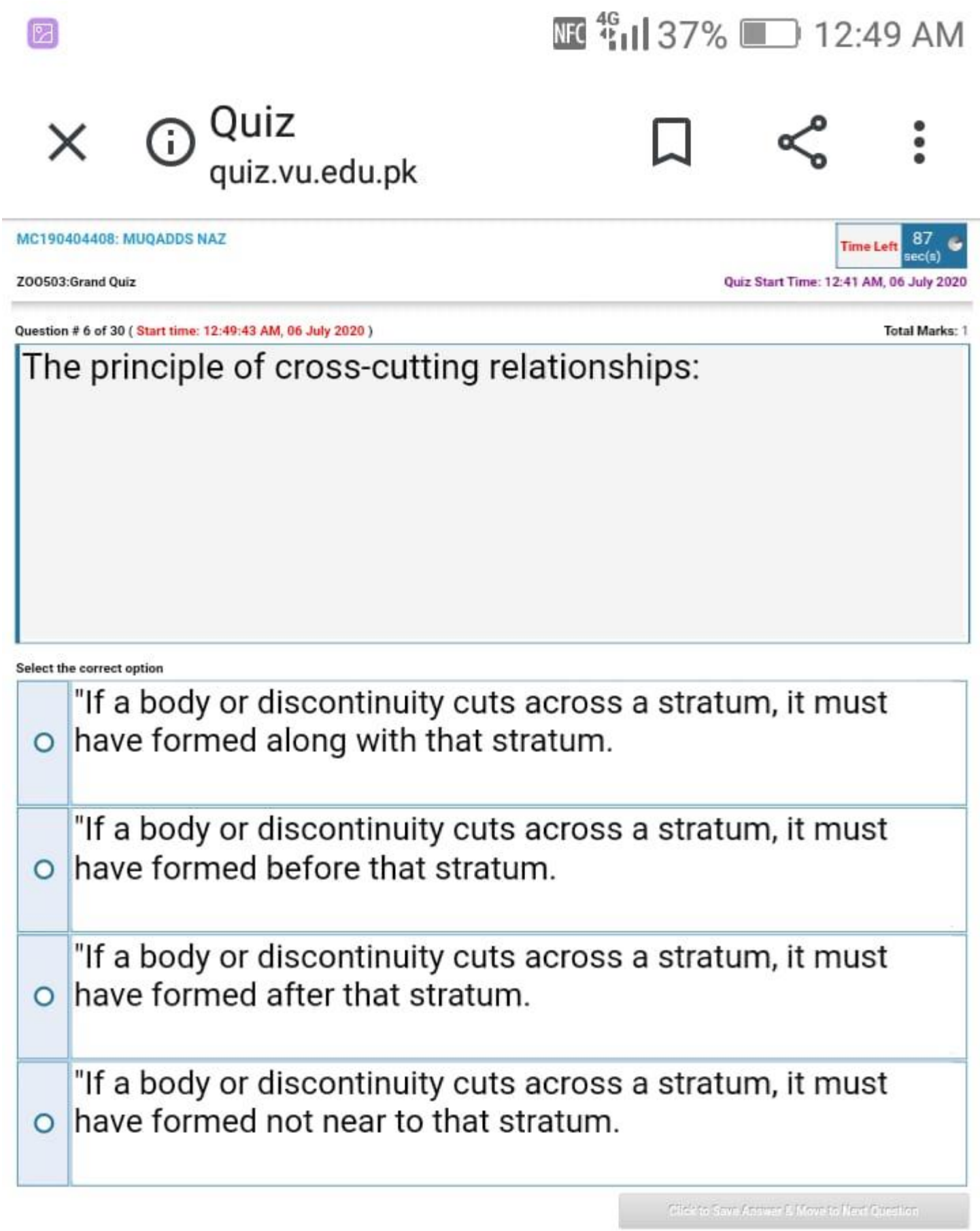

66% 12:43
ZOO503-Zoogeograp...5-final-term-syllabus - Read-only
Sign in to edit and save changes to this file.
sister 1 / 1
Mollusks
During the Mesozoic developed a number of protective strategies
Robust armor
Deep infaunal life
Multiformity of shape and color
Annelid worms
Annelid worms were a sister group to the mollusks
62 Spiralians 2: mollusks : Mollusks: in
Phylum Mollusca

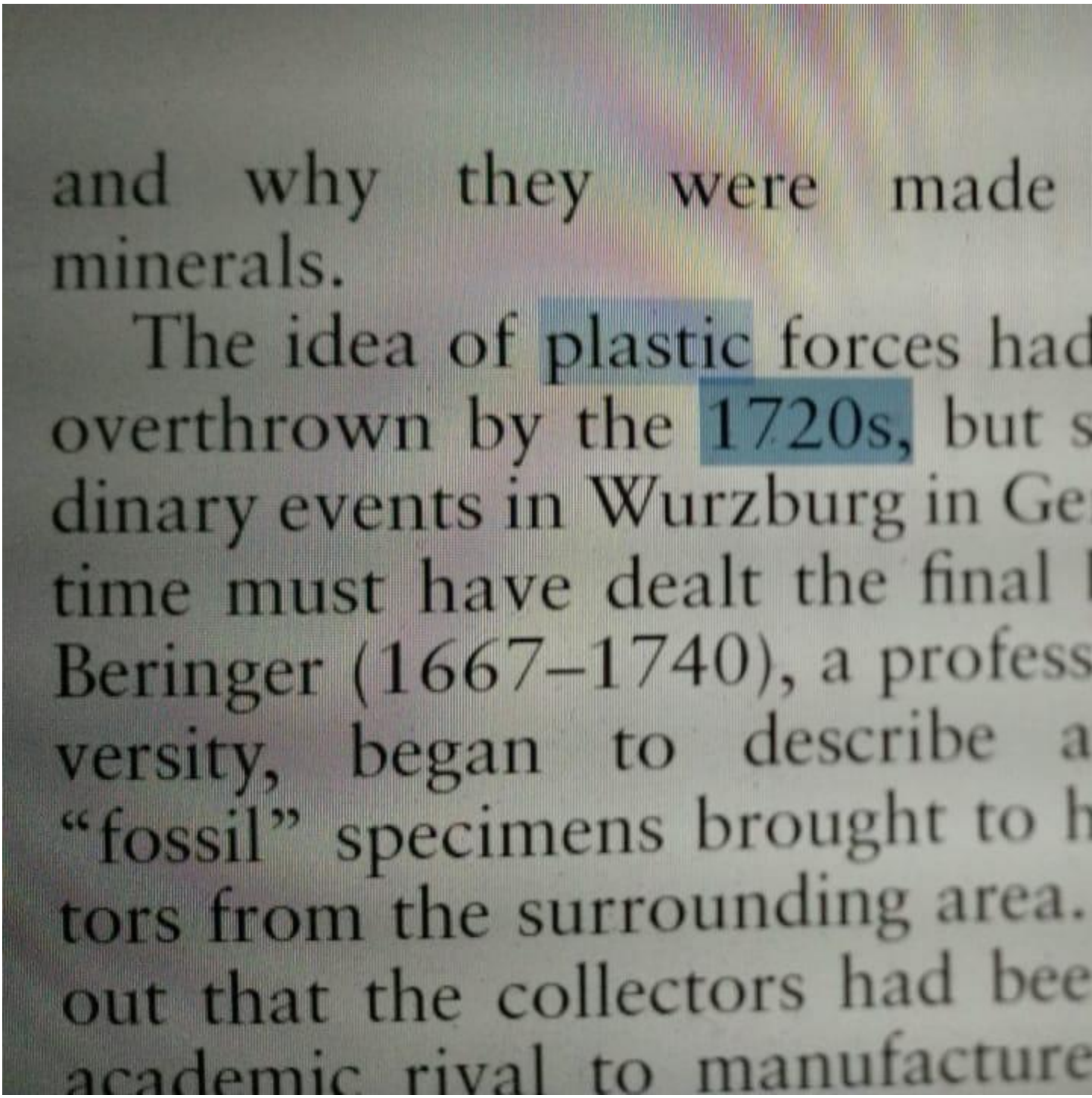

The Phylum Mollusca includes
The slugs, snails, squids, cuttlefish and octopuses
All manner of marine shellfish such as clams, mussels and oysters


NFC 4G 39% 12:44 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 87 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 3 of 30 (Start time: 12:44:30 AM, 06 July 2020)
Total Marks: 1
Insects had probably already evolved flight before the _____, when giant dragonflies patrolled the forests.
Select the correct option
Late Devonian
Mid Carboniferous
Jurassic
Mesozoic
Click to Save Answer & Move to Next Question


38% 12:45 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 89 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 4 of 30 (Start time: 12:45:42 AM, 06 July 2020)
Total Marks: 1
The first pennate diatoms appeared during Paleogene colonizing freshwater and this group reached an acme during the _____.
Select the correct option
Jurassic
Miocene
Pleistocene
Curastacian
Click to Save Answer & Move to Next Question

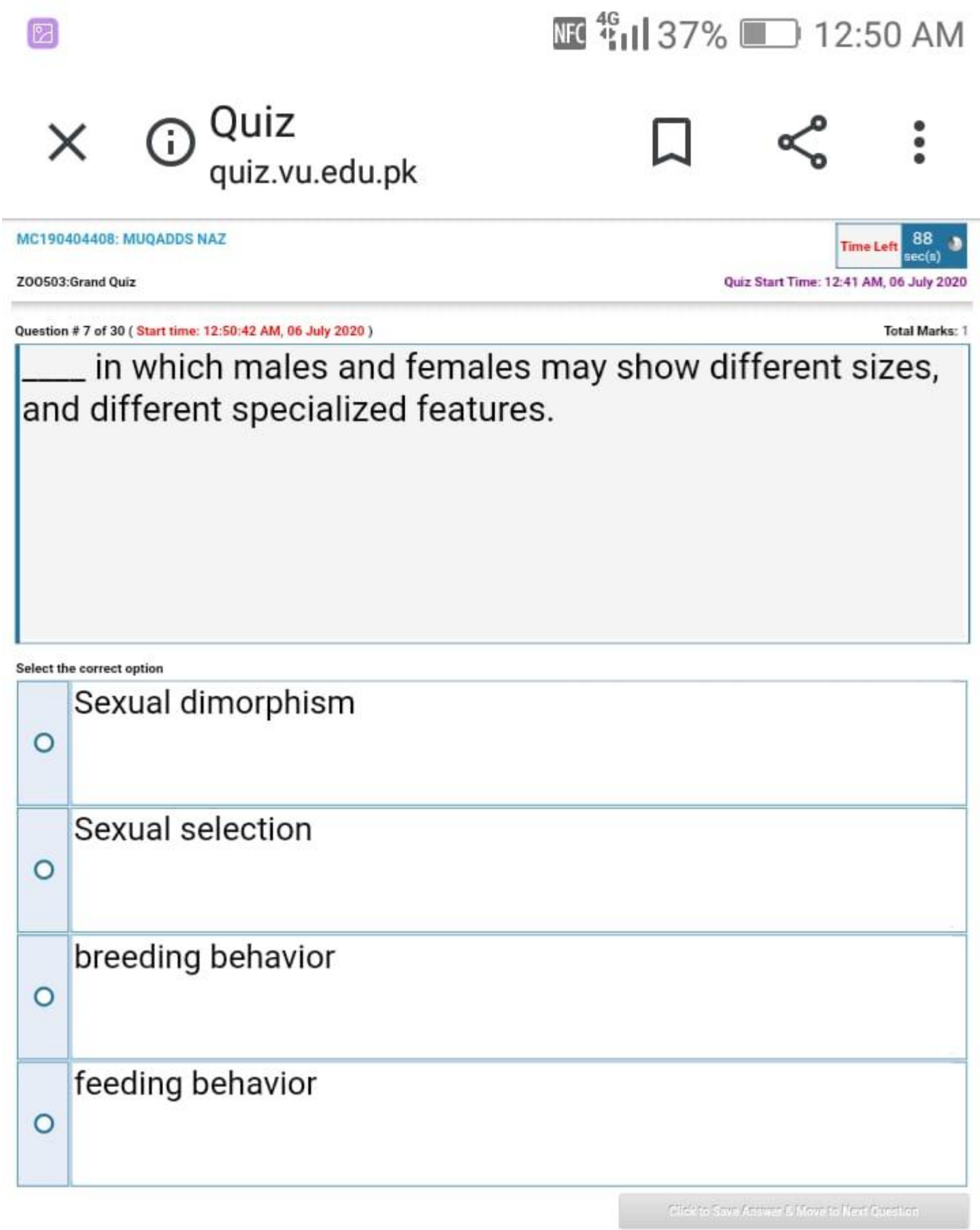

38% 12:47 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 89 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 5 of 30 (Start time: 12:47:10 AM, 06 July 2020)
Total Marks: 1
The idea of plastic forces had been largely overthrown by the
Select the correct option
1720s
1820s
1920s
2020s
Click to Save Answer & Move to Next Question

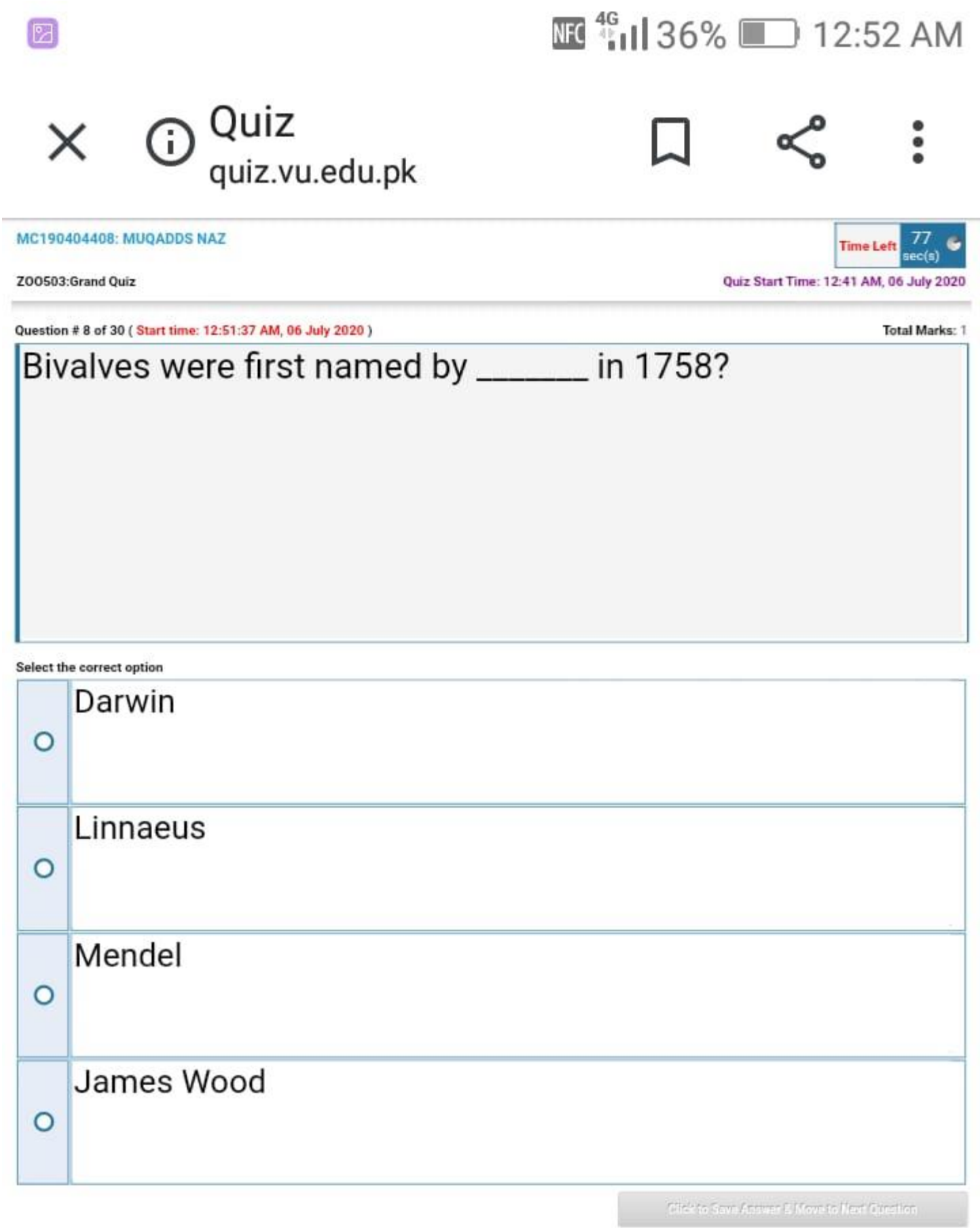

NFC 4G 37% 12:49 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 87 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 6 of 30 (Start time: 12:49:43 AM, 06 July 2020)
Total Marks: 1
The principle of cross-cutting relationships:
Select the correct option
"If a body or discontinuity cuts across a stratum, it must have formed along with that stratum.
"If a body or discontinuity cuts across a stratum, it must have formed before that stratum.
"If a body or discontinuity cuts across a stratum, it must have formed after that stratum.
"If a body or discontinuity cuts across a stratum, it must have formed not near to that stratum.
Click to Save Answer & Move to Next Question

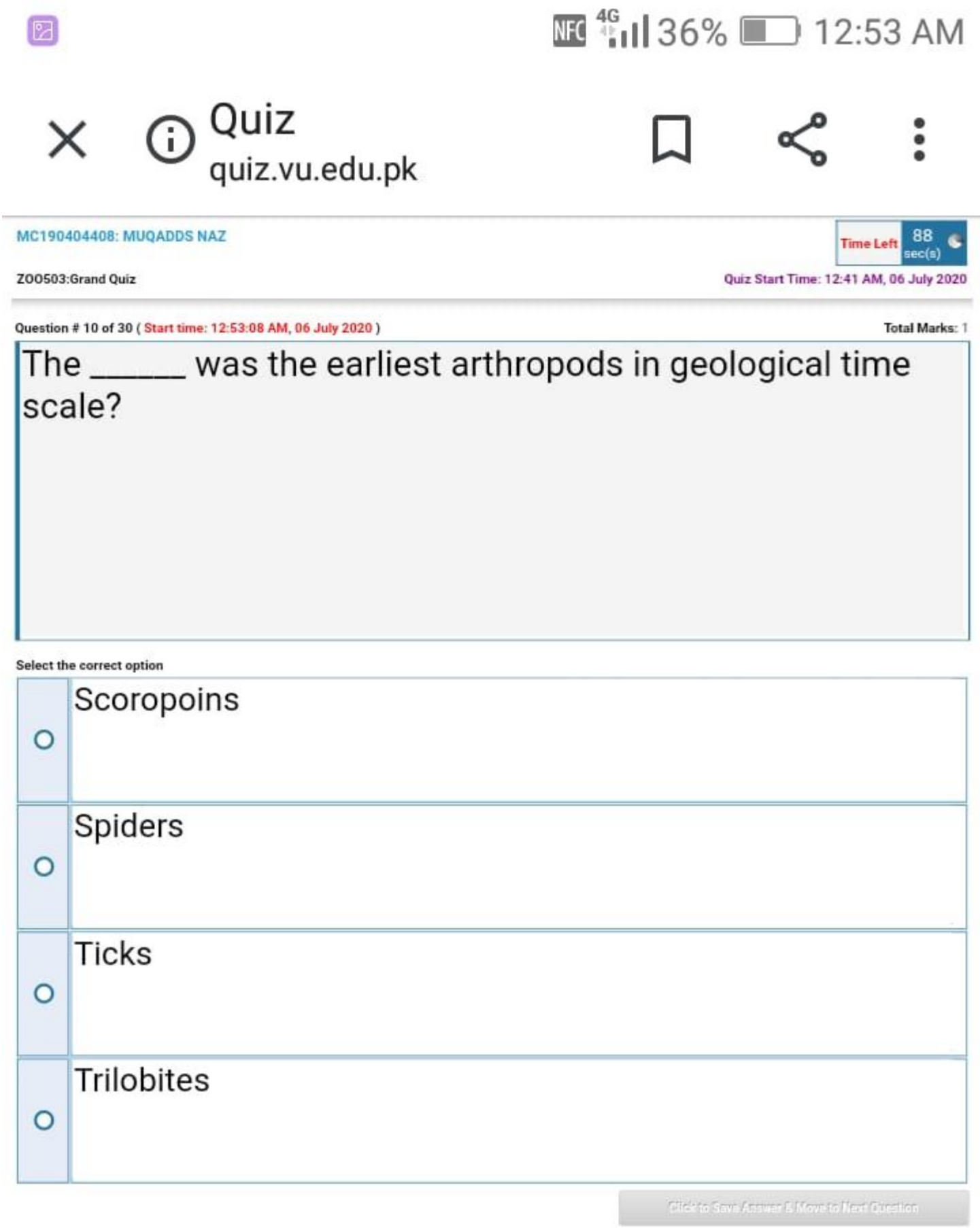


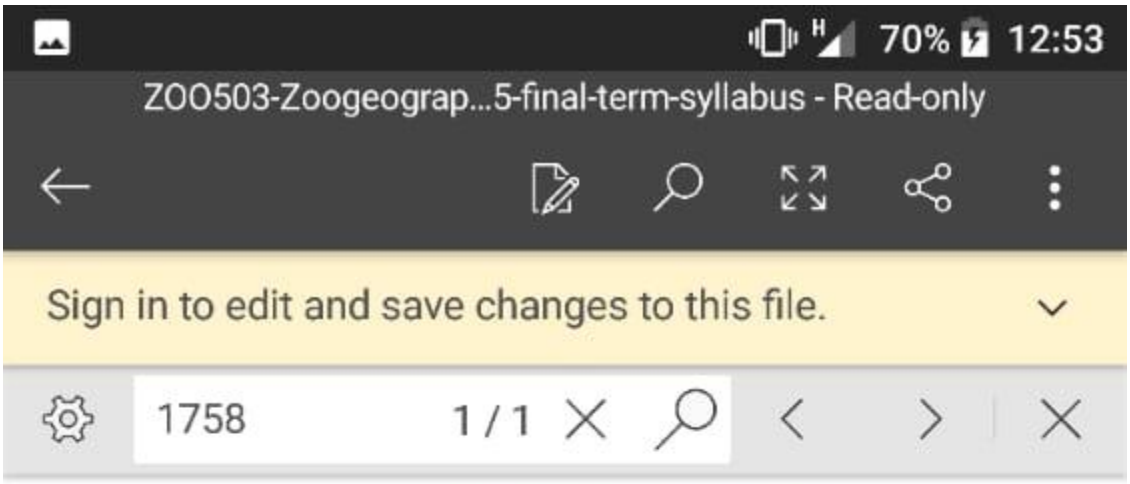
and why they were made
minerals.

The idea of plastic forces had
overthrown by the 1720s, but s
dinary events in Wurzburg in Ge
time must have dealt the final
Beringer (1667–1740), a profess
versity, began to describe a
"fossil" specimens brought to h
tors from the surrounding area.
out that the collectors had bee
academic rival to manufacture


NFC 4G 37% 12:50 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 7 of 30 (Start time: 12:50:42 AM, 06 July 2020)
Total Marks: 1
____ in which males and females may show different sizes, and different specialized features.
Select the correct option
Sexual dimorphism
Sexual selection
breeding behavior
feeding behavior
Click to Save Answer & Move to Next Question


NFC 4G 36% 12:52 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 77 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 8 of 30 (Start time: 12:51:37 AM, 06 July 2020)
Total Marks: 1
Bivalves were first named by _____ in 1758?
Select the correct option
Darwin
Linnaeus
Mendel
James Wood
Click to Save Answer & Move to Next Question


NFC 4G 36% 12:53 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 10 of 30 (Start time: 12:53:08 AM, 06 July 2020)
Total Marks: 1
The _____ was the earliest arthropods in geological time scale?
Select the correct option
Scoropoins
Spiders
Ticks
Trilobites
Click to Save Answer & Move to Next Question



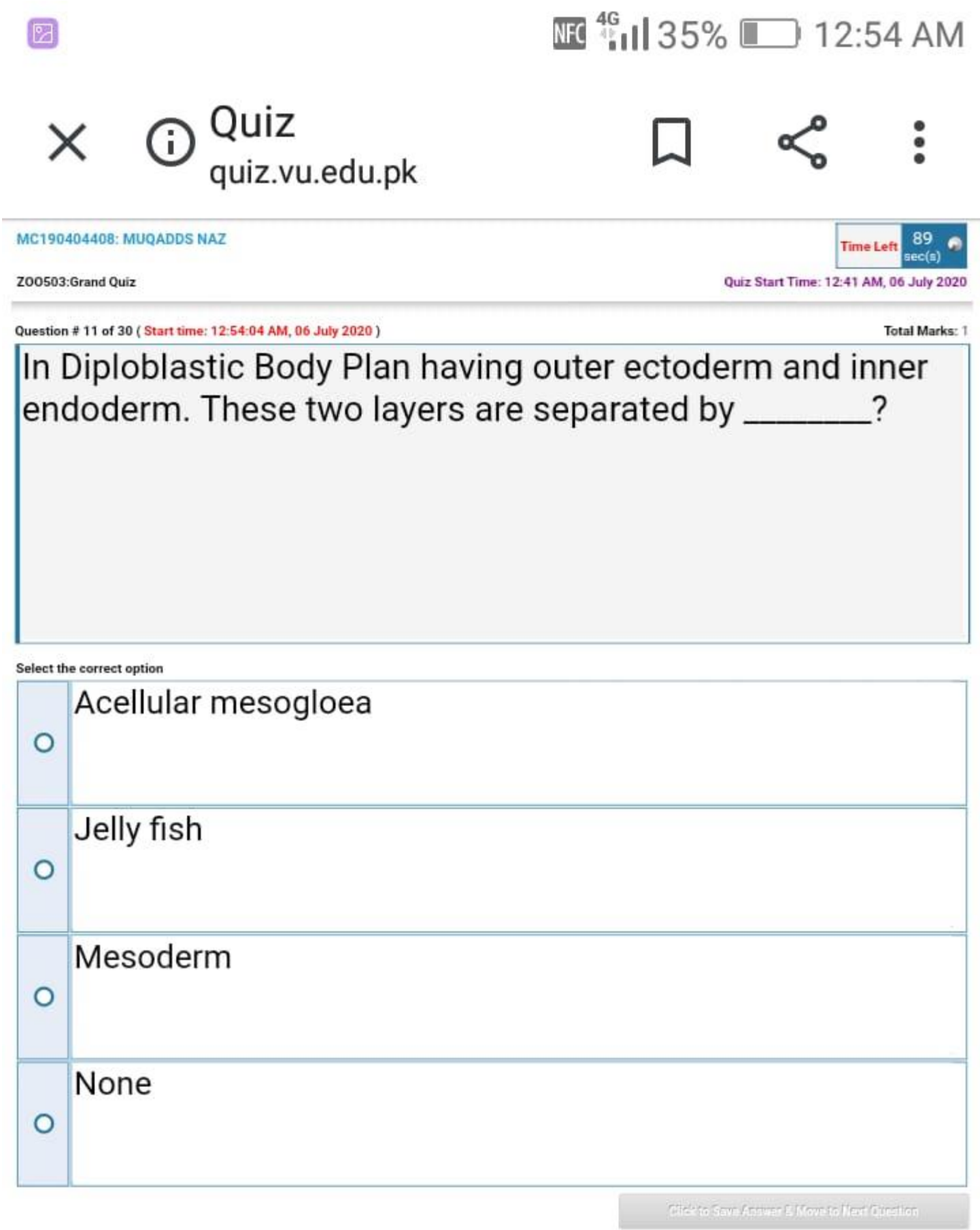
n superficially resembling the brachiopods. Common in modern
y aragonite. Two valves are virtually mirror images of each o
they were first named Bivalvia by Linnaeus in 1758

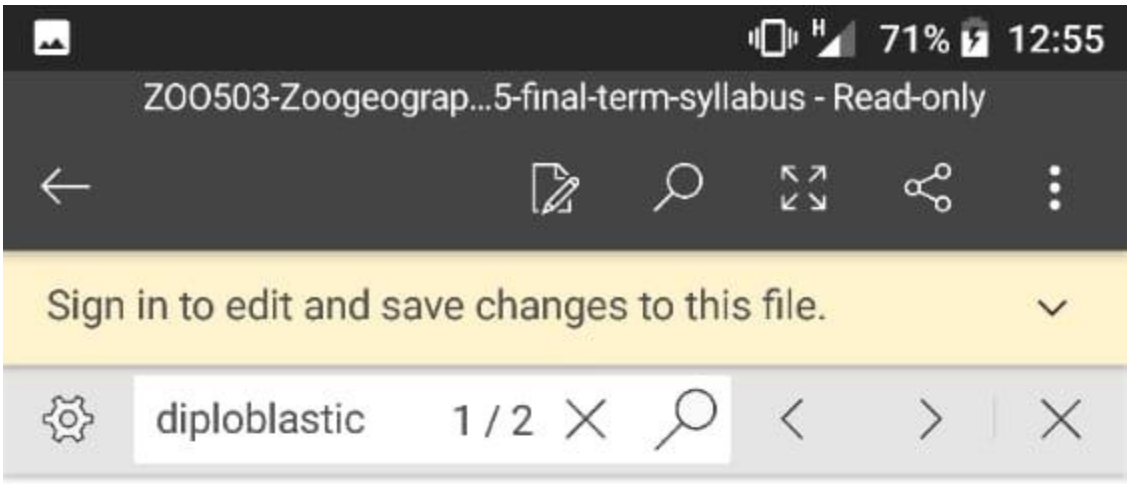
rian genera

d China
orth America and Siberia

lass evolved rapidly in the Early Ordovician to include basal forms
l Floian radiation. Major diversification: Paleozoic evolved exte
s. Early Mesozoic radiation featured siphonate forms

MUHAMMAD


NFC 4G 35% 12:54 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 89 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 11 of 30 (Start time: 12:54:04 AM, 06 July 2020)
Total Marks: 1
In Diploblastic Body Plan having outer ectoderm and inner endoderm. These two layers are separated by _____?
Select the correct option
Acellular mesogloea
Jelly fish
Mesoderm
None
Click to Save Answer & Move to Next Question



y Plan

aracterized by cells organized in two layers separated by jelly-like materi
lls or amoebocytes

ody Plan

tenophorans Two layers are separated by acellular mesogloea

oderm
loderm

Body Plan

nimal. Has three layers of tissues from the outside in: the ectoderm, mesoc

y

ns. The development of the celom or body cavity characterizes Most of the

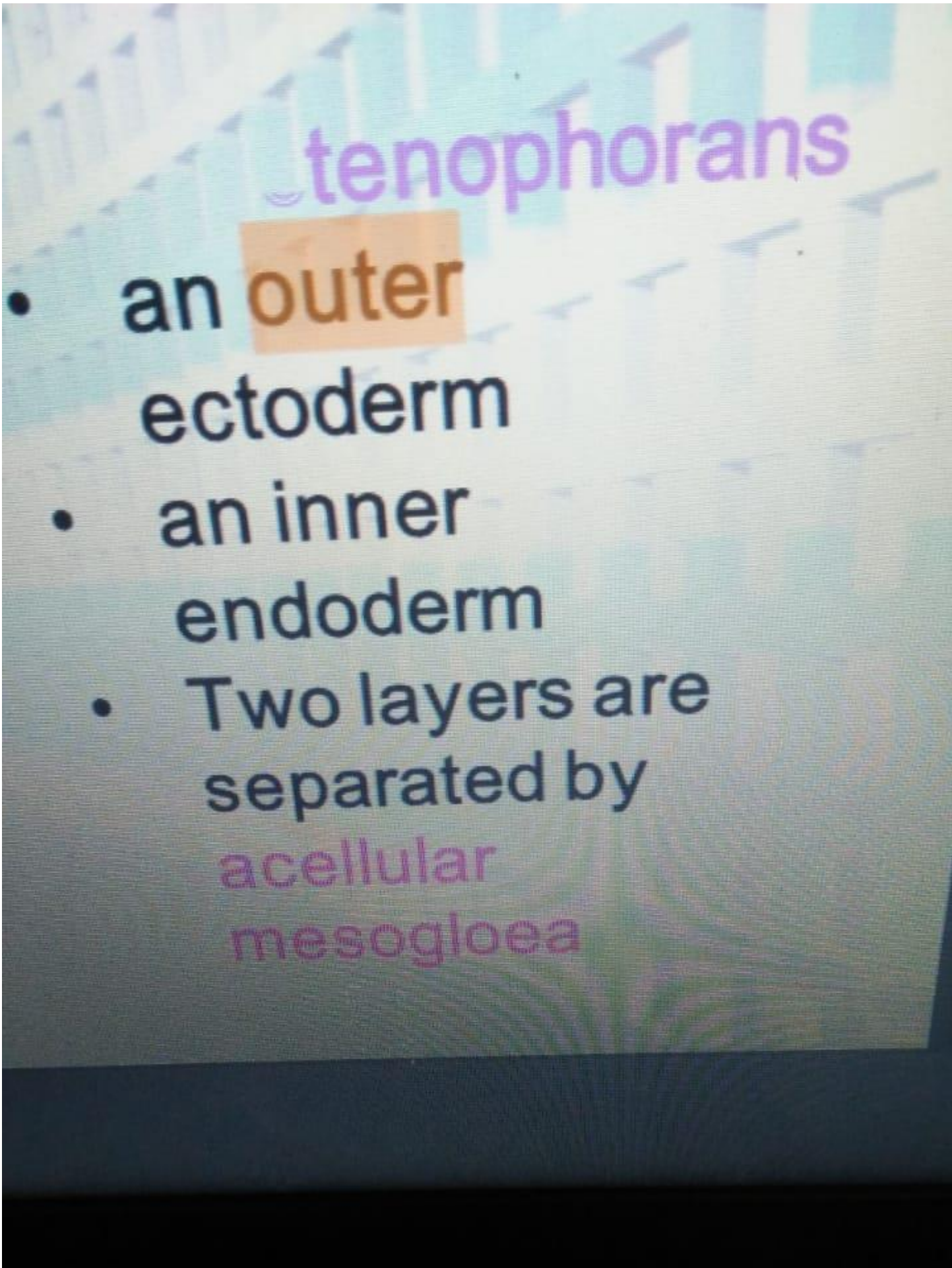
integral part of the body plan of an animal, providing support, protection

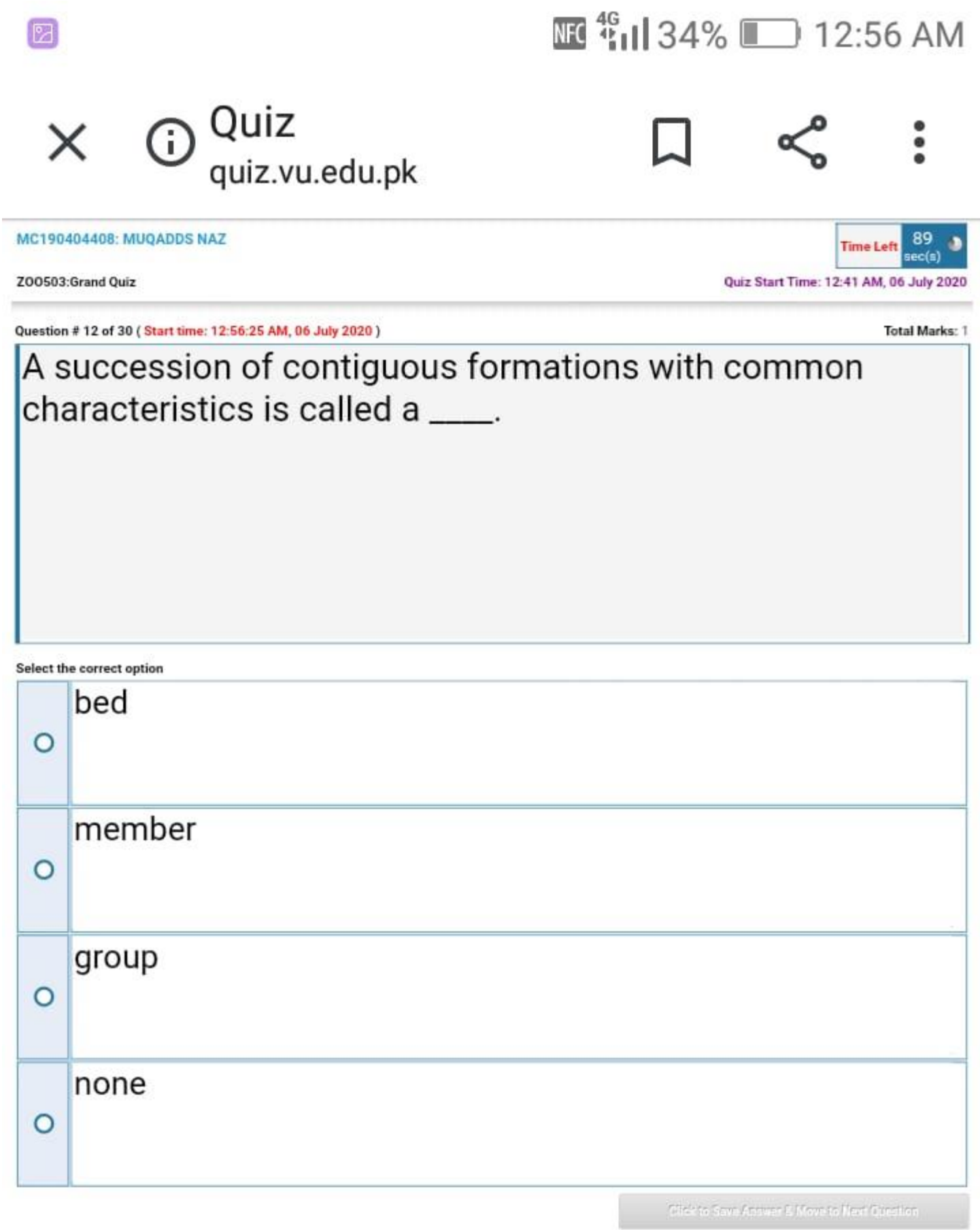
Skeleton

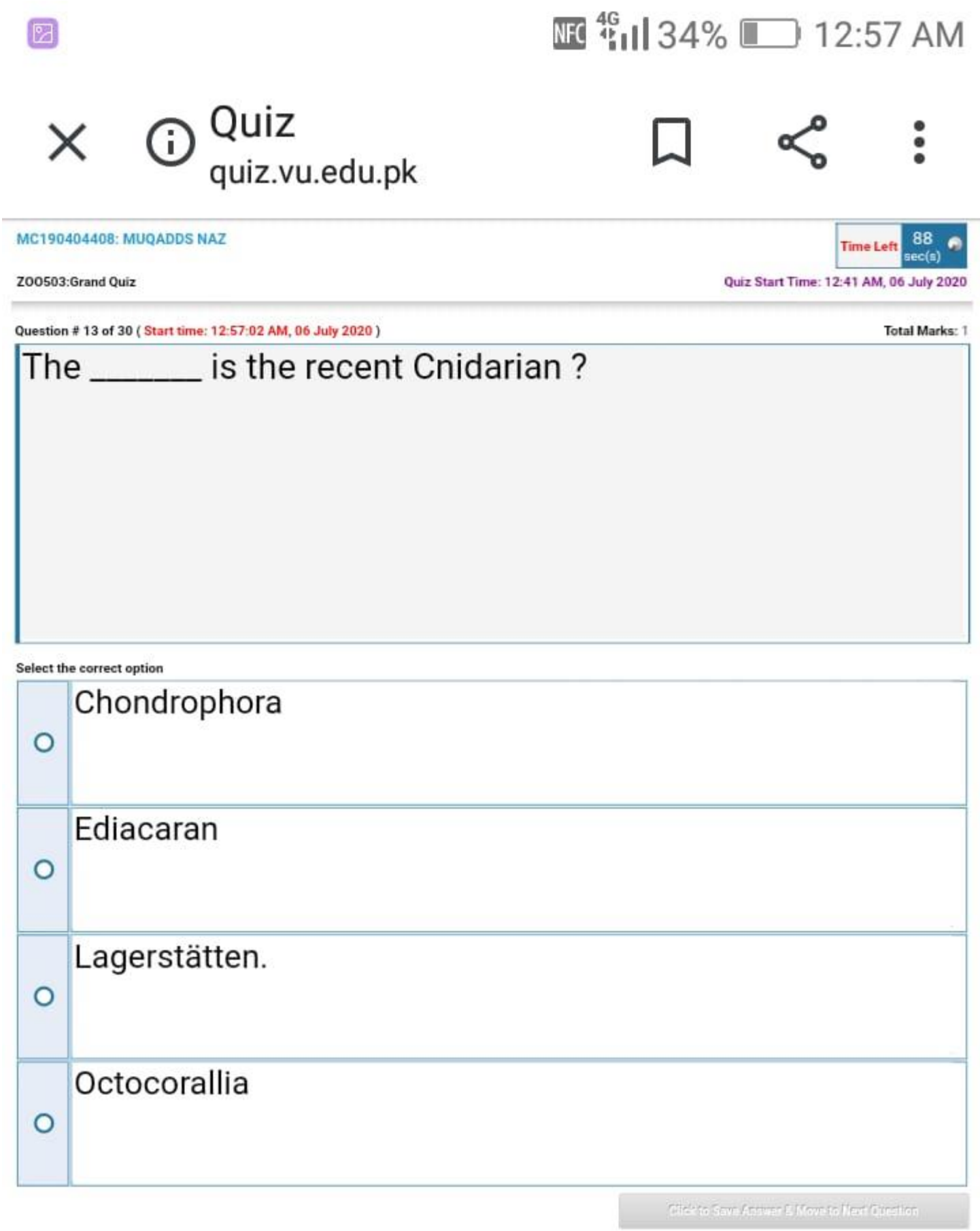
as the soft-bodied mollusks (slugs) possess a hydraulic skeleton in which th

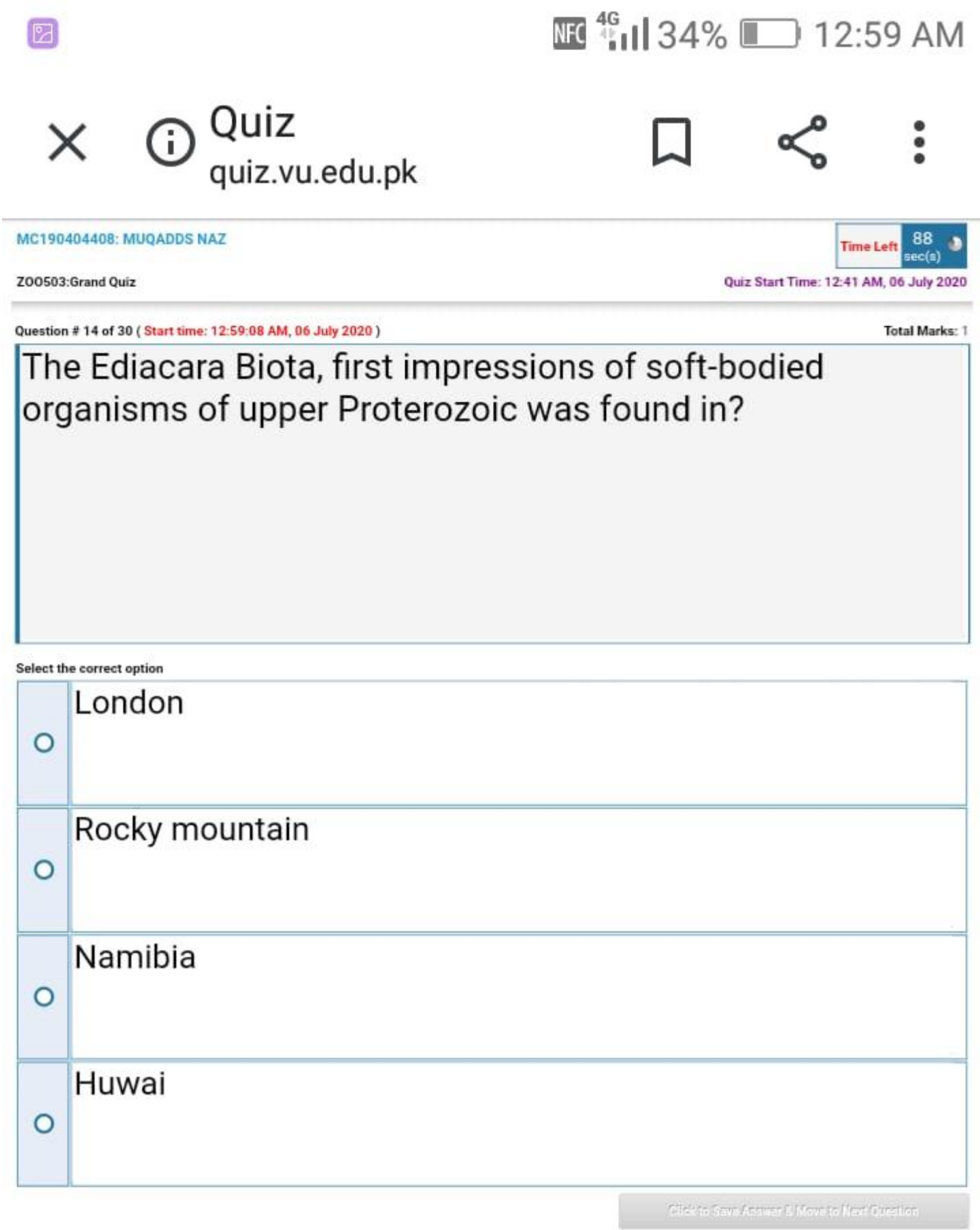
l

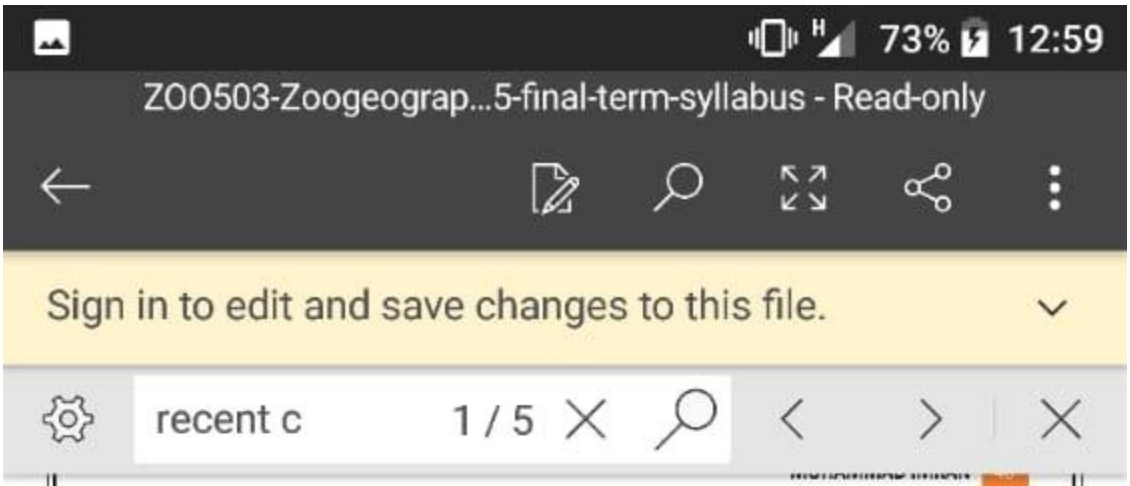
rnal (exoskeleton) in the case of most invertebrates
doskeleton) structures, in the case of a few mollusks


tenophorans
• an outer ectoderm
• an inner endoderm
• Two layers are separated by acellular mesogloea


NFC 4G 34% 12:56 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 89 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 12 of 30 (Start time: 12:56:25 AM, 06 July 2020)
Total Marks: 1
A succession of contiguous formations with common characteristics is called a ____.
Select the correct option
bed
member
group
none
Click to Save Answer & Move to Next Question


NFC 4G 34% 12:57 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 13 of 30 (Start time: 12:57:02 AM, 06 July 2020)
Total Marks: 1
The _____ is the recent Cnidarian ?
Select the correct option
Chondrophora
Ediacaran
Lagerstätten.
Octocorallia
Click to Save Answer & Move to Next Question


NFC 4G 34% 12:59 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 14 of 30 (Start time: 12:59:08 AM, 06 July 2020)
Total Marks: 1
The Ediacara Biota, first impressions of soft-bodied organisms of upper Proterozoic was found in?
Select the correct option
London
Rocky mountain
Namibia
Huwai
Click to Save Answer & Move to Next Question



Medusoids and polyps appear different, they are essentially same inverted structures. Cnidarian life cycles generalized view of the life of the hydrozoan Obelia, alternating between the conspicuous polyp and medusa stages

Class Hydrozoa

Polymorphic forms. Undivided enteron, solid tentacles, colonies. Chondrophora oldest cnidarians. Ediacaran to Recent **Class Scyphozoa**

Jellyfish, Lagerstätten. The extinct Conulata. Appeared in Cambrian. Extinct in Mid Triassic. Ediacaran to Recent

Cl

Marine, sessile, colonial forms. Mobile planula l vae. Octocorallia class includes corals, sea anemones. Produce spicules microfossils. Ediacaran to Recent

Cl

The sea wasps and box jellyfish. Both medusae and polyps. Restricted to tropical and subtropical latitudes. Carboniferous to Recent

Corals

Most diverse and most threatened ecosystems. The coral reef. Shallow-water coral reefs in zone extending 30° north and south of equator. 30m or 18 °C

57 Spiralians 1: lophophorates

The Spiralians

Morphologically diverse clade of protostome animals

- Molluscs
- Annelids
- Platyhelminths and other taxa

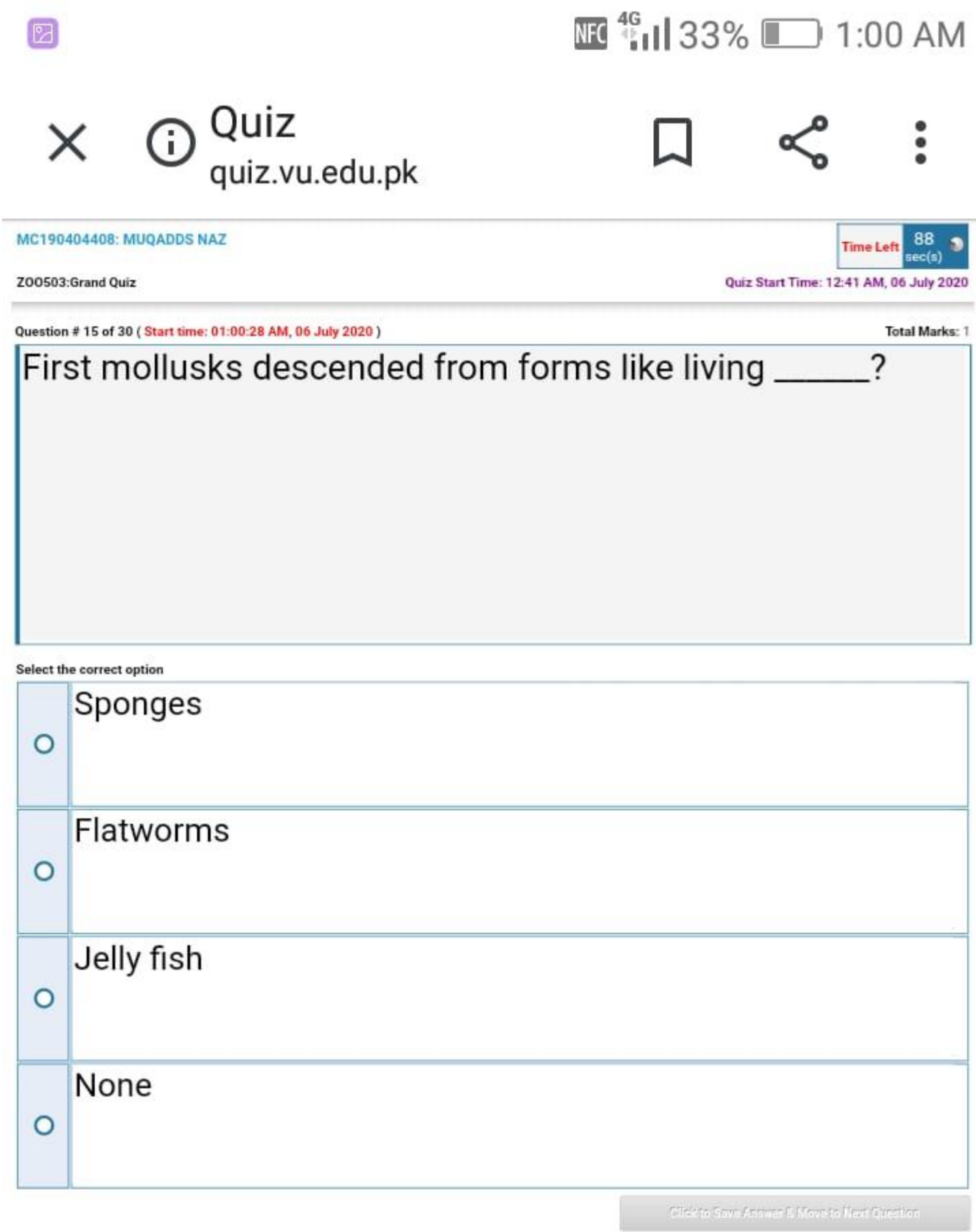
Lophophores

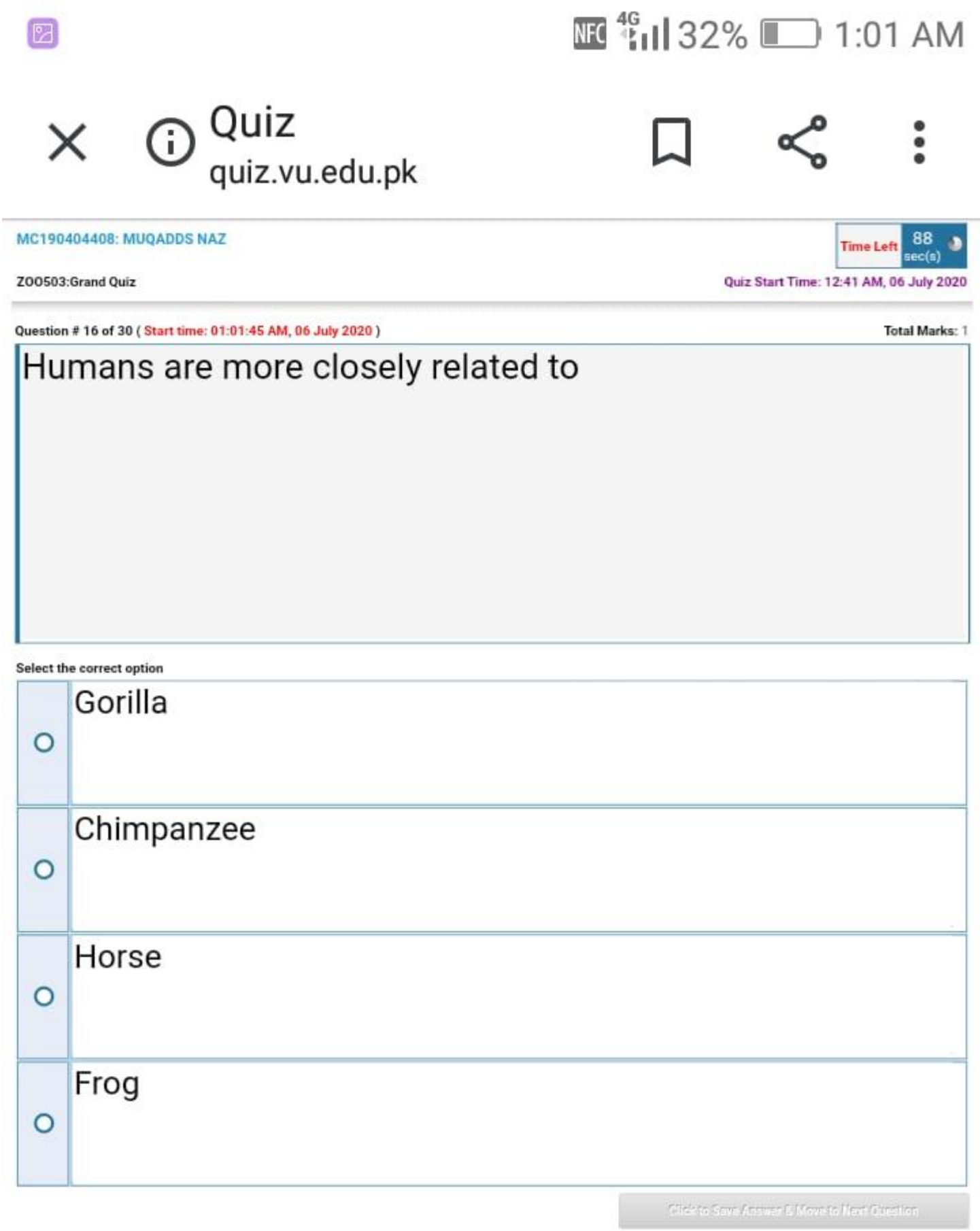
Lophophores, a filamentous feeding organ. Three spiralian invertebrate groups

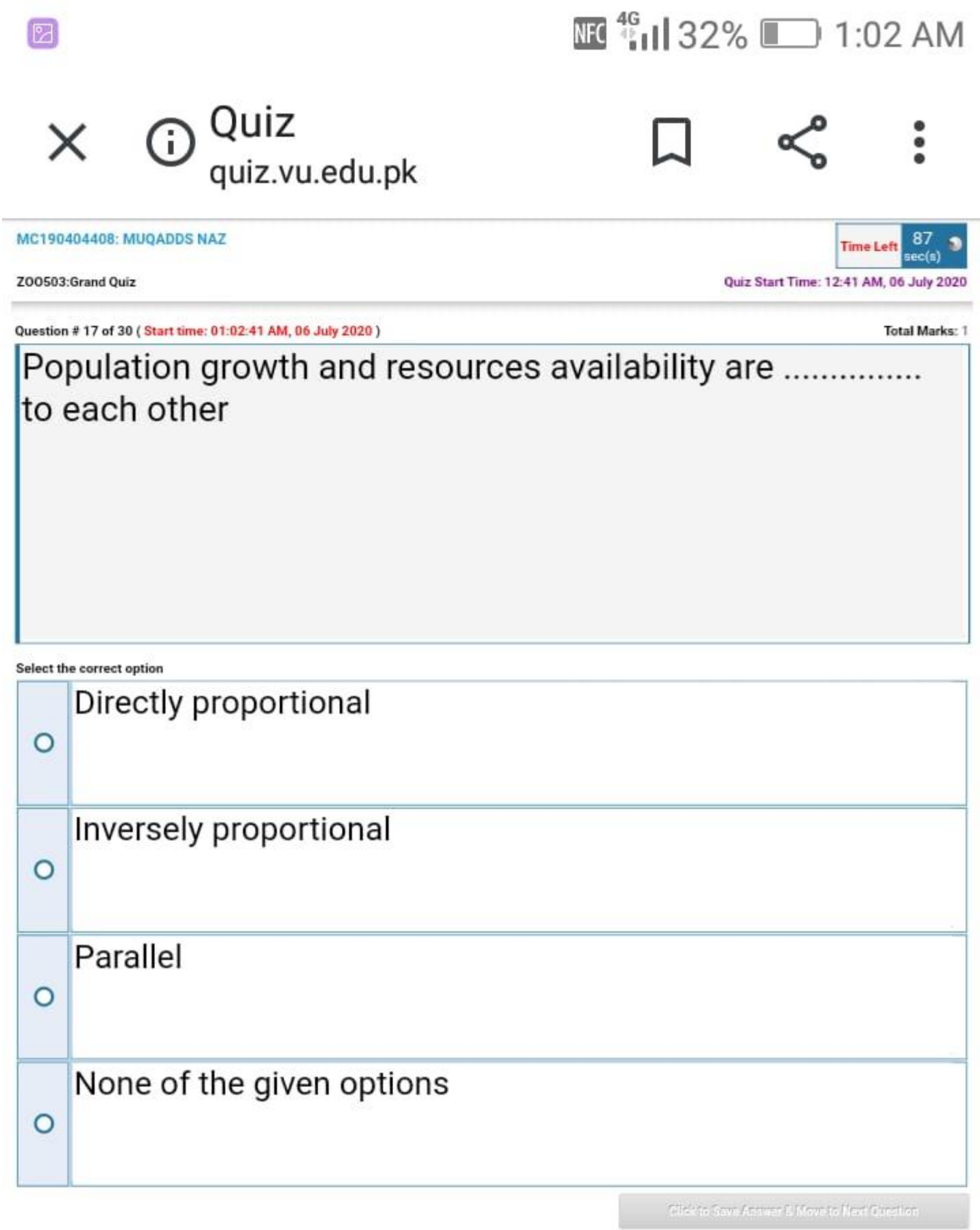
- Brachiopods
- Bryozoans
- Phoronids

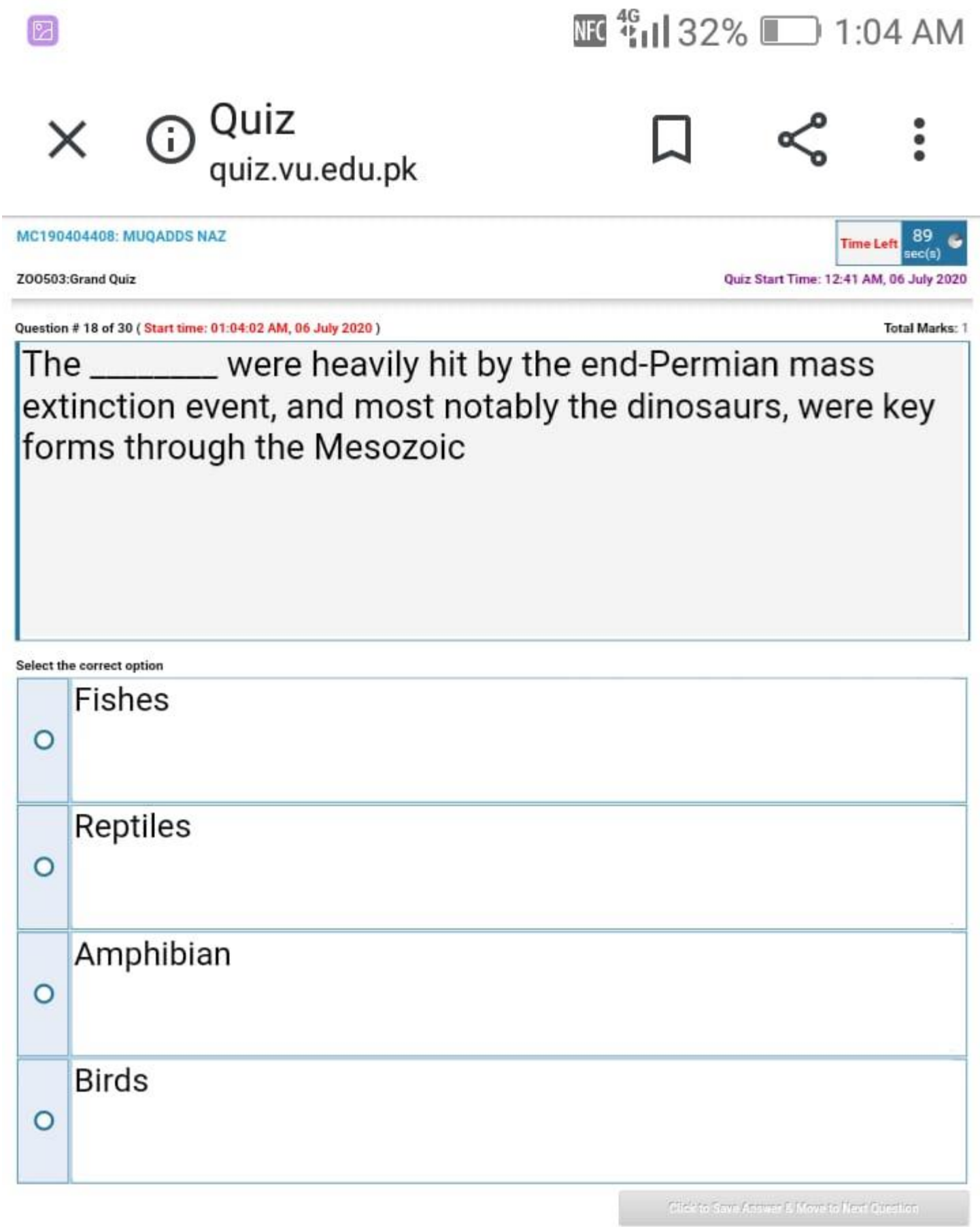
Brachiopods

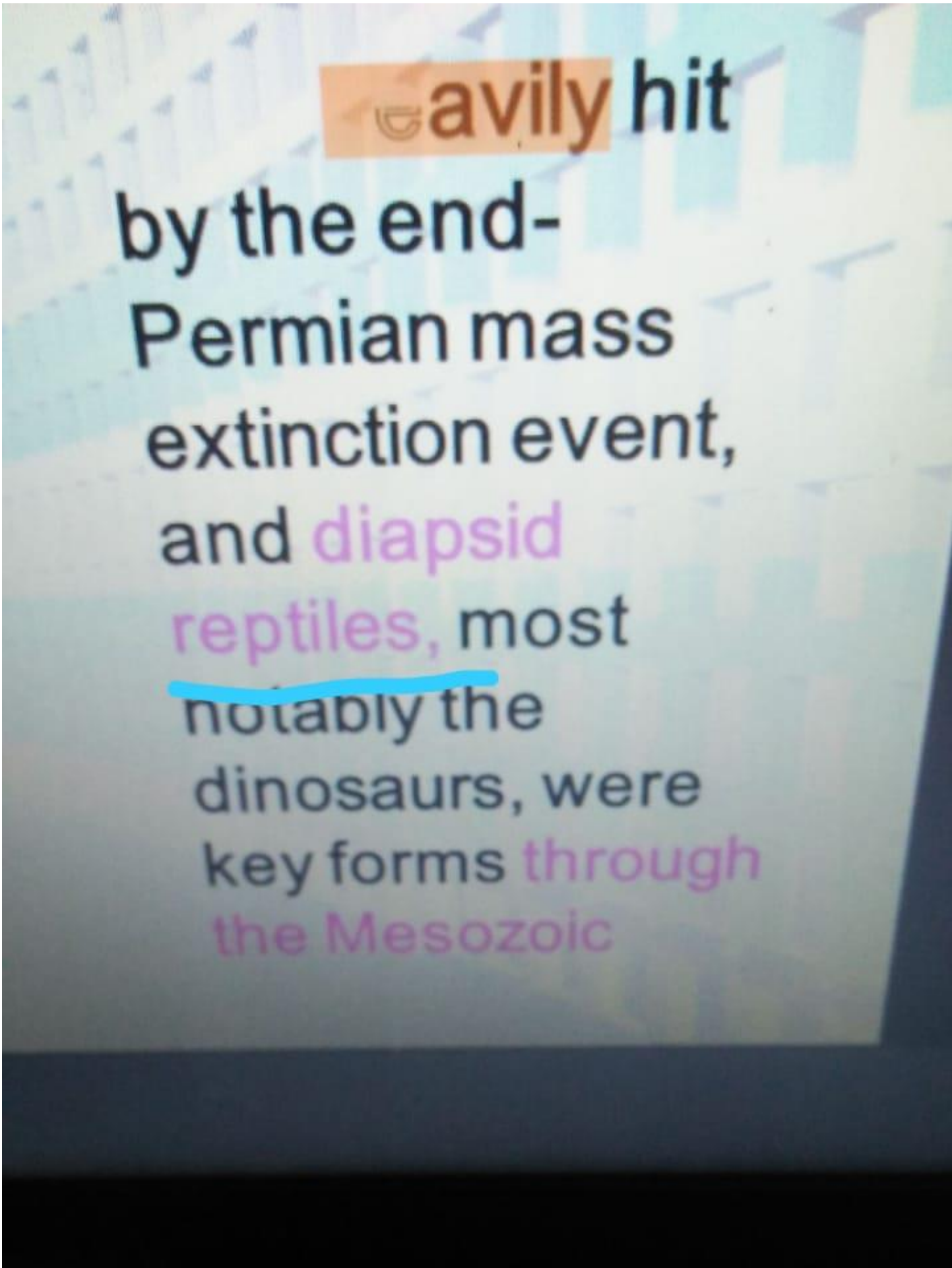
- Twin-valved shellfish,
- Lophophore and a pedicle,
- Linguliformeans: organophosphatic shells
- Craniiformeans & rhynchonelliformeans: calcareous shells

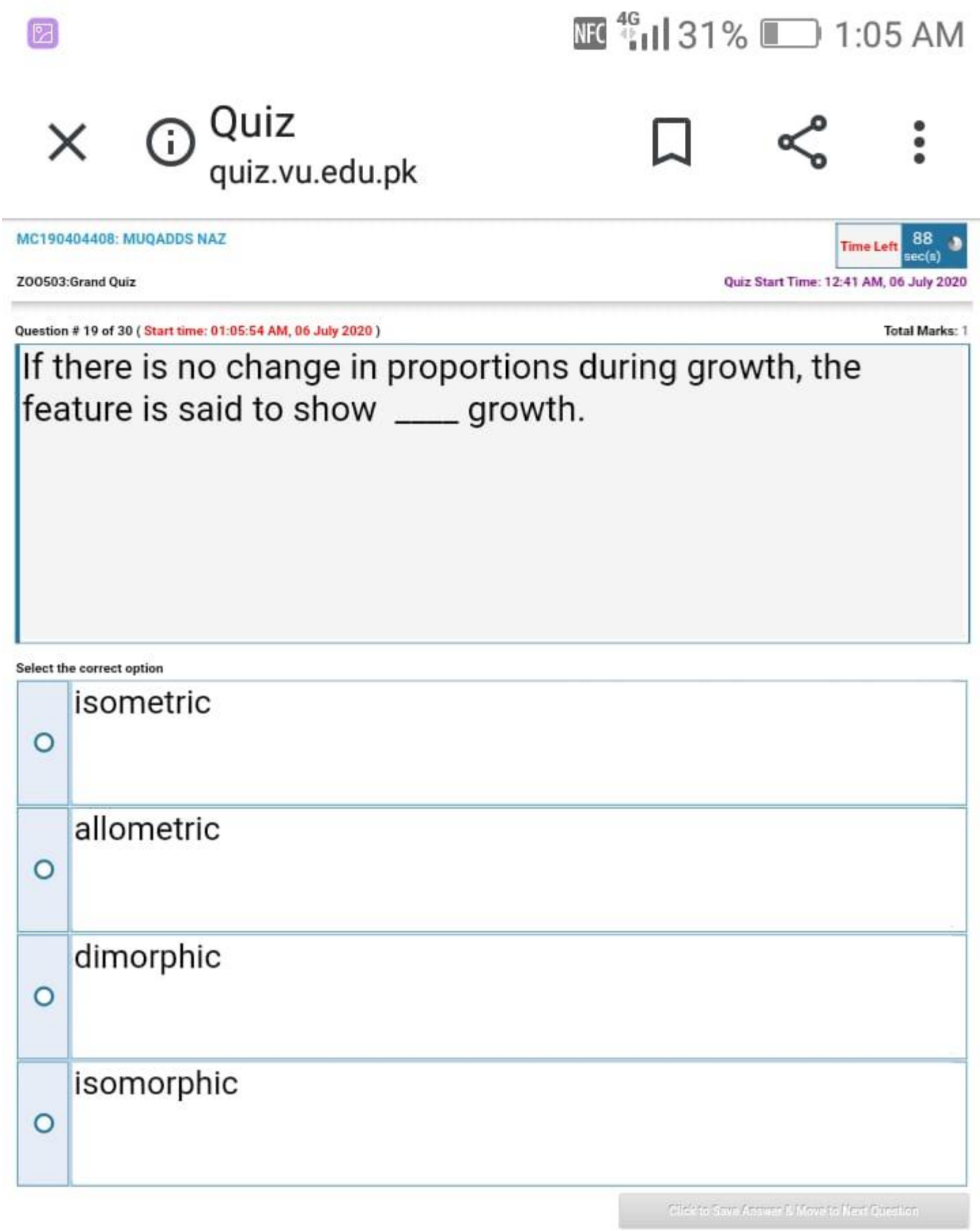

NFC 4G 33% 1:00 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 15 of 30 (Start time: 01:00:28 AM, 06 July 2020)
Total Marks: 1
First mollusks descended from forms like living _____?
Select the correct option
Sponges
Flatworms
Jelly fish
None
Click to Save Answer & Move to Next Question

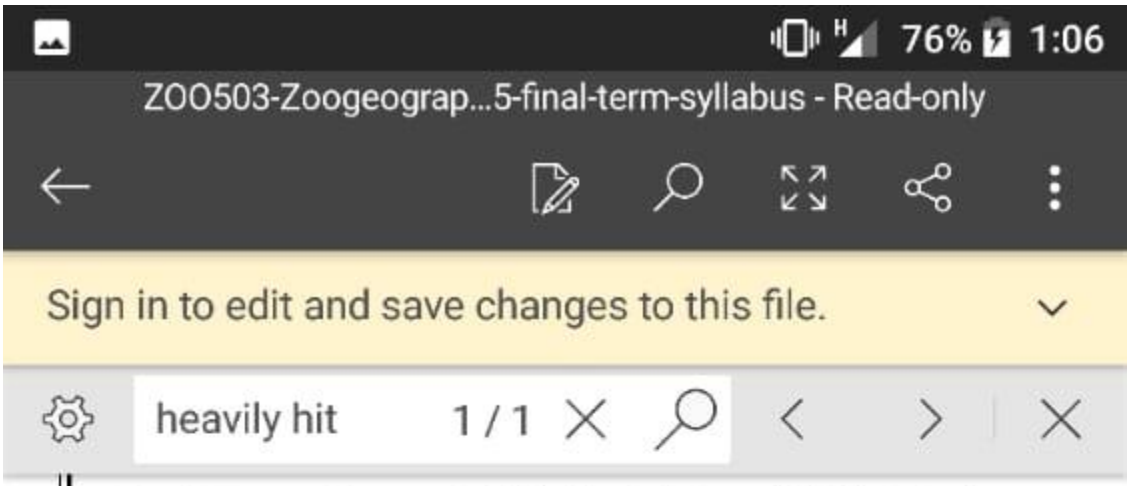

NFC 4G 32% 1:01 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 16 of 30 (Start time: 01:01:45 AM, 06 July 2020)
Total Marks: 1
Humans are more closely related to
Select the correct option
Gorilla
Chimpanzee
Horse
Frog
Click to Save Answer & Move to Next Question


NFC 4G 32% 1:02 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 87 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 17 of 30 (Start time: 01:02:41 AM, 06 July 2020)
Total Marks: 1
Population growth and resources availability are ..... to each other
Select the correct option
Directly proportional
Inversely proportional
Parallel
None of the given options
Click to Save Answer & Move to Next Question


NFC 4G 32% 1:04 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 89 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 18 of 30 (Start time: 01:04:02 AM, 06 July 2020)
Total Marks: 1
The _____ were heavily hit by the end-Permian mass extinction event, and most notably the dinosaurs, were key forms through the Mesozoic
Select the correct option
Fishes
Reptiles
Amphibian
Birds
Click to Save Answer & Move to Next Question


avily hit
by the end-
Permian mass
extinction event,
and diapsid
reptiles, most
notably the
dinosaurs, were
key forms through
the Mesozoic


NFC 4G 31% 1:05 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 19 of 30 (Start time: 01:05:54 AM, 06 July 2020)
Total Marks: 1
If there is no change in proportions during growth, the feature is said to show ____ growth.
Select the correct option
isometric
allometric
dimorphic
isomorphic
Click to Save Answer & Move to Next Question



Conodonts commonly occur as tooth-like elements that are useful in biostratigraphy, as are some teeth and scales (ichthyoliths). **Tetrapods**

Arose during the Devonian from lobe-finned fish ancestors, and fish-eating amphibians.

Diversified in the Carboniferous.

Reptiles

The first reptiles were small insect eaters. Synapsids dominated ecosystems on land during Triassic. The non-mammalian members are described as mammal-like reptiles in classical systems be called stem mammals or proto-mammals. **Synapsids** evolved from basal amniotes and are on groups of the later amniotes; the other is the sauropsids, a group that includes modern reptiles groups were heavily hit by the end-Permian mass extinction event, and diapsid reptiles, most not were key forms through the Mesozoic. Diapsids ("two arches") are a group of amniote tetrapods t holes (temporal fenestra) in each side of their skulls about 300 million years ago during the l period. The diapsids are extremely diverse, and include all crocodiles, lizards, snakes, tuatara, tur

Importance of Vertebrates

Humans are such a successful species. The huge diversity and abundance of species of bony mammals. Insects and microbes, arealso abundant and diverse. Vertebrates include the largest a the sea and in the air.

Final term 81 to 165

ZOO503 - Zoogeography and Paleontology

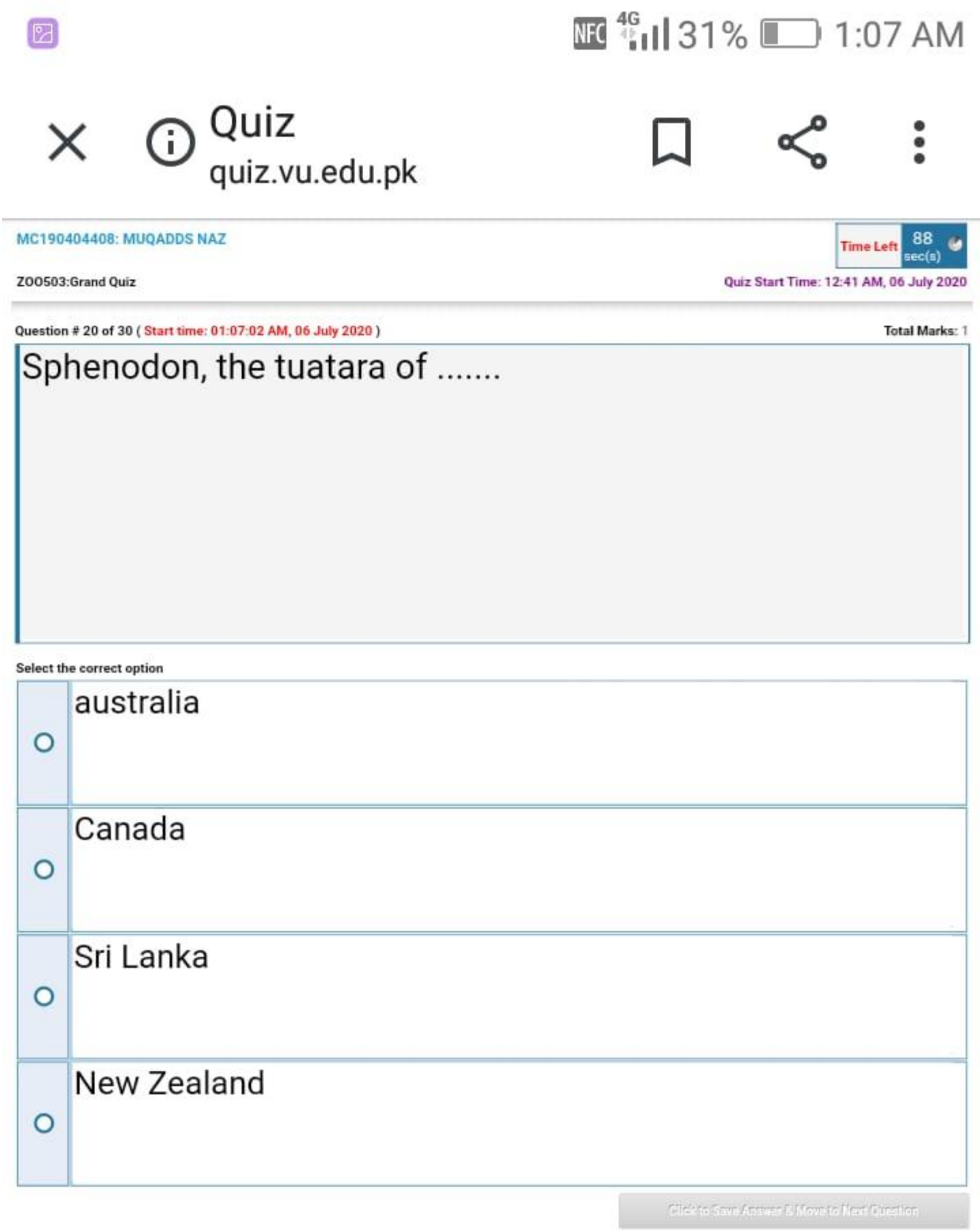
81 Fishes and basal tetrapod's: Origin of the vertebrates

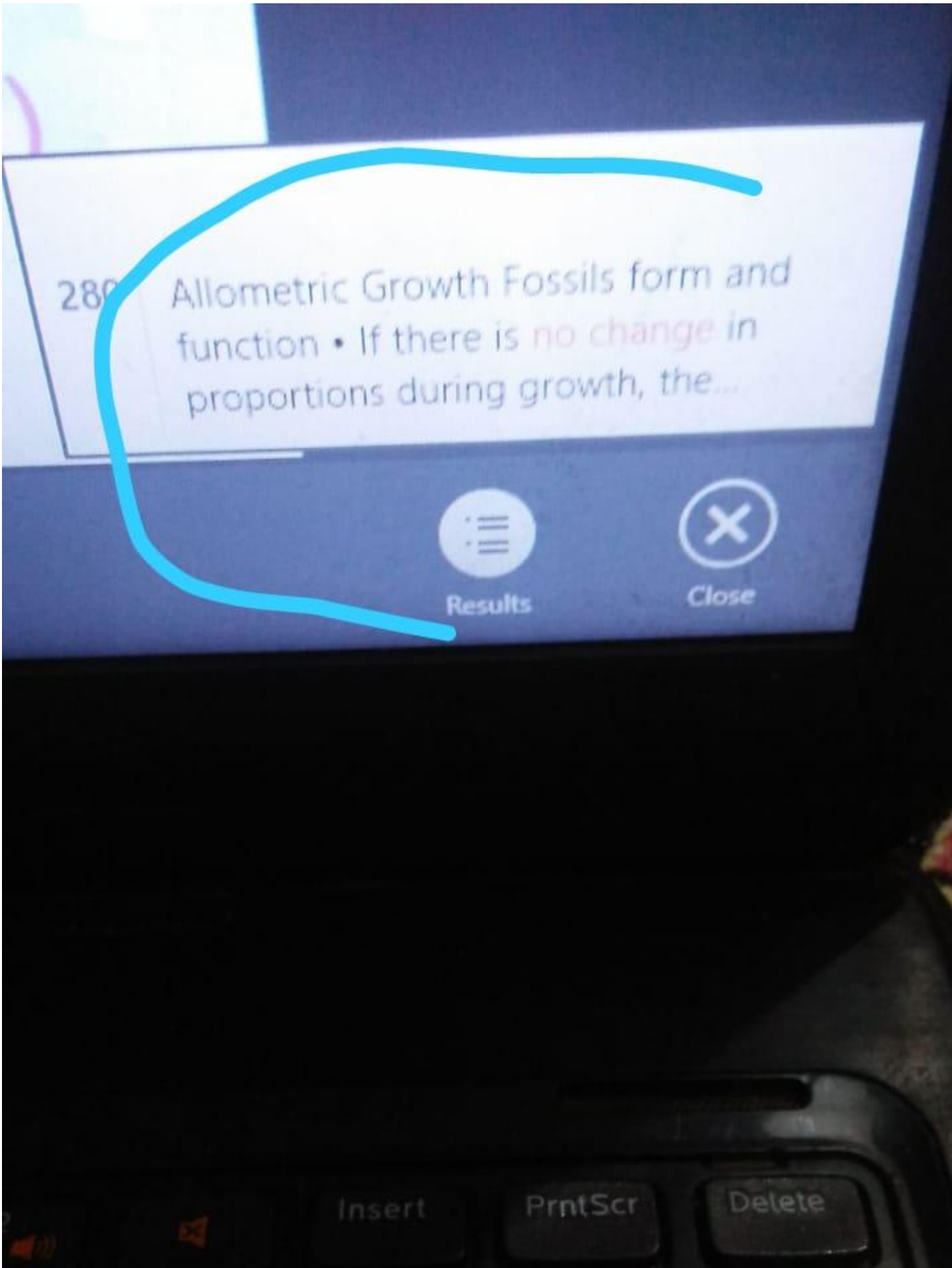
The Skeleton

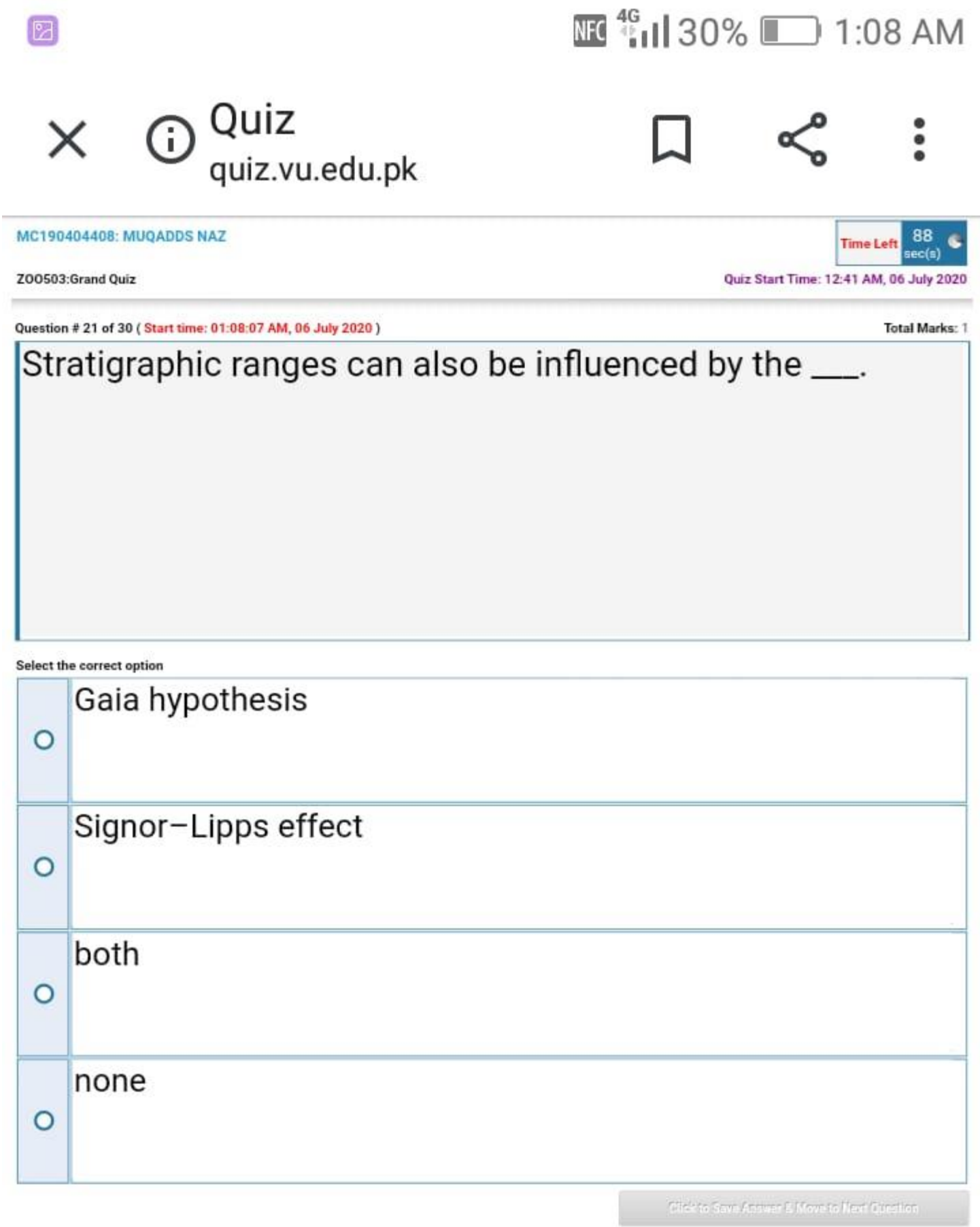
Bone and cartilage

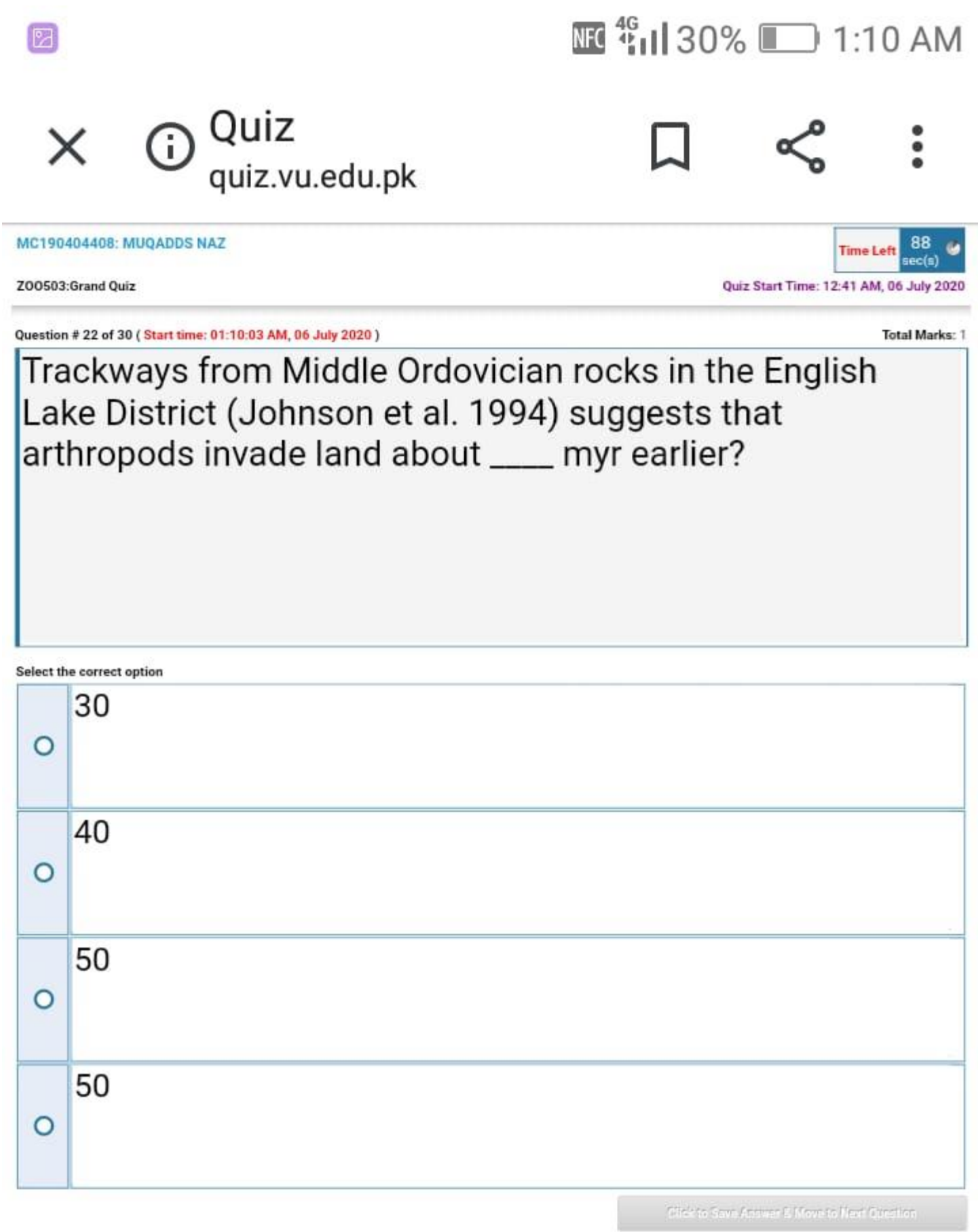
- **Bone:** A network of collagen fibers on which needle-like crystals of hydroxyapatite CaPO4) accumulate
- **Cartilage**: Flexible tissue, usually unmineralized, contains collagen and elastic tissue

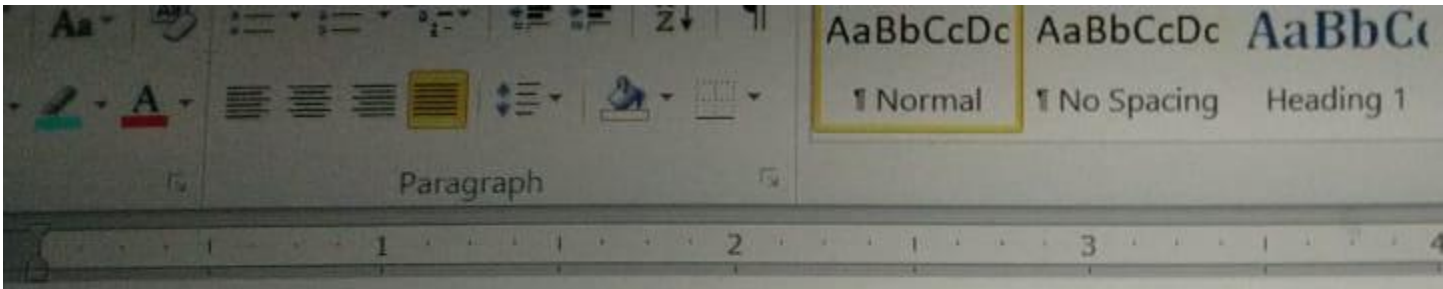
In humans; early skeleton, cartilage, progressively mineralizes by apatite. The first ve


NFC 4G 31% 1:07 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 20 of 30 (Start time: 01:07:02 AM, 06 July 2020)
Total Marks: 1
Sphenodon, the tuatara of .....
Select the correct option
australia
Canada
Sri Lanka
New Zealand
Click to Save Answer & Move to Next Question


Allometric Growth Fossils form and function • If there is no change in proportions during growth, the...
Results
Close
Insert
PrntScr
Delete


NFC 4G 30% 1:08 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 21 of 30 (Start time: 01:08:07 AM, 06 July 2020)
Total Marks: 1
Stratigraphic ranges can also be influenced by the ___.
Select the correct option
Gaia hypothesis
Signor–Lipps effect
both
none
Click to Save Answer & Move to Next Question


30% 1:10 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 22 of 30 (Start time: 01:10:03 AM, 06 July 2020)
Total Marks: 1
Trackways from Middle Ordovician rocks in the English Lake District (Johnson et al. 1994) suggests that arthropods invade land about ____ myr earlier?
Select the correct option
30
40
50
50
Click to Save Answer & Move to Next Question



➢**Allometric Growth**

The ratio of eye diameter to body length diminishes as th
of allometric ("different measure") growth.

➢**Isometric Growth**

If there is no change in proportions during growth, the fe
growth.

BIOLOGICAL SCALING PRINCIPLE:

Some organs and functions relate to the mass of an anin
body length or body outline (one- and two-dimensional me

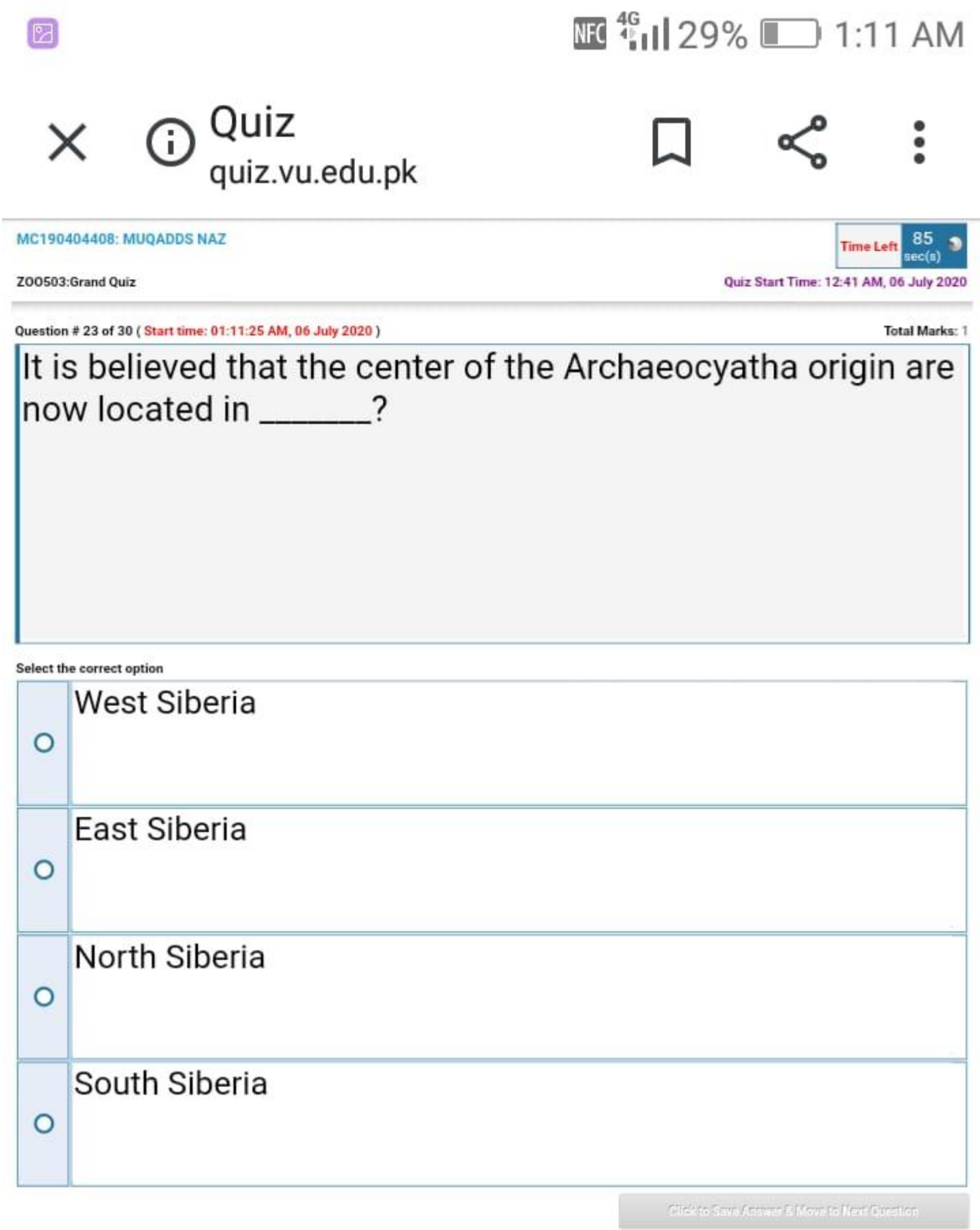
FOSSIL FORM AND FUNCTION : EVO

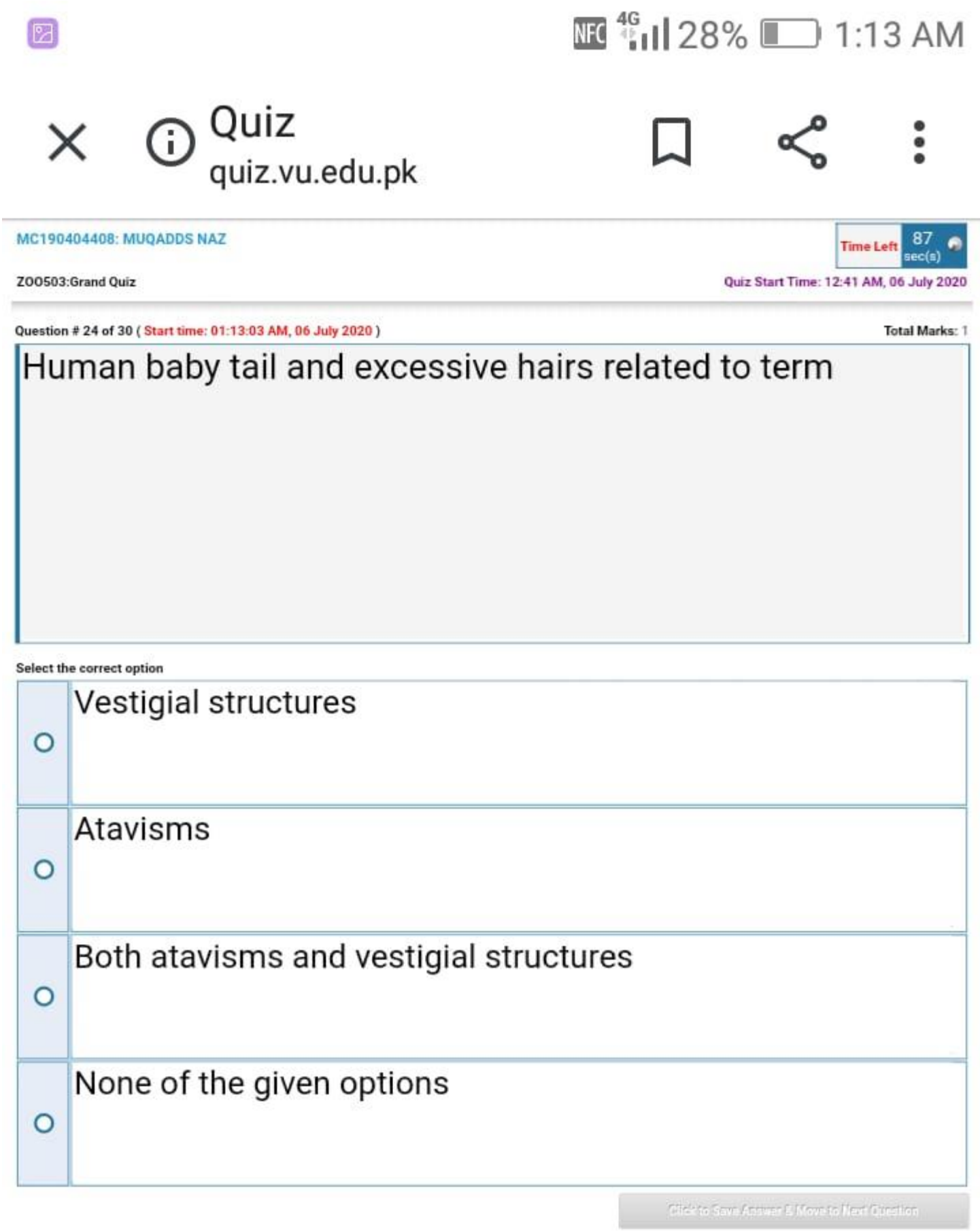
ONTOGENY AND PHYLOGENY

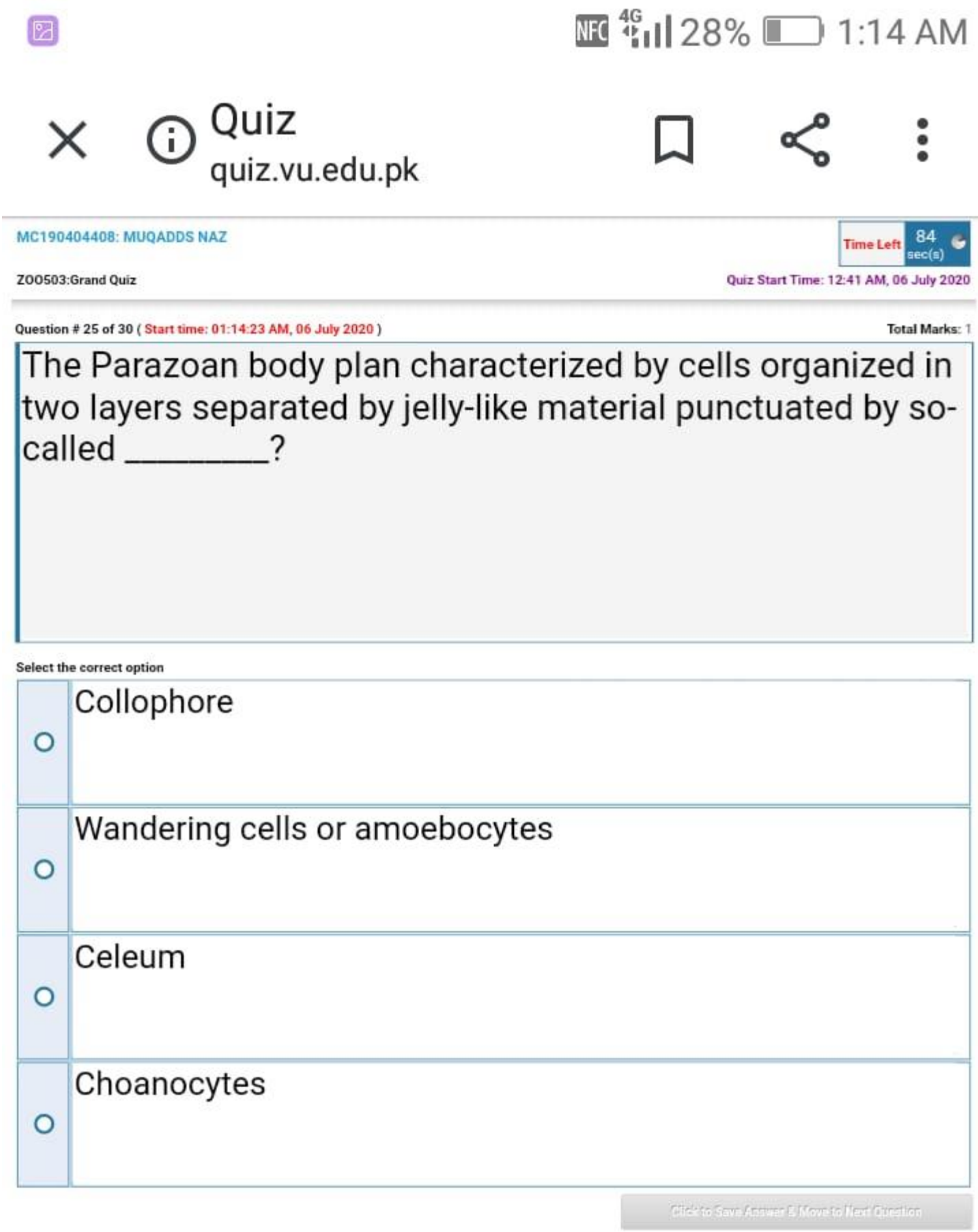
Biologists have long sought a link between ontogeny (deve

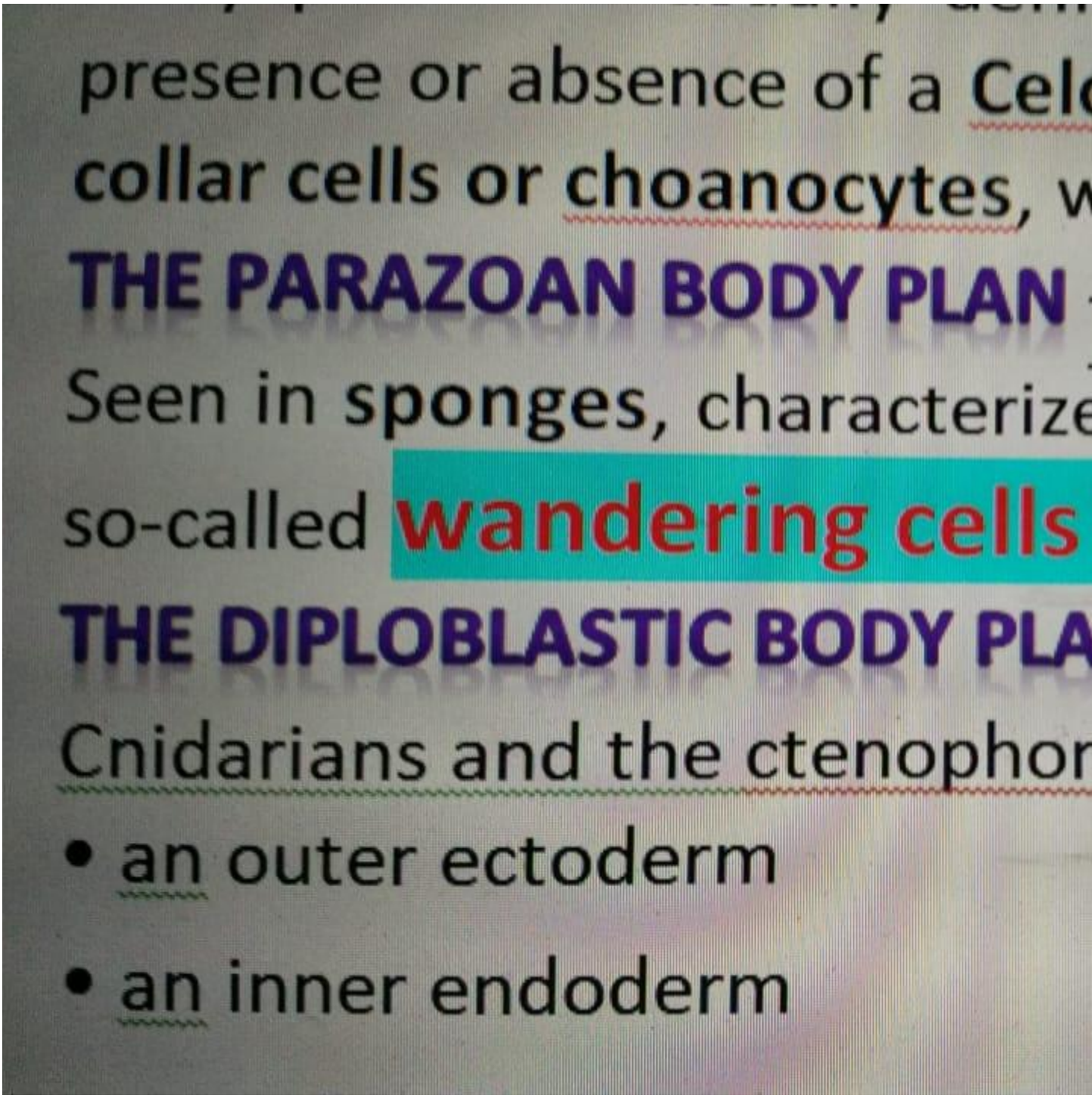
BIOGENETIC LAW

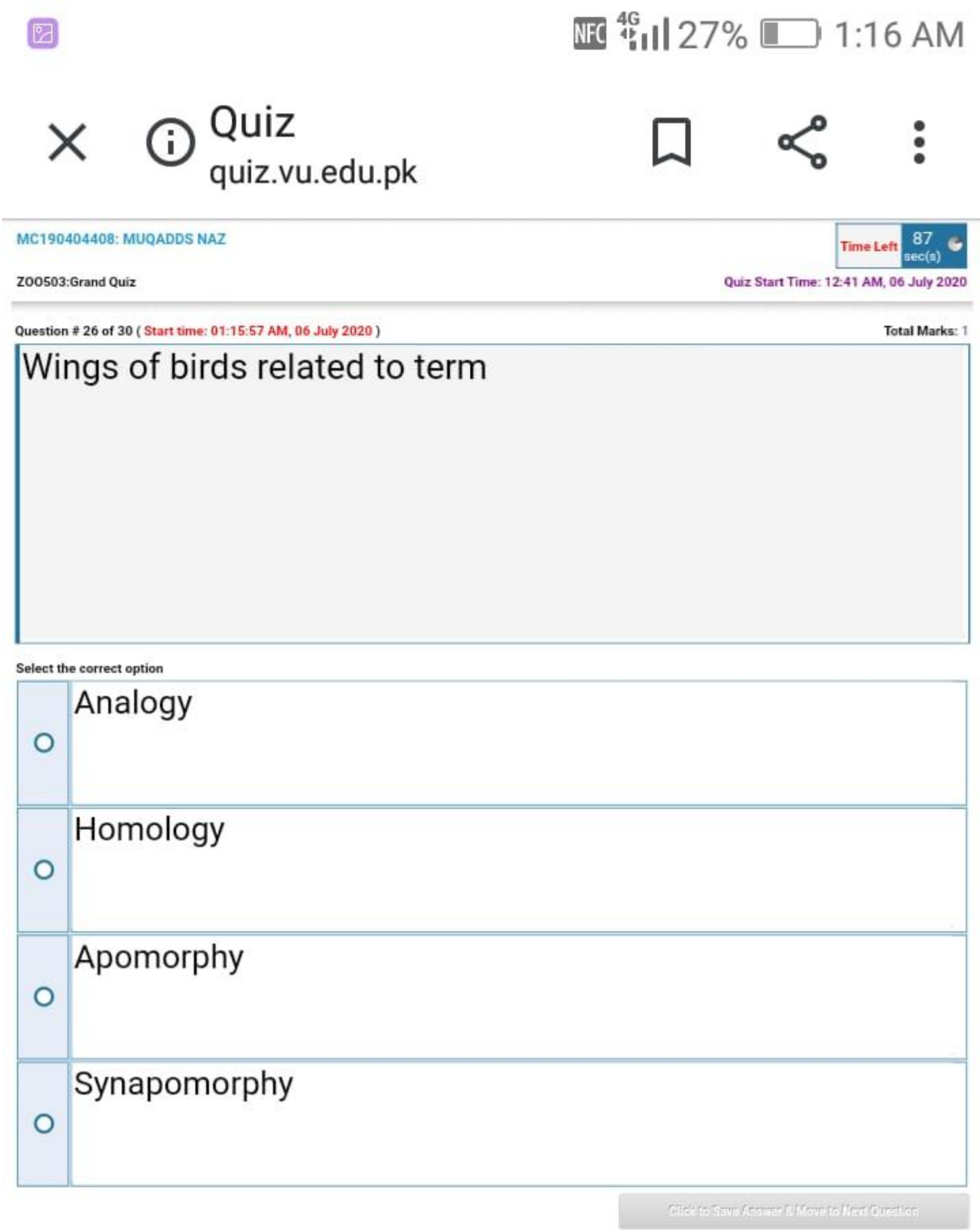
In 1866, Ernst Haeckel, a German evolutionist, announce
phylogeny". Ernst Haeckel's idea was that the sequence o

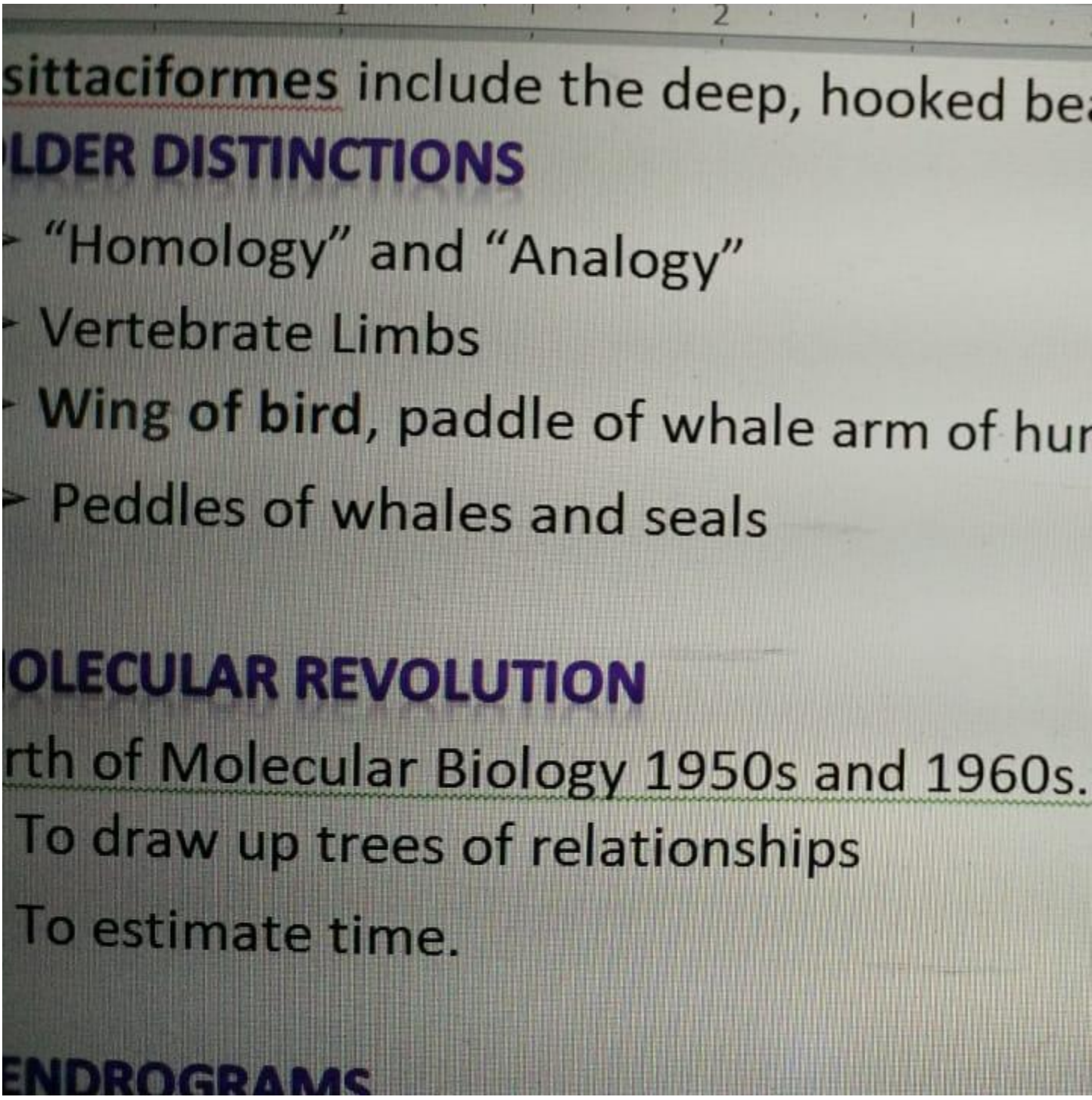

NFC 4G 29% 1:11 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 85 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 23 of 30 (Start time: 01:11:25 AM, 06 July 2020)
Total Marks: 1
It is believed that the center of the Archaeocyatha origin are now located in _____?
Select the correct option
West Siberia
East Siberia
North Siberia
South Siberia
Click to Save Answer & Move to Next Question

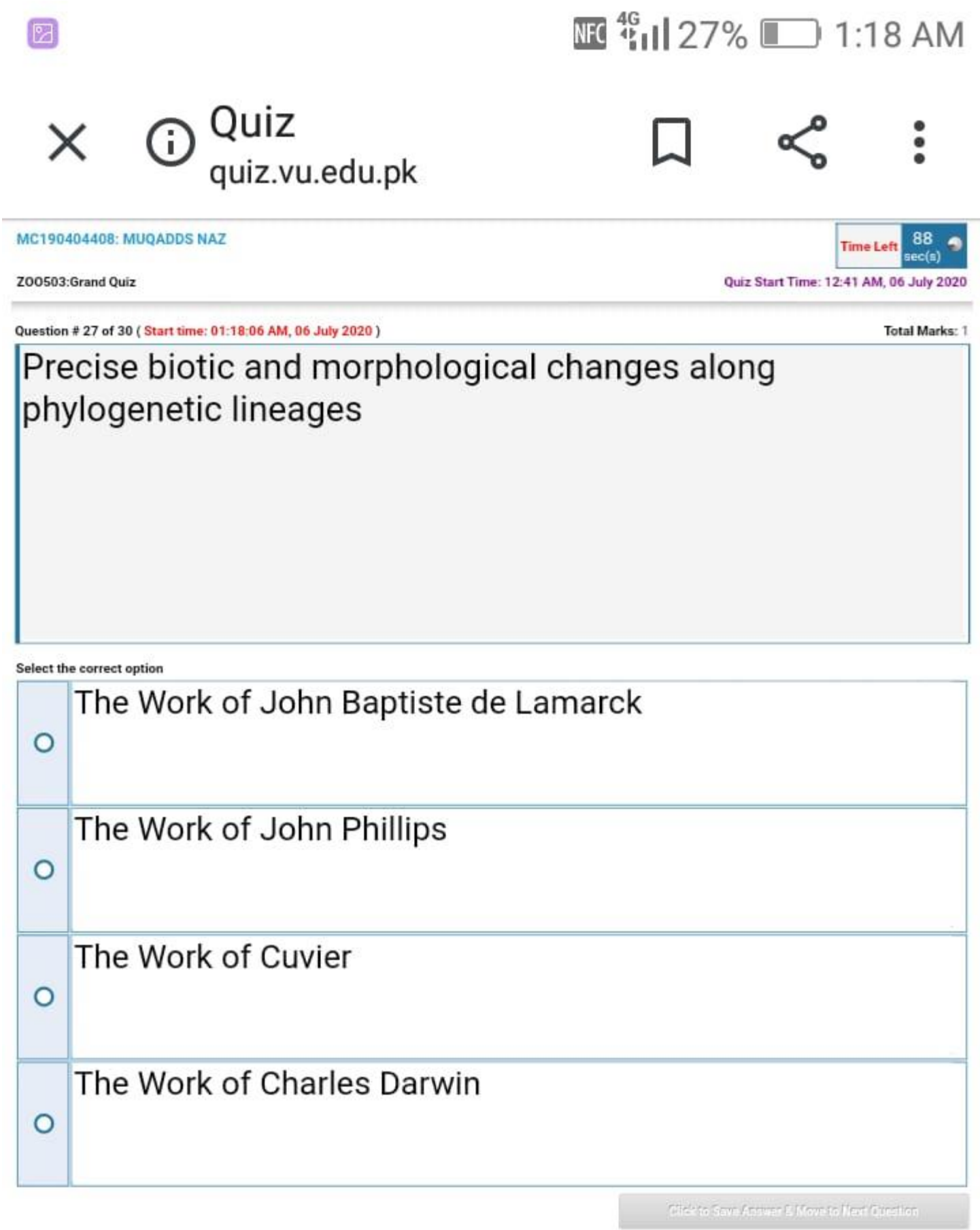

NFC 4G 28% 1:13 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 87 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 24 of 30 (Start time: 01:13:03 AM, 06 July 2020)
Total Marks: 1
Human baby tail and excessive hairs related to term
Select the correct option
Vestigial structures
Atavisms
Both atavisms and vestigial structures
None of the given options
Click to Save Answer & Move to Next Question

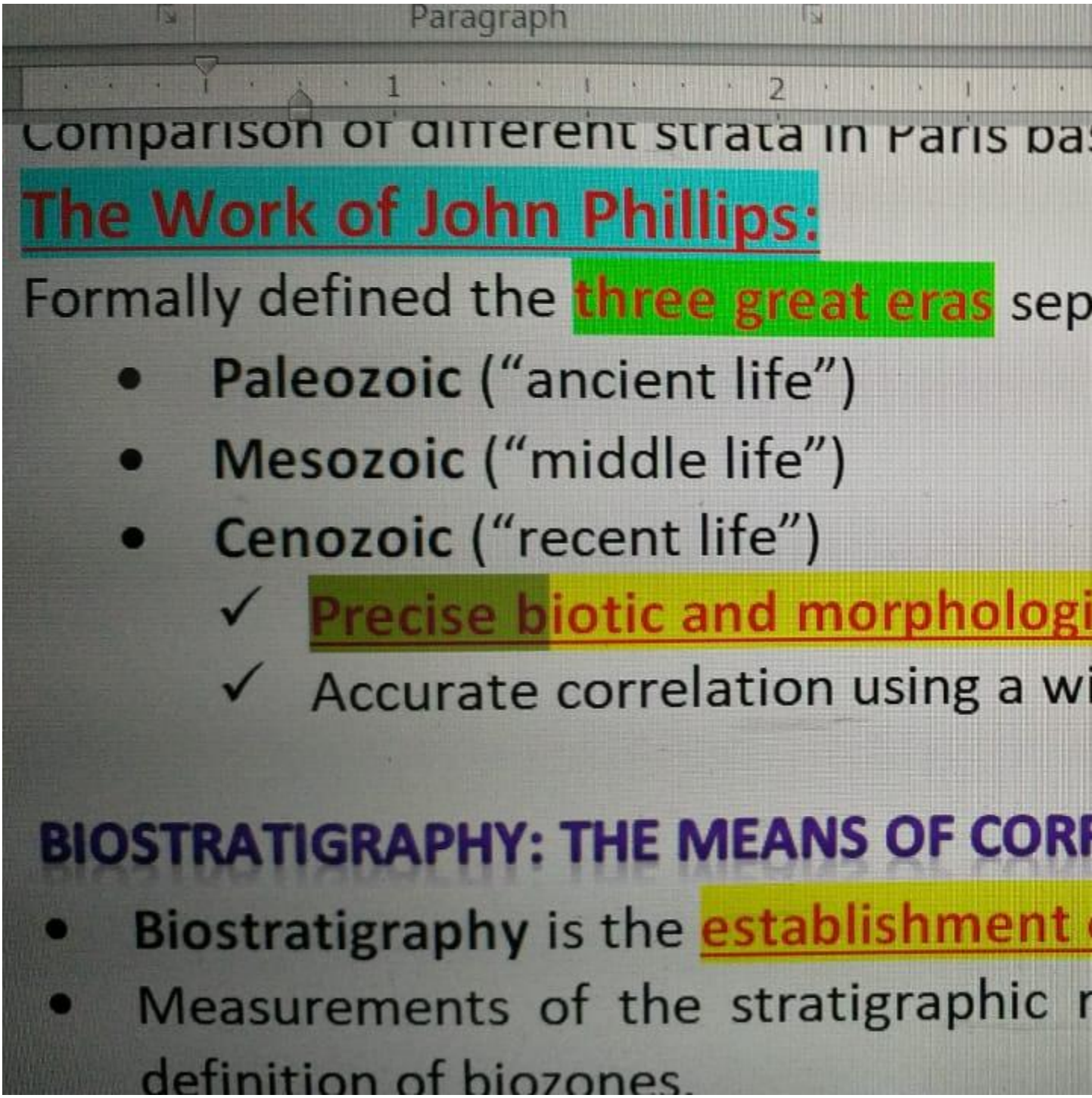

NFC 4G 28% 1:14 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 84 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 25 of 30 (Start time: 01:14:23 AM, 06 July 2020)
Total Marks: 1
The Parazoan body plan characterized by cells organized in two layers separated by jelly-like material punctuated by so-called _____?
Select the correct option
Collophore
Wandering cells or amoebocytes
Celeum
Choanocytes
Click to Save Answer & Move to Next Question

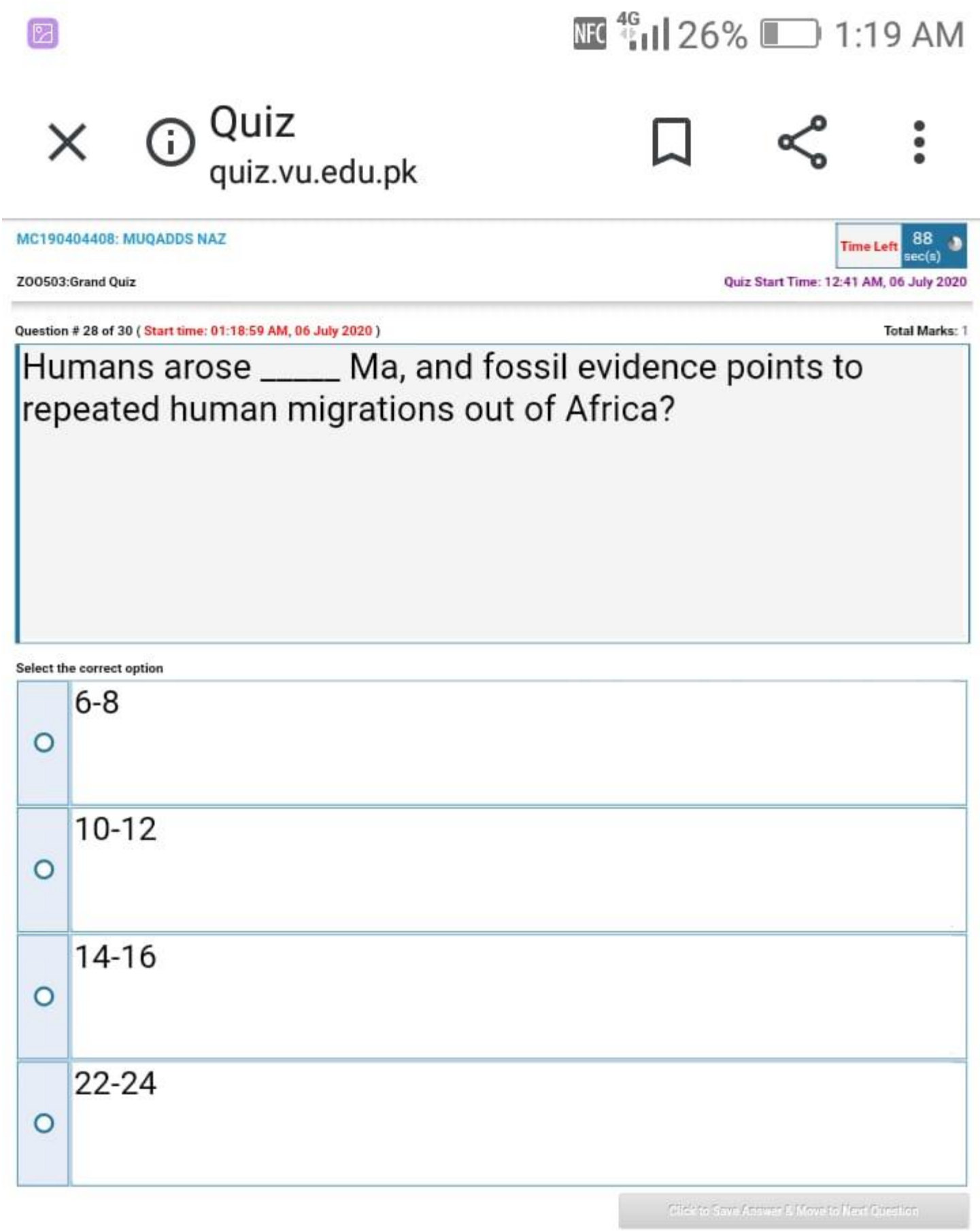

presence or absence of a Cel
collar cells or choanocytes, w
THE PARAZOAN BODY PLAN
Seen in sponges, characterize
so-called wandering cells
THE DIPLOBLASTIC BODY PLA
Cnidarians and the ctenophor
• an outer ectoderm
• an inner endoderm

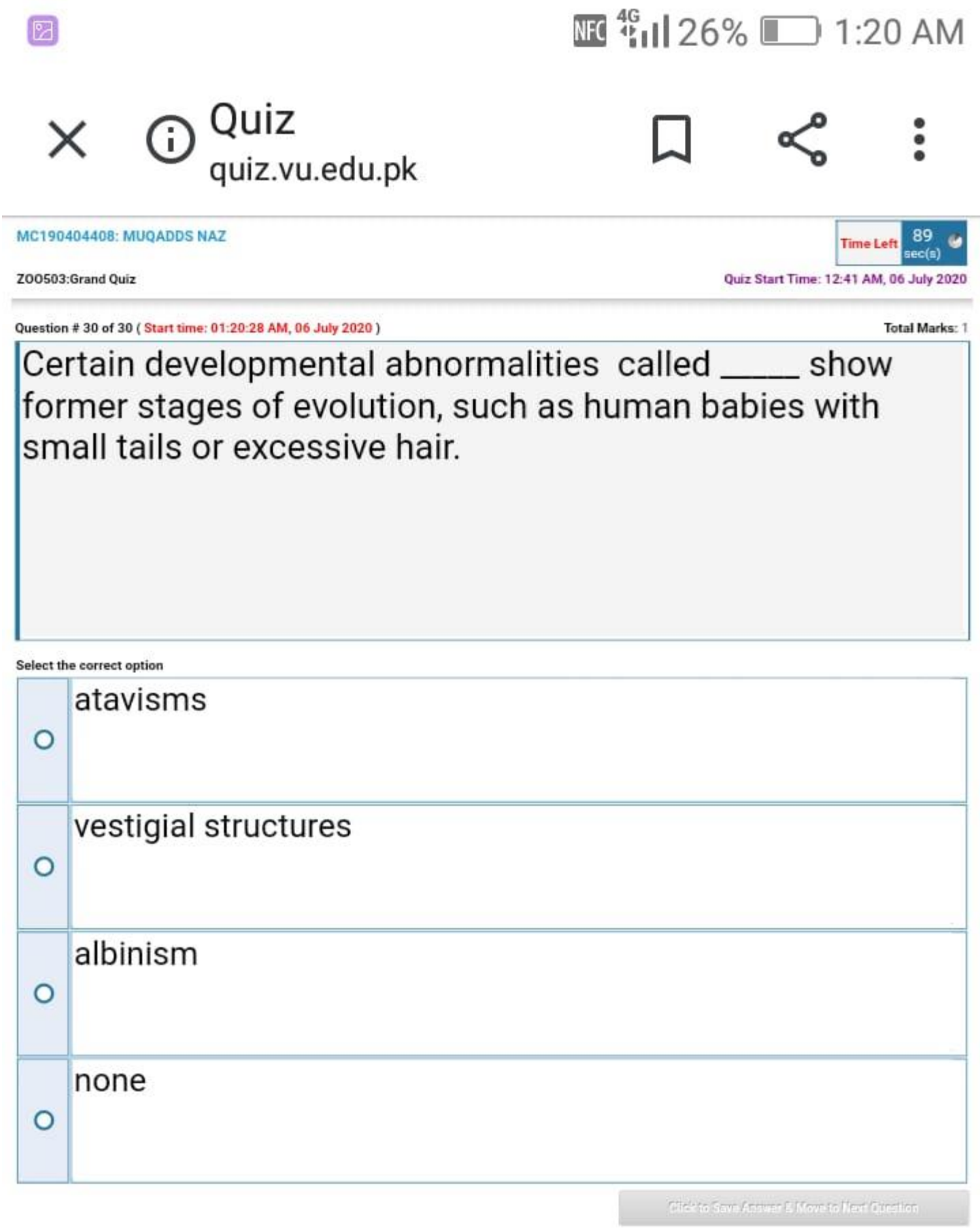

NFC 4G 27% 1:16 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 87 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 26 of 30 (Start time: 01:15:57 AM, 06 July 2020)
Total Marks: 1
Wings of birds related to term
Select the correct option
Analogy
Homology
Apomorphy
Synapomorphy
Click to Save Answer & Move to Next Question


sittaciformes include the deep, hooked be
LDER DISTINCTIONS
"Homology" and "Analogy"
Vertebrate Limbs
Wing of bird, paddle of whale arm of hu
Peddles of whales and seals
OLECULAR REVOLUTION
rth of Molecular Biology 1950s and 1960s.
To draw up trees of relationships
To estimate time.
ENDROGRAMS


NFC 4G 27% 1:18 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 27 of 30 (Start time: 01:18:06 AM, 06 July 2020)
Total Marks: 1
Precise biotic and morphological changes along phylogenetic lineages
Select the correct option
The Work of John Baptiste de Lamarck
The Work of John Phillips
The Work of Cuvier
The Work of Charles Darwin
Click to Save Answer & Move to Next Question


Paragraph
Comparison of different strata in Paris ba
The Work of John Phillips:
Formally defined the three great eras sep
Paleozoic ("ancient life")
Mesozoic ("middle life")
Cenozoic ("recent life")
✓ Precise biotic and morphologi
✓ Accurate correlation using a wi
BIOSTRATIGRAPHY: THE MEANS OF CORR
Biostratigraphy is the establishment
Measurements of the stratigraphic r
definition of biozones.


NFC 4G 26% 1:19 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 88 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 28 of 30 (Start time: 01:18:59 AM, 06 July 2020)
Total Marks: 1
Humans arose _____ Ma, and fossil evidence points to repeated human migrations out of Africa?
Select the correct option
6-8
10-12
14-16
22-24
Click to Save Answer & Move to Next Question


NFC 4G 26% 1:20 AM
Quiz
quiz.vu.edu.pk
MC190404408: MUQADDS NAZ
Time Left 89 sec(s)
ZOO503:Grand Quiz
Quiz Start Time: 12:41 AM, 06 July 2020
Question # 30 of 30 (Start time: 01:20:28 AM, 06 July 2020)
Total Marks: 1
Certain developmental abnormalities called _____ show former stages of evolution, such as human babies with small tails or excessive hair.
Select the correct option
atavisms
vestigial structures
albinism
none
Click to Save Answer & Move to Next Question

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 5 of 30 (Start time: 01:00:07 AM, 06 July 2020)

The first vertebrates fish had no _____. Fossil sites at Chengjiang in China: Early Cambrian?

Select the correct option

○	Lungs
○	Jaws
○	Kidney
○	Heart

ZOO503:Grand Quiz

Question # 7 of 30 (Start time: 01:01:47 AM, 06 July 2020)

The Sub phylum Myriapods first appeared during the _____?

Select the correct option

○	Devonion
○	Mid Silurian
○	Jurassic
○	Mesozoic

ZOO503:Grand Quiz

Question # 8 of 30 (Start time: 01:03:04 AM, 06 July 2020)

The _____ diversification generated witnessed a staggering increase in biodiversity at the family, genus and species levels.

Select the correct option

○	Devonian
○	Jurassic
○	Mesozoic
●	Ordovician

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 10 of 30 (Start time: 01:04:38 AM, 06 July 2020)

The Subphylum _____ can be divided into pterygotes (with wings) and apterygotes (without wings)?

Select the correct option

- Myripods
- Crustacea
- Hexapods
- None

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 12 of 30 (Start time: 01:06:47 AM, 06 July 2020)

___ means the rock has been covered by younger rock.

Select the correct option

- Burial
- Deposition
- Decay
- Mineralization

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 16 of 30 (Start time: 01:11:54 AM, 06 July 2020)

Paleontology lies on the border

Select the correct option

- between Biology and Geology
- between Ecology and Geology
- between Biology and Geography
- between Time and Space

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 17 of 30 (Start time: 01:12:30 AM, 06 July 2020)

Scottish scientist ____ (1726-1797) challenged
"no vestige of a beginning,- no prospect of an end.

Select the correct option

○	Robert Hooke
●	James Hutton
○	Robert Brown
○	Watson Crick

ZOO503:Grand Quiz

Question # 18 of 30 (Start time: 01:13:15 AM, 06 July 2020)

species help shape the ecosystem and that can trigger large-scale changes if they disappear.

Select the correct option

●	Keystone
○	Classic
○	both
○	none

Question # 19 of 30 (Start time: 01:14:11 AM, 06 July 2020)

The German scientist _____ suggested that the continents moved across the Earth's surface on a lic

Select the correct option

◉	Alfred Wegener
○	Alfred Russel
○	George Cuvier
○	Charles Darwin

Question # 20 of 30 (Start time: 01:14:56 AM, 06 July 2020)

A classic keystone species is the _____.

Select the correct option

○	camel
○	goat
○	lion
●	elephant

ZOO503:Grand Quiz

Question # 21 of 30 (Start time: 01:15:55 AM, 06 July 2020)

The functions of single fossil organisms ____.

Select the correct option

○	paleoautecology
○	paleosynecology
○	both
○	none

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 22 of 30 (Start time: 01:16:40 AM, 06 July 2020)

Best amount of fossils that are present at sediment-water interface

Select the correct option

○	Epifaunal organisms
○	Infaunal organisms
○	Both epifaunal and infaunal organisms
○	None of the given options

ZOO503:Grand Quiz

Question # 23 of 30 (Start time: 01:17:21 AM, 06 July 2020)

The _____, successfully built reefs in unstable environments

Select the correct option

○	Sponges
○	Archaeocyaths
○	Jelly fish
○	None

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 24 of 30 (Start time: 01:17:58 AM, 06 July 2020)

By the Late Carboniferous a very diverse _____ fauna had evolved, with forms such as the dragonflies and mayflies capab

Select the correct option

○	Ticks
●	Insect
○	Arachnids
○	Scorpion

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 25 of 30 (Start time: 01:18:30 AM, 06 July 2020)

The _____ group may prove to have as many as 10 million living species when the rich faunas of the tropics have been comp

Select the correct option

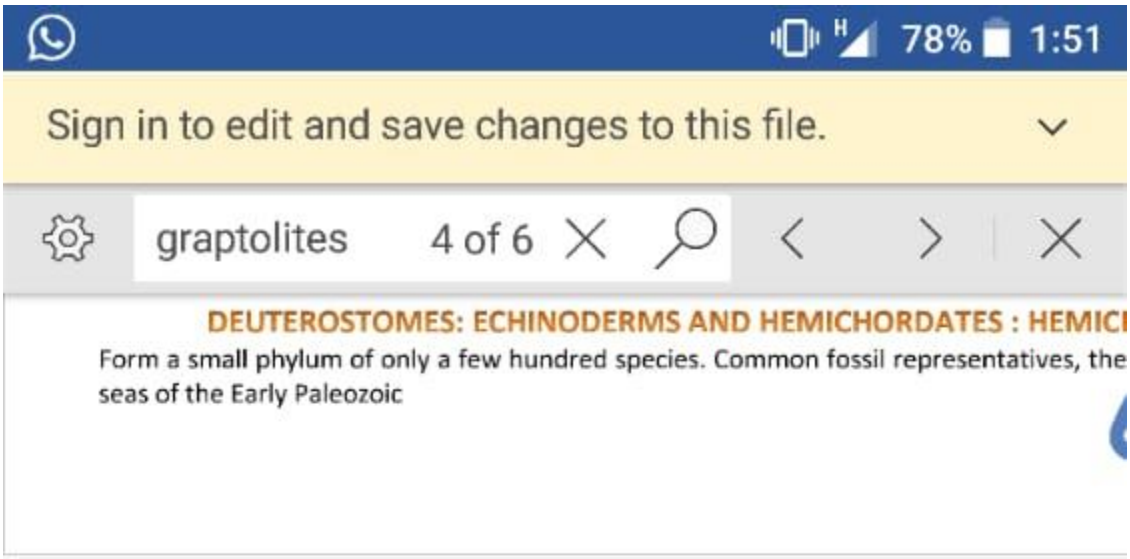
○	Myripods
○	Chelipods
○	Hexapods
○	Crustacea

Question # 26 of 30 (Start time: 01:19:27 AM, 06 July 2020)

What is the living status of Graptolites which belong to phylum Hemichordate?

Select the correct option

- Endangered
- Critically Endangered
- Least Concern
- Extinct



GRAPTOLITES

- Extinct
- life styles are difficult to interpret
- small, soft-bodied animals with bilateral symmetry and a lack of segmentation

HEMICHORDATA Two classes

- First, the tiny, mainly colonial, pterobranchs in sessile benthos
- Second, the larger infaunal acorn or tongue worms, the enteropneusts

Closely related to cephalochordates and urochordates or tunicates. Molecular data sugg groups are more closely related to the chordates. Graptolites probably pursued benthic (dendroids and graptoloids) and automobile (graptoloids) lifestyles. Graptoloids evol widespread. The ideal zone fossils in rocks of Ordovician-Silurian and Early Devonian age.

- Class Graptolithina
- Order Dendroidea Cambrian to Carboniferous
- Order Tuboidea Ordovician to Silurian
- Order Camaroidea Ordovician
- Order Crustoidea Ordovician
- Order Dithecoidea Cambrian to Silurian
- Order Graptoloidea Ordovicianto Devonian

FISHES AND BASAL TETRAPODS: INTRODUCTION

VERTEBRATES

Subgroup of the Phylum Chordata, a major deuterostome clade. The evolution of fishes the present day, and the Paleozoic tetrapods. Characterized by a skeleton made from b is a group of phosphate minerals, usually referring to hydroxylapatite, fluorapatite and c concentrations of OH−, F− and Cl− ions, respectively, in the crystal.) The oldest vertebr creatures from the Early Cambrian of China.

ANCIENT FISHES

Armored fishes were abundant in Devonian seas and lakes. After the Devonian, the c fishes radiated in several phases.

CONODONTS

Conodonts commonly occur as tooth-like elements that are useful in biostratigraphy, a

MC190202027: FAIZAN MAMOON

ZOO503:Grand Quiz

Question # 27 of 30 (Start time: 01:20:48 AM, 06 July 2020)

Life on Earth, has been evolving for up to ___ billion years.

Select the correct option

○	2
○	3
○	4
○	5